CHAPTER 02

Isomers

Content and rendering review packet

Original app content and actual Chromium screenshots. No textbook passages or textbook figures are embedded. Source labels and curriculum headings are not evidence of factual correctness.

Source snapshot SHA-256: 22b996e8be8e705c37407c92ba9f1d04c2b2e61066f7e3bf0bbd650678d6ab96

Review all claims, molecule graphs, lessons, exercises, hints and displayed states. Keys are author claims, not ground truth. Screenshots show static states with reduced motion: they do not prove animation timing or persistence behavior.

For every finding: item ID/page; severity (critical/major/minor); exact claim; derivation or reference evidence; proposed correction; confidence. List unreviewed/uncertain items. Model agreement is not scientific verification.

Coverage checklist

ch02:exercise:isomer-pair | isomer-pair:stage-1, isomer-pair:stage-2, isomer-pair:stage-3

ch02:exercise:butane-rotation | butane-rotation:stage-1, butane-rotation:stage-2, butane-rotation:stage-3

ch02:exercise:stereocenters | stereocenters:stage-1, stereocenters:stage-2, stereocenters:stage-3

ch02:exercise:fischer-cip | fischer-cip:stage-1, fischer-cip:stage-2, fischer-cip:stage-3

ch02:exercise:ez-chirality | ez-chirality:stage-1, ez-chirality:stage-2, ez-chirality:stage-3

ch02:exercise:specific-rotation | specific-rotation:stage-1, specific-rotation:stage-2, specific-rotation:stage-3

ch02:exercise:optical-mixture | optical-mixture:stage-1, optical-mixture:stage-2, optical-mixture:stage-3

ch02:exercise:chiral-resolution | chiral-resolution:stage-1, chiral-resolution:stage-2, chiral-resolution:stage-3

ch02:lesson:connectivity | connectivity:rule, connectivity:cue, connectivity:trap, connectivity:app, connectivity:step-1, connectivity:step-2, connectivity:step-3

ch02:lesson:conformations | conformations:rule, conformations:cue, conformations:trap, conformations:app, conformations:step-1, conformations:step-2, conformations:step-3

ch02:lesson:chirality | chirality:rule, chirality:cue, chirality:trap, chirality:app, chirality:step-1, chirality:step-2

ch02:lesson:configuration | configuration:rule, configuration:cue, configuration:trap, configuration:app, configuration:step-1, configuration:step-2, configuration:step-3

ch02:lesson:alkene | alkene:rule, alkene:cue, alkene:trap, alkene:app, alkene:step-1, alkene:step-2

ch02:lesson:optical | optical:rule, optical:cue, optical:trap, optical:app, optical:step-1, optical:step-2, optical:step-3

ch02:molecule:ethanol | ethanol

ch02:molecule:ether | ether

ch02:molecule:isobutane | isobutane

ch02:molecule:butane | butane

ch02:molecule:cyclohexane | cyclohexane

ch02:molecule:butanol2 | butanol2

ch02:molecule:butanediol | butanediol

ch02:molecule:e-butene | e-butene

ch02:molecule:z-butene | z-butene

ch02:curriculum | scope-1, scope-2, scope-3

Curriculum scope

scope-1: 2.1 Structural Isomers

scope-2: 2.2 Stereoisomers

scope-3: 2.3 Relative and Absolute Configurations

Objective: Classify isomer relationships

Objective: Assign CIP and E/Z descriptors

Objective: Analyze conformations, symmetry, and optical mixtures

Objective: Calculate specific and observed rotation with explicit units

Objective: Distinguish optical sign, enantiomeric excess, and chiral resolution

Supplied Kaplan curriculum: Chapter 2; section scope checked against local body and summary | organic_chemistry_kaplan/OEBPS/chapter02b.xhtml

IUPAC optical rotatory power definition | https://old.goldbook.iupac.org/html/O/O04313.html

IUPAC stereochemical terminology: Fischer projection and conventions | https://iupac.qmul.ac.uk/stereo/FM.html

No source passage is included. Request relevant authoritative excerpts when a claim cannot be checked by derivation.

connectivity | Connectivity vs. spatial arrangement

Same formula is only the first test

connectivity:rule | Core Rules

Constitutional isomers share a molecular formula but differ in connectivity. Stereoisomers have the same connectivity and differ in spatial arrangement. Conformers interconvert by rotations about single bonds.

Schematic summary: Same formula is only the first test

Formula — Compare elemental composition

Connectivity — Compare the bonded atom pairs

Space — Compare spatial arrangement

connectivity:cue | Memory Cues

Formula → connections → space. Compare in that order.

Schematic summary: Choose the case that applies

Same formula, different connectivity — Constitutional isomers

Same connectivity, different configuration — Stereoisomers

Only drawing orientation changes — Not evidence of a different isomer

connectivity:trap | Common Mistakes / MCAT Traps

A different drawing orientation does not prove isomerism. Rotating the whole molecule changes neither connectivity nor configuration.

Schematic summary: Shortcut versus a better check

Avoid — Calling every rotated drawing a new isomer

Check — Connectivity and configuration, not the page orientation

connectivity:app | MCAT Applications

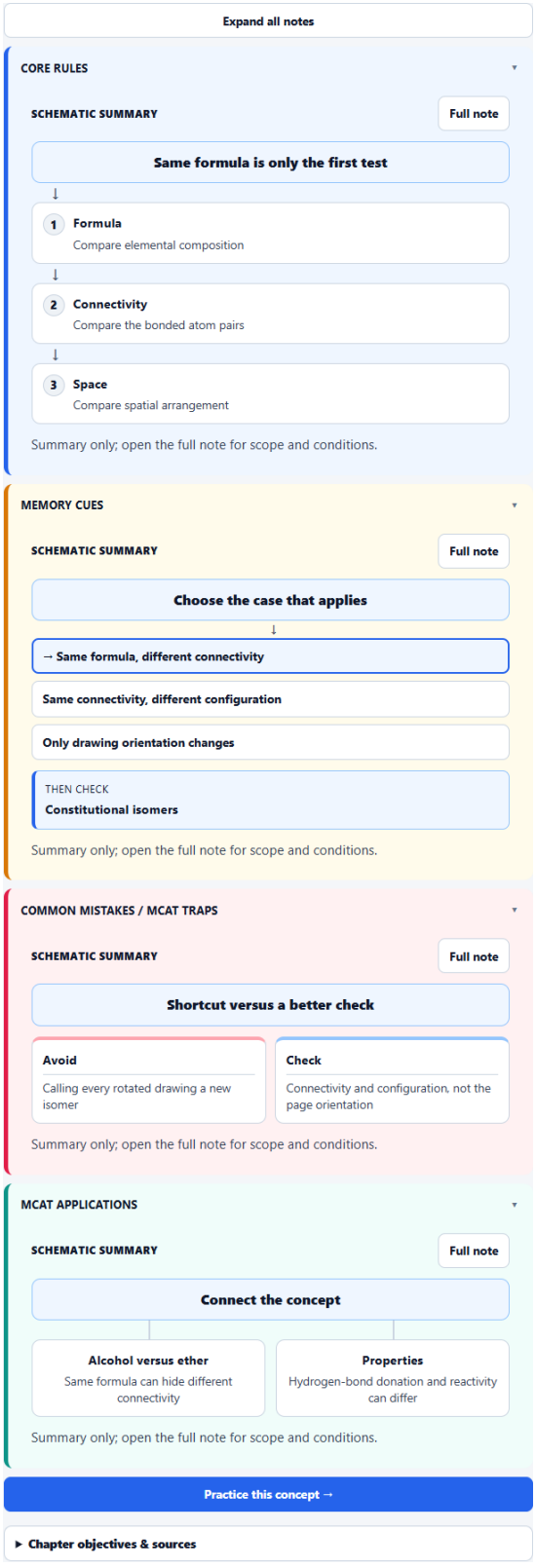
An alcohol and an ether can share a formula but differ in hydrogen-bond donation, boiling point, and reactivity.

Schematic summary: Connect the concept

Alcohol versus ether — Same formula can hide different connectivity

Properties — Hydrogen-bond donation and reactivity can differ

Screenshot: connectivity-visuals



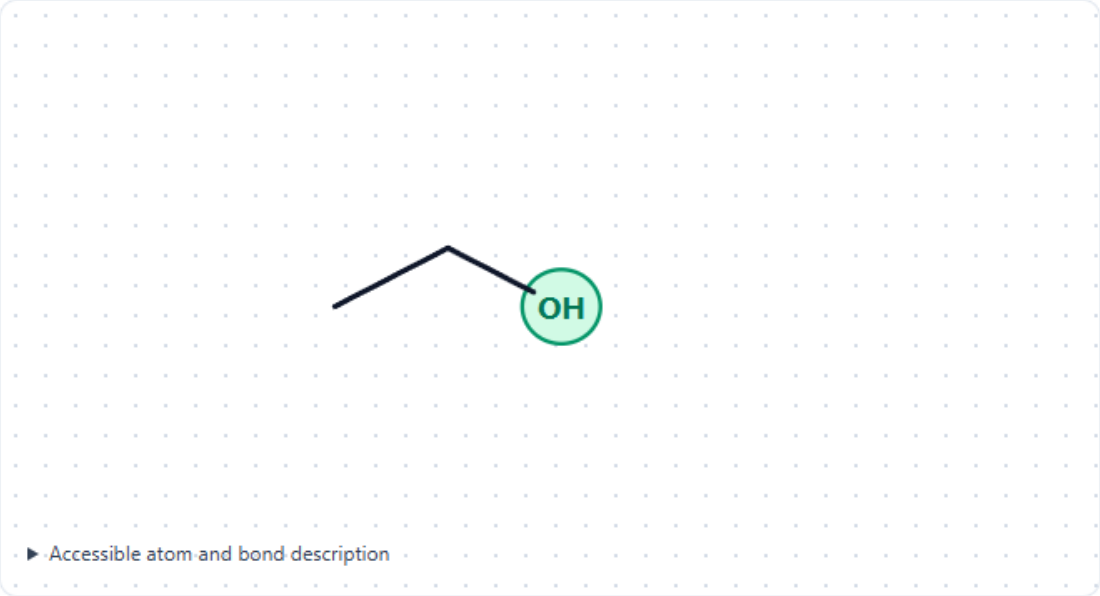
connectivity:step-1 | 1. Alcohol connectivity

Count two carbons, six hydrogens, and one oxygen. The oxygen bears H and can donate a hydrogen bond.

Displayed molecule ID: ethanol

{"highlights": {"prin": [2]}, "revealName": true}

Screenshot: connectivity-step-1



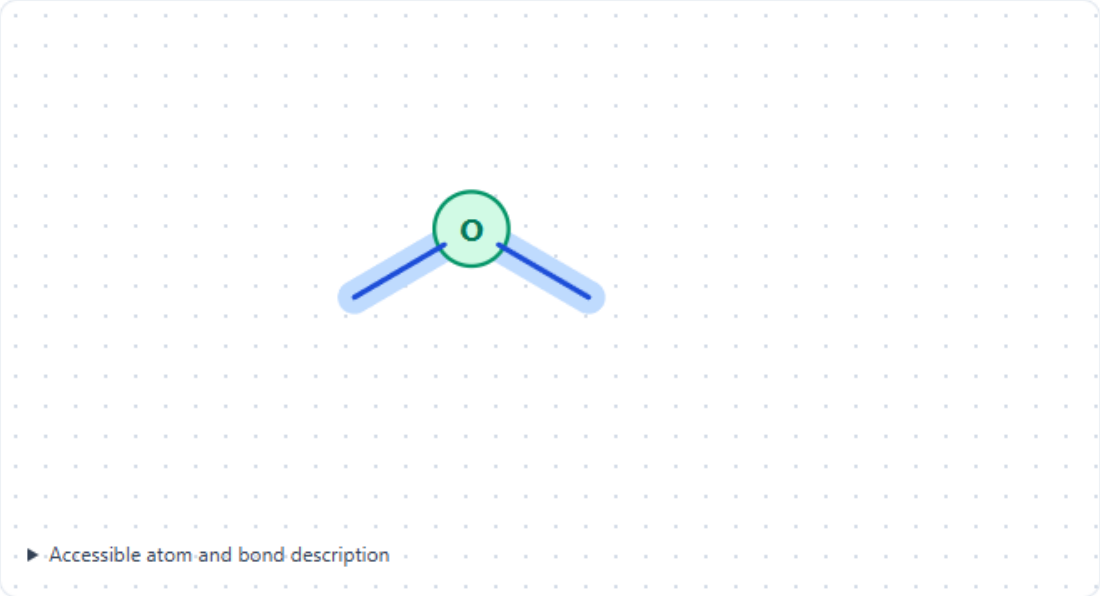
connectivity:step-2 | 2. Ether connectivity

The same formula now has oxygen between two carbons. No O–H bond is present. This is a constitutional isomer, not a different conformation.

Displayed molecule ID: ether

{"highlights": {"path": [0, 1, 2], "prin": [1]}, "revealName": true}

Screenshot: connectivity-step-2



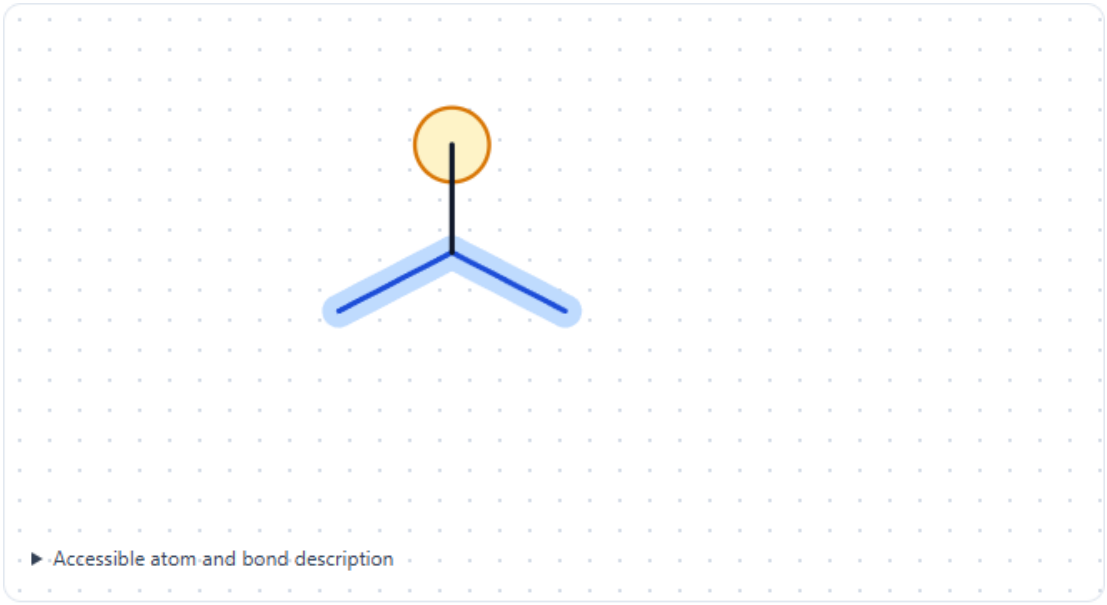
connectivity:step-3 | 3. Branched framework

Butane and 2-methylpropane are another constitutional-isomer pair: identical formula, different carbon skeletons.

Displayed molecule ID: isobutane

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Screenshot: connectivity-step-3



conformations | Conformations and ring strain

Rotation changes energy, not identity

conformations:rule | Core Rules

Angle strain is deviation from preferred bond angles; torsional strain comes from eclipsing bonds; nonbonded (steric) strain comes from crowding of atoms not bonded to each other. Butane anti and gauche conformations are staggered, but gauche has greater methyl–methyl crowding. A cyclohexane chair has nearly ideal angles and staggered bonds; bulky substituents usually prefer equatorial sites to reduce 1,3-diaxial crowding.

Schematic summary: Rotation changes energy, not identity

Angle strain — Deviation from preferred angles

Torsional strain — Eclipsing interactions

Steric strain — Crowding of nonbonded atoms

conformations:cue | Memory Cues

For the butane methyl–methyl dihedral shown: anti = 180°, gauche = 60°, ordinary eclipsed = 120°, and totally (fully) eclipsed = 0°. At 0° the methyl groups eclipse one another, giving the highest energy of these conformations.

Schematic summary: Choose the case that applies

Butane at 180° — Anti, staggered

Butane at 60° — Gauche, staggered

Butane at 120° / 0° — Eclipsed / fully eclipsed

conformations:trap | Common Mistakes / MCAT Traps

Axial is not synonymous with up: axial directions alternate. A chair flip swaps axial/equatorial positions while preserving up/down and cis/trans. Boat flagpole contacts are steric interactions; do not call every unfavorable contact angle strain.

Schematic summary: Shortcut versus a better check

Avoid — Equating axial with up

Check — Chair flip swaps axial/equatorial but retains up/down and cis/trans

conformations:app | MCAT Applications

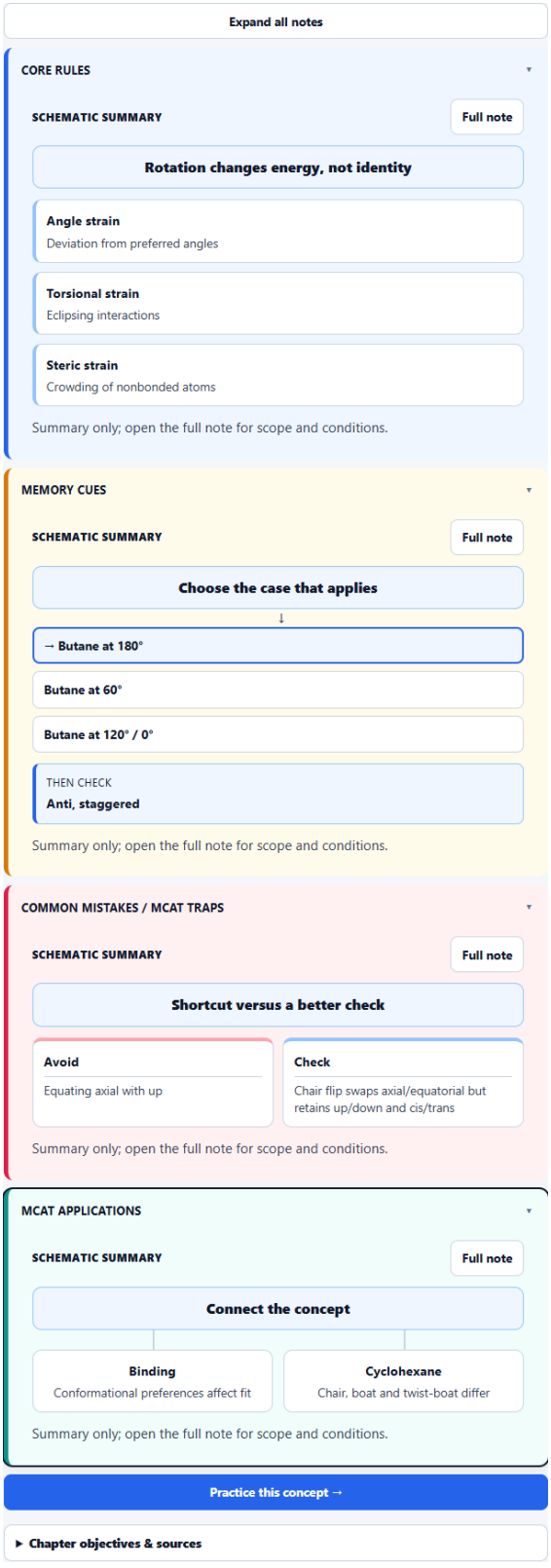
Conformational preferences affect binding. Cyclobutane puckers and cyclopentane can adopt envelope-like shapes to reduce eclipsing; cyclohexane has higher-energy boat and twist-boat forms, and its chair-flip pathway passes through half-chair transition states.

Schematic summary: Connect the concept

Binding — Conformational preferences affect fit

Cyclohexane — Chair, boat and twist-boat differ

Screenshot: conformations-visuals



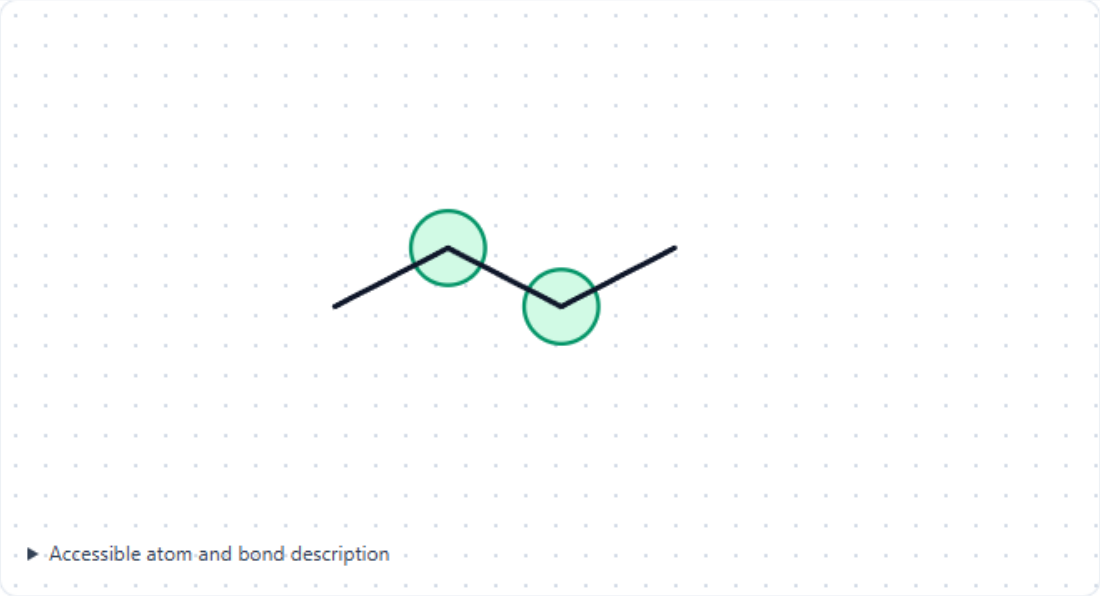
conformations:step-1 | 1. Rotate a single bond

Look down the middle C–C bond of butane. Adjust the angle to compare torsional and methyl–methyl interactions.

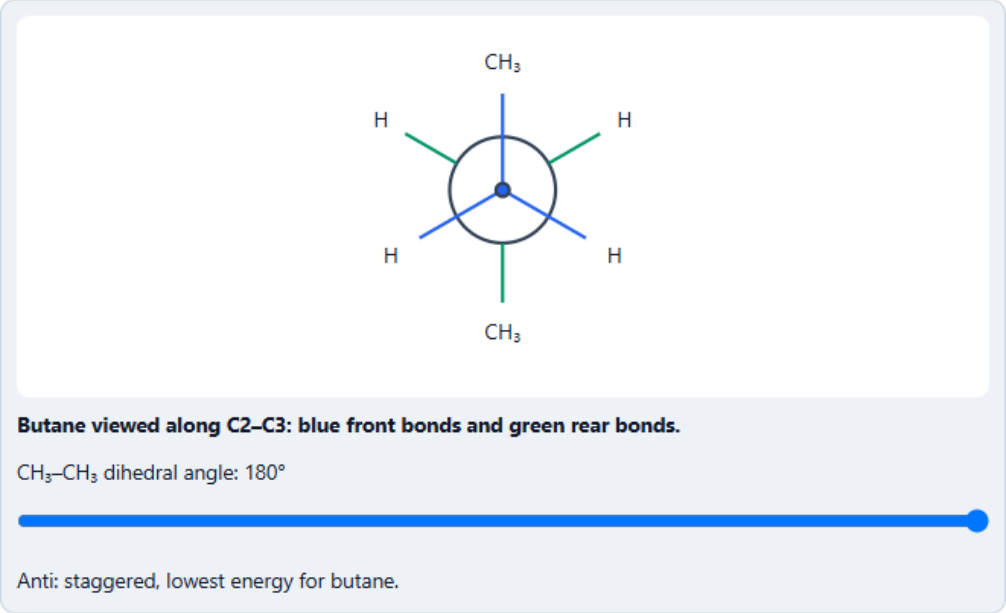
Displayed molecule ID: butane

{"highlights": {"prin": [1, 2]}, "revealName": true}

Screenshot: conformations-step-1



Screenshot: conformations-step-1-diagram



{"diagramSpecification": {"type": "newman", "caption": "Butane viewed along C2–C3: blue front bonds and green rear bonds."}}

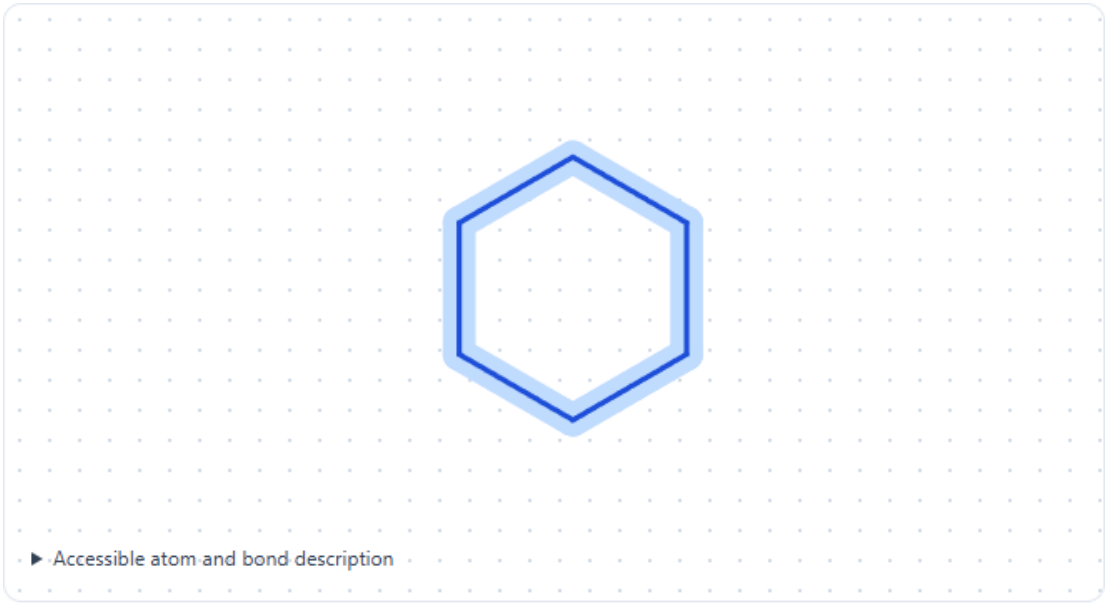
conformations:step-2 | 2. Ring constraints

This hexagon shows connectivity only, not a flat equilibrium geometry. Cyclohexane normally adopts a chair. An axial substituent interacts with axial groups at C3 and C5; an equatorial position avoids much of this crowding.

Displayed molecule ID: cyclohexane

{"highlights": {"path": [0, 1, 2, 3, 4, 5]}, "revealName": true}

Screenshot: conformations-step-2



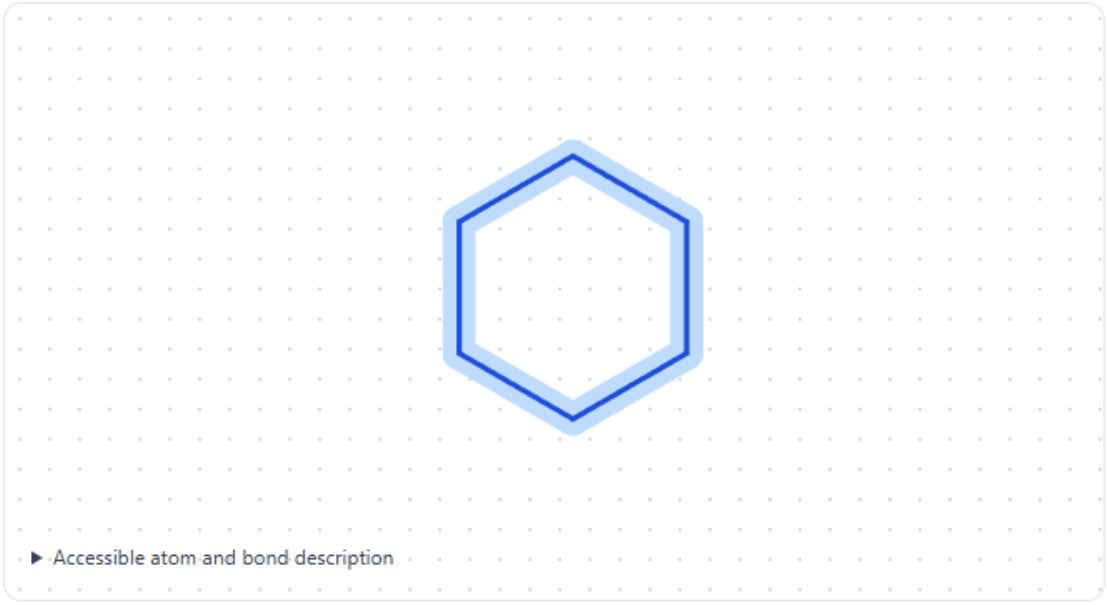
conformations:step-3 | 3. Ring-flip pathway

This hexagon is a connectivity map, not a chair drawing. A chair flip passes through higher-energy shapes: half-chair transition states and twist-boat regions; the boat has flagpole crowding and eclipsing interactions. The final chair exchanges axial/equatorial sites without changing up/down.

Displayed molecule ID: cyclohexane

{"highlights": {"path": [0, 1, 2, 3, 4, 5]}, "revealName": true}

Screenshot: conformations-step-3



chirality | Chirality, enantiomers, and meso compounds

Mirror image ≠ always a different molecule

chirality:rule | Core Rules

A tetrahedral carbon with four different substituents is a common stereogenic center. Enantiomers are nonsuperimposable mirror images. Diastereomers are stereoisomers that are not mirror images; E/Z pairs qualify too.

Schematic summary: Mirror image ≠ always a different molecule

Four different groups — Common tetrahedral stereogenic-center test

Mirror image — Check superimposability

Whole molecule — Compare every center and internal symmetry

chirality:cue | Memory Cues

One inverted center in a one-center molecule gives its enantiomer. With several centers, compare the entire molecule.

Schematic summary: Choose the case that applies

Nonsuperimposable mirror images — Enantiomers

Stereoisomers, not mirror images — Diastereomers

Internal symmetry — Check for a meso compound

chirality:trap | Common Mistakes / MCAT Traps

2^n is a maximum number of stereoisomers, not a guarantee. Internal symmetry can make a molecule with stereocenters achiral (meso). R/S does not predict the sign of optical rotation.

Schematic summary: Shortcut versus a better check

Avoid — Treating 2^n as a guaranteed stereoisomer count

Check — It is a maximum; symmetry can reduce the count

chirality:app | MCAT Applications

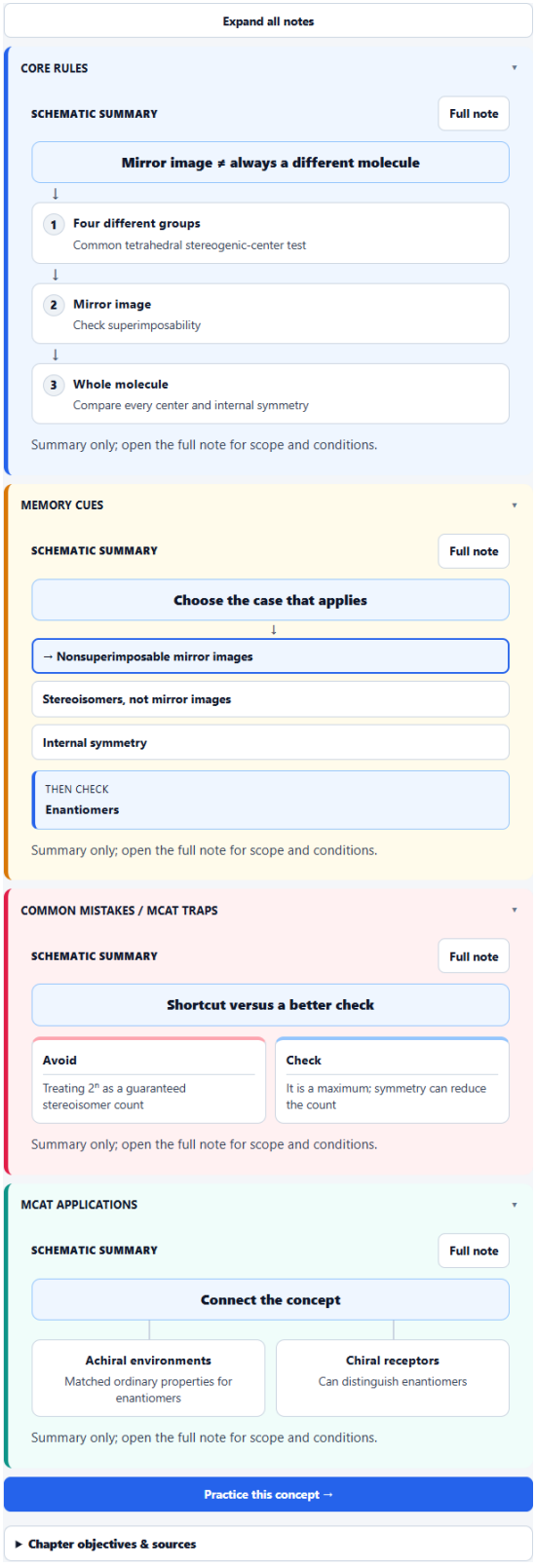
Enantiomers have the same ordinary properties in achiral environments but opposite optical rotations under matched conditions; chiral receptors can distinguish them. A suitable single-enantiomer resolving agent can convert a racemate into diastereomeric salts or derivatives. Their different properties permit separation, followed by regeneration of the original enantiomers. Chiral chromatography is another approach.

Schematic summary: Connect the concept

Achiral environments — Matched ordinary properties for enantiomers

Chiral receptors — Can distinguish enantiomers

Screenshot: chirality-visuals



chirality:step-1 | 1. Identify four groups

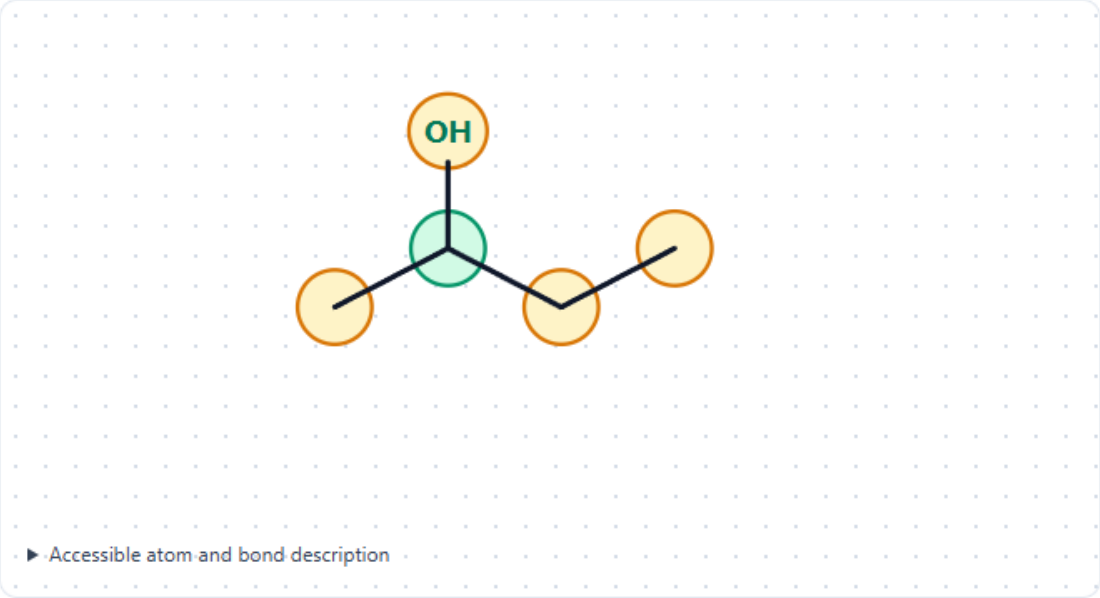
The alcohol-bearing carbon has OH, H, methyl, and ethyl as four distinct groups. Both carbons of the ethyl group are highlighted. The flat graph does not assign R or S. Try It Yourself removes these walkthrough highlights before asking for an

atom selection.

Displayed molecule ID: butanol2

{"highlights": {"prin": [1], "sub": [0, 2, 3, 4]}, "revealName": false}

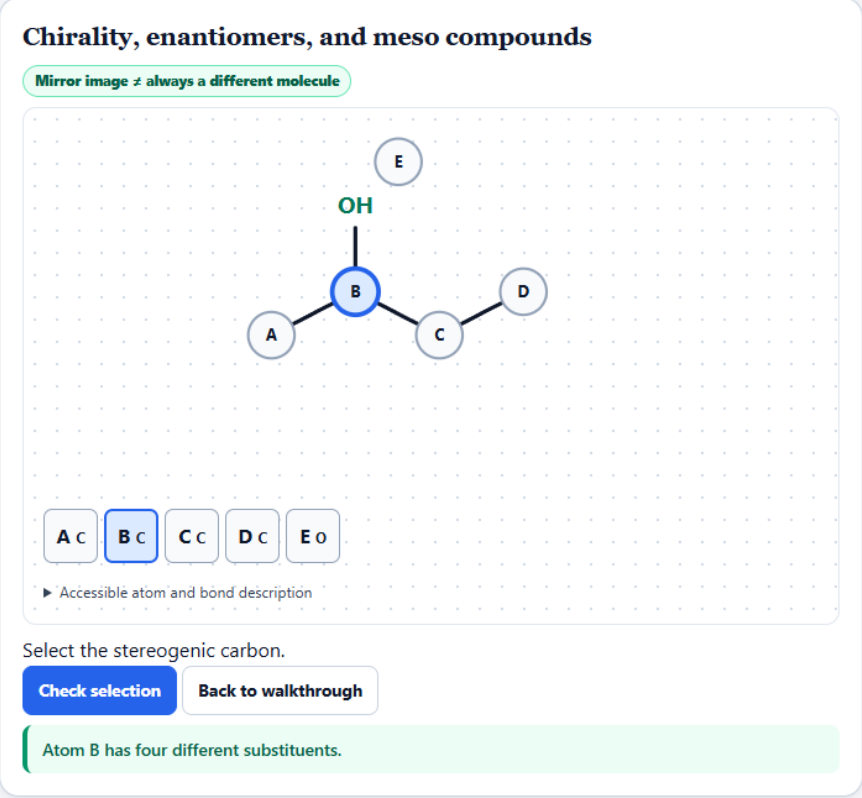
Screenshot: chirality-step-1



Embedded challenge: Select the stereogenic carbon.

{"prompt": "Select the stereogenic carbon.", "answer": [1], "explanation": "Atom B has four different substituents."}

Screenshot: chirality-step-1-challenge



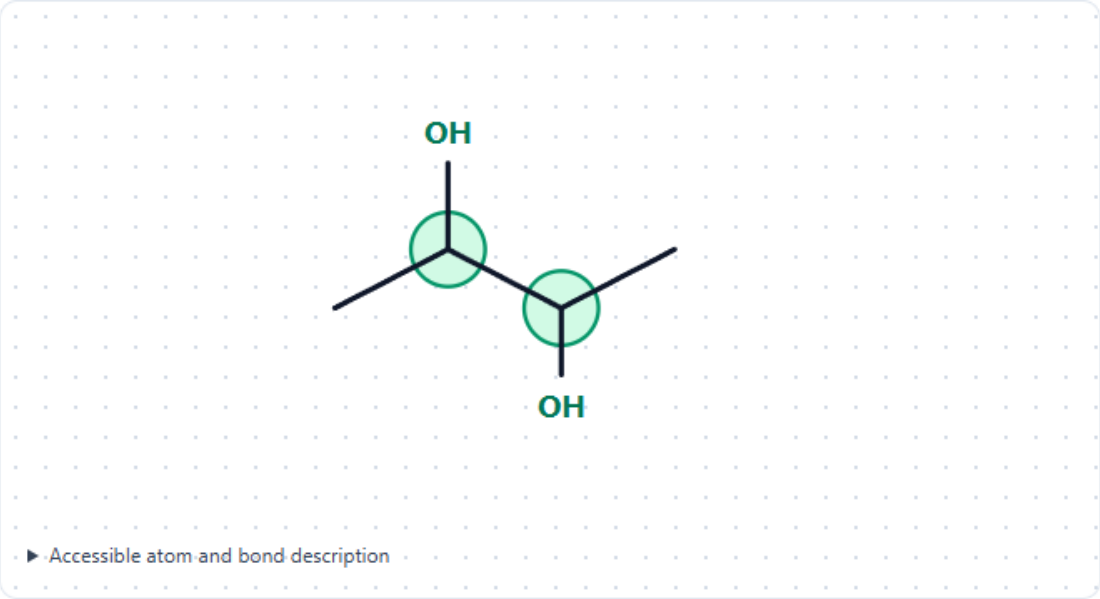
chirality:step-2 | 2. Count with symmetry

Butane-2,3-diol has two potential stereocenters. Its (2R,3S) and (2S,3R) descriptions refer to the same meso compound because of symmetry; the RR and SS forms are an enantiomeric pair. Total: three stereoisomers.

Displayed molecule ID: butanediol

{"highlights": {"prin": [1, 2]}, "revealName": true}

Screenshot: chirality-step-2



configuration | CIP priorities and Fischer projections

Assign priority before tracing direction

configuration:rule | Core Rules

Rank directly attached atoms by atomic number; isotopes by mass. Resolve ties one sphere at a time using sorted atom lists. Treat multiple bonds with duplicate-atom bookkeeping. With priority 4 away, clockwise 1→2→3 is R and counterclockwise is S. Absolute configuration assigns a stereochemical arrangement independently (for example R/S). Relative configuration describes a relationship to another stereocenter or reference compound. A remote chemical change can retain the local spatial arrangement while changing CIP priorities and therefore its R/S label; reassess priorities instead of equating a letter change with physical inversion.

Schematic summary: Assign priority before tracing direction

Rank atoms — Atomic number; isotope mass for isotopes

Resolve ties — Sorted lists, one sphere at a time

Group 4 away — Clockwise is R; counterclockwise is S

configuration:cue | Memory Cues

Fischer: horizontal bonds come forward; vertical bonds go back. If group 4 is horizontal, reverse the apparent result. We draw C1 at the top here. A 180° rotation of the entire Fischer drawing preserves its configuration.

Schematic summary: Choose the case that applies

Fischer group 4 vertical — Points back: use the apparent direction

Fischer group 4 horizontal — Points forward: reverse the apparent direction

configuration:trap | Common Mistakes / MCAT Traps

A 180° in-plane Fischer rotation preserves configuration; a 90° rotation does not. D/L is a relative-configuration convention, not a synonym for R/S or +/−.

Schematic summary: Shortcut versus a better check

Avoid — Rotating a Fischer drawing by 90°

Check — A 180° whole-drawing rotation preserves configuration

configuration:app | MCAT Applications

Most proteinogenic L amino acids are S; L cysteine is R because sulfur changes CIP ranking. Glycine has no stereogenic alpha carbon.

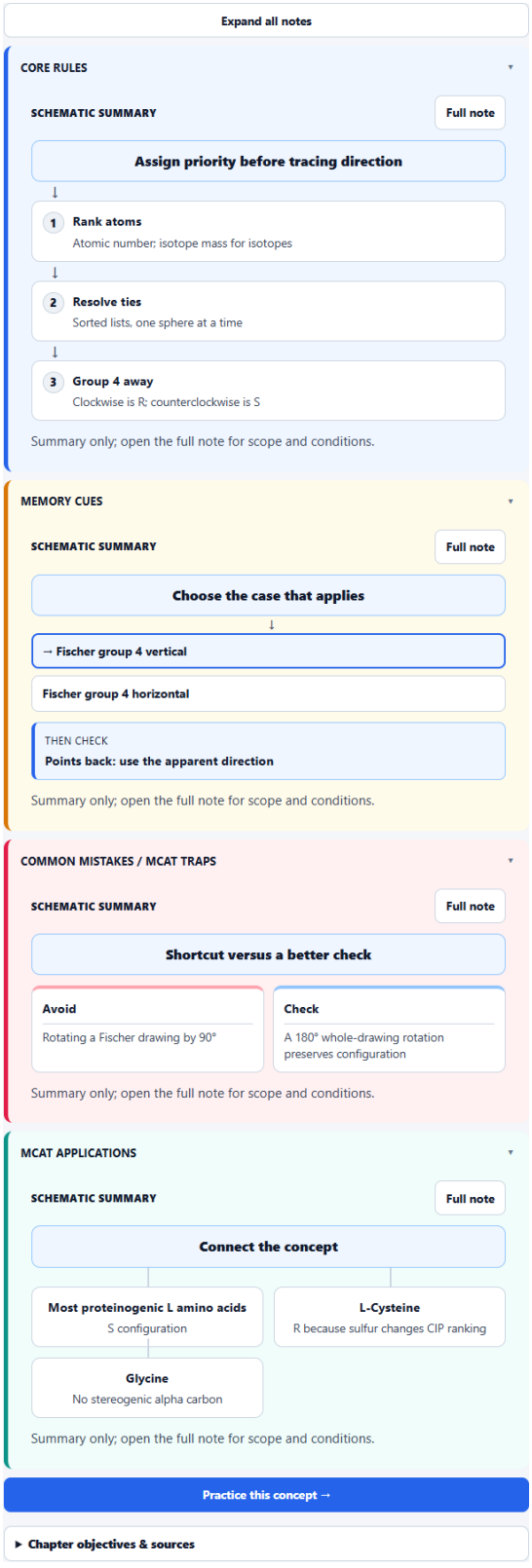
Schematic summary: Connect the concept

Most proteinogenic L amino acids — S configuration

L-Cysteine — R because sulfur changes CIP ranking

Glycine — No stereogenic alpha carbon

Screenshot: configuration-visuals



configuration:step-1 | 1. Rank the substituents

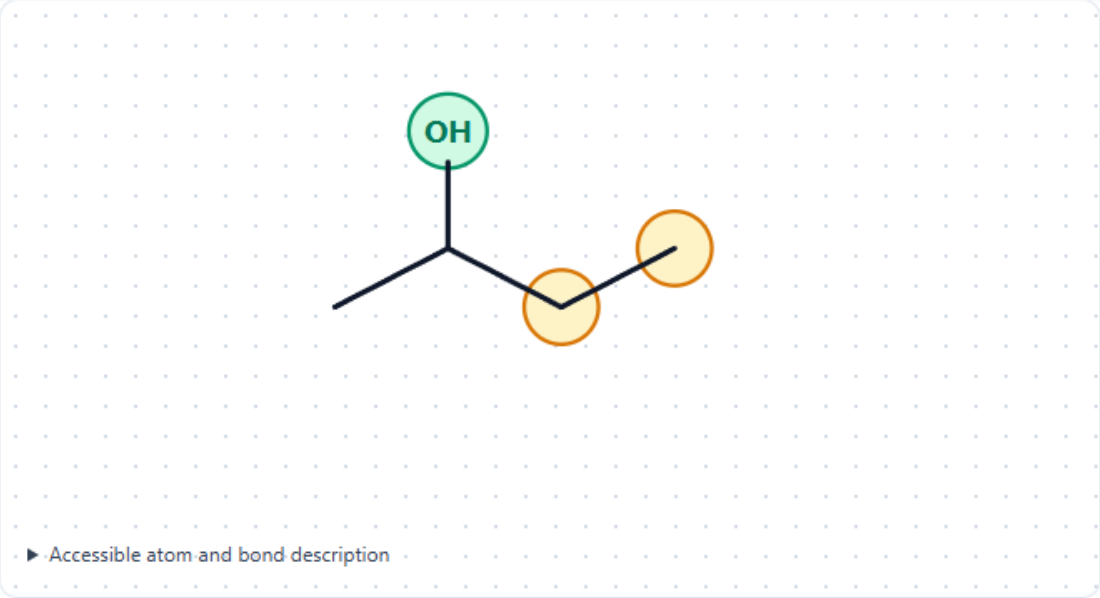
OH ranks first. Ethyl beats methyl at the next sphere: [C,H,H] beats [H,H,H]. The terminal CH_3 carbon of the ethyl group supplies the C that breaks the tie; both carbons of ethyl are highlighted in orange, while the competing methyl carbon is

unhighlighted. H ranks fourth.

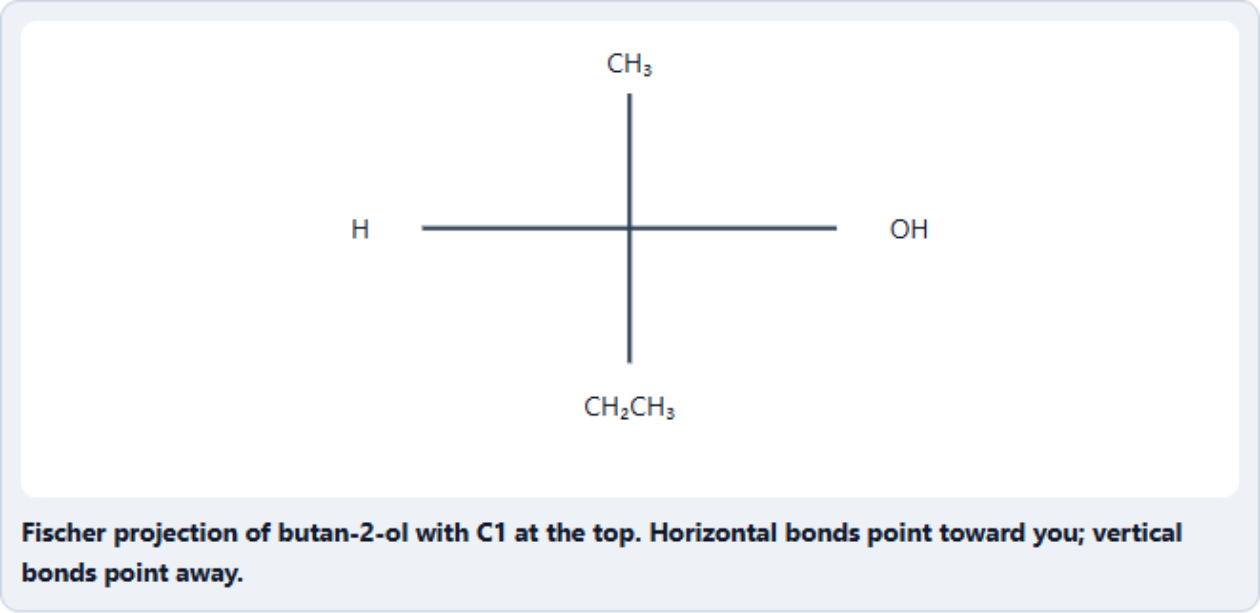
Displayed molecule ID: butanol2

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Screenshot: configuration-step-1



Screenshot: configuration-step-1-diagram



{"diagramSpecification": {"type": "fischer", "top": "CH_3", "bottom": "CH_2CH_3", "left": "H", "right": "OH", "caption": "Fischer projection of butan-2-ol with C1 at the top. Horizontal bonds point toward you; vertical bonds point away."}}

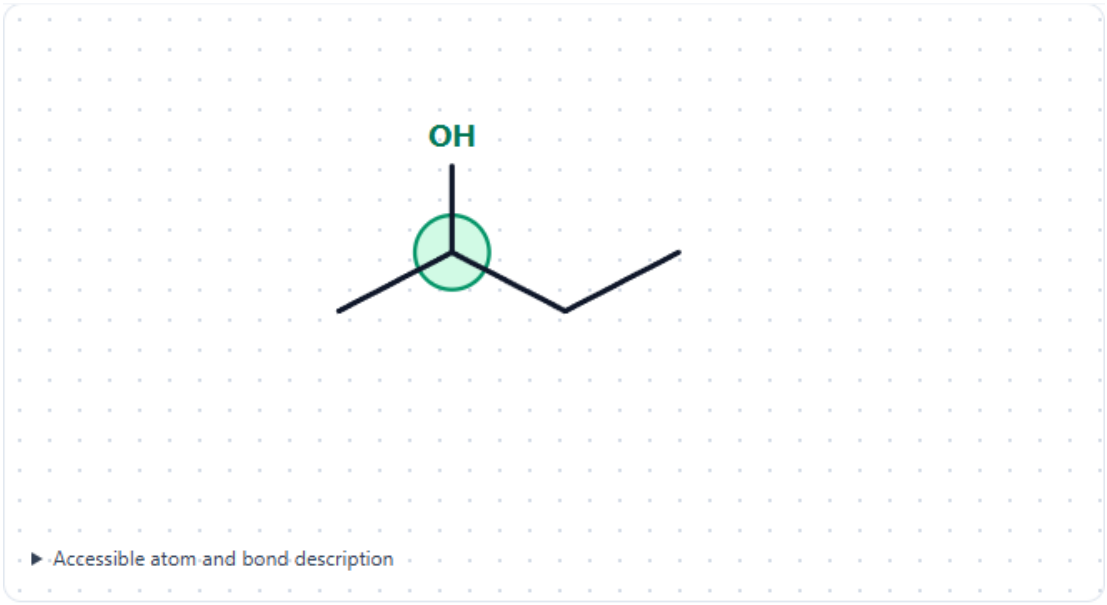
configuration:step-2 | 2. Orient then assign

Use the same Fischer projection with C1 at the top. Priorities are OH (right) > ethyl (bottom) > methyl (top) > H (left). The apparent 1→2→3 order is clockwise; H is horizontal and toward you, so reverse it: S.

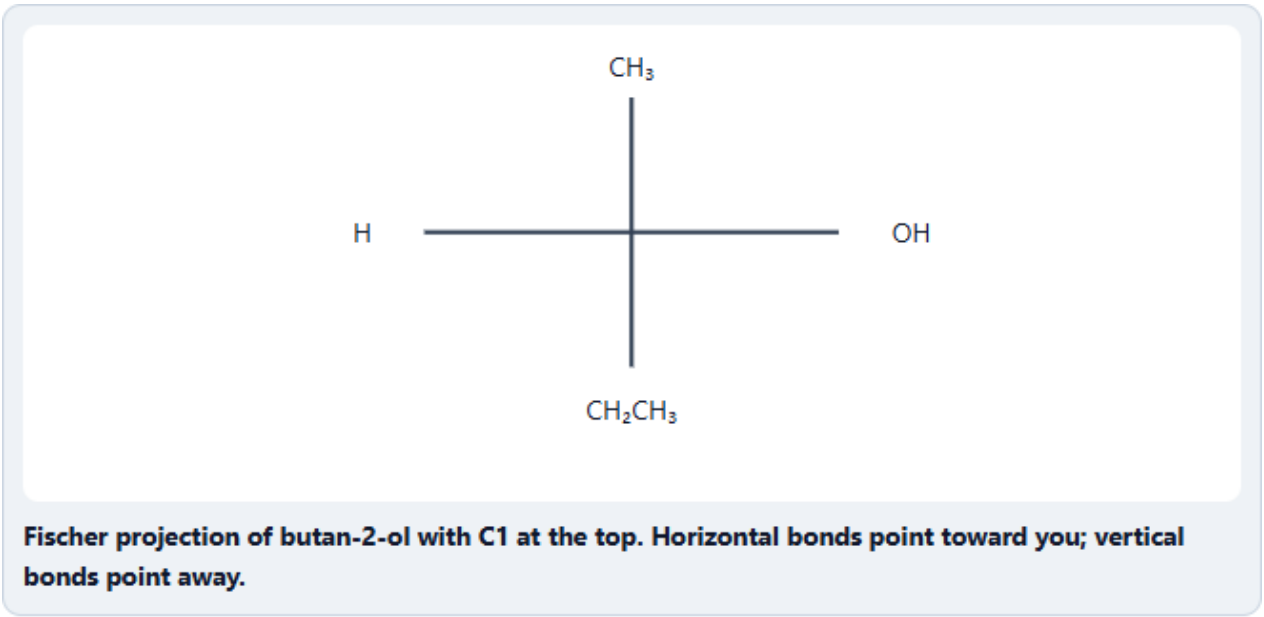
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Screenshot: configuration-step-2



Screenshot: configuration-step-2-diagram



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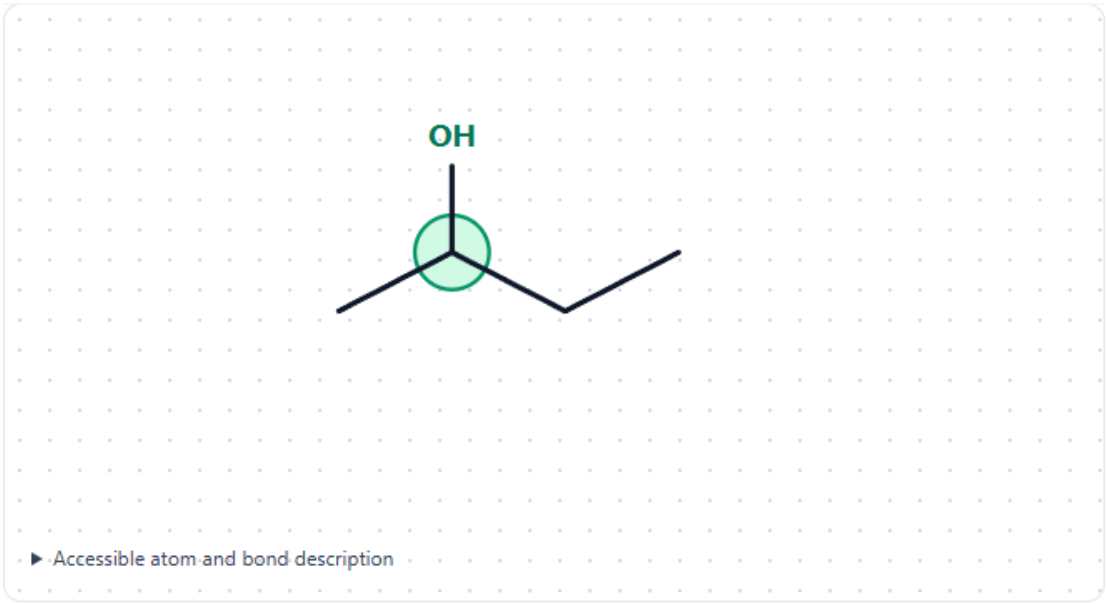
configuration:step-3 | 3. Mirror the projection

Interchanging H and OH inverts this single center. The mirrored projection is R and represents the enantiomer.

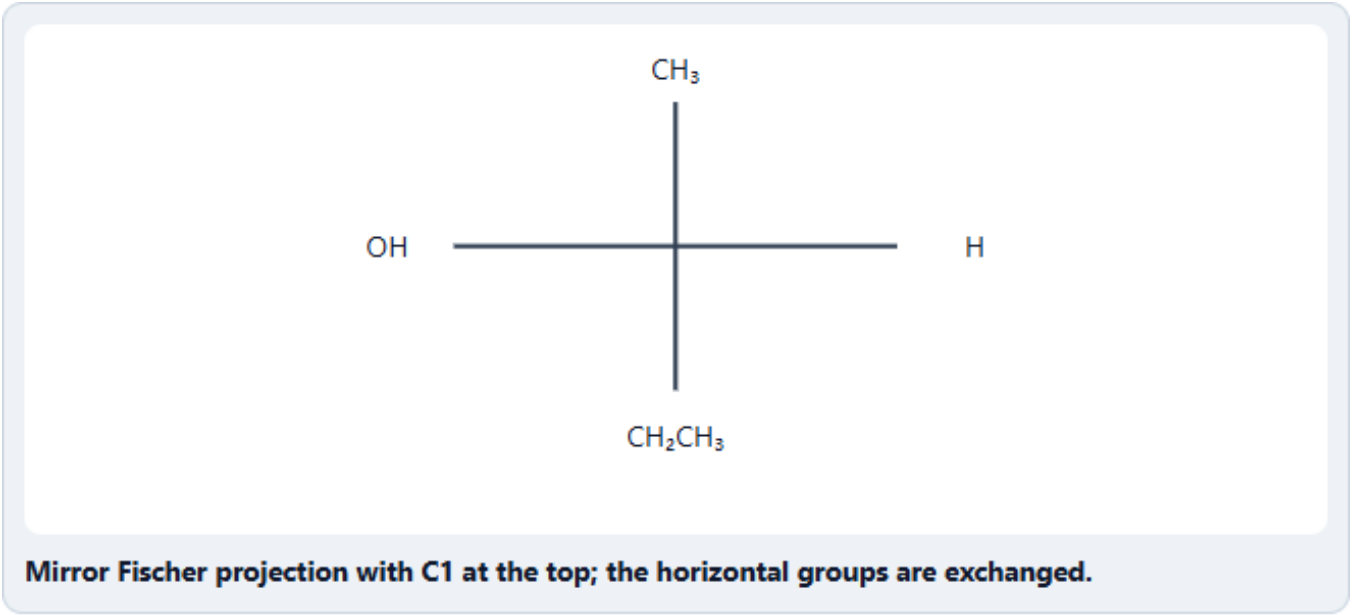
Displayed molecule ID: butanol2

{"highlights": {"prin": [1]}, "revealName": false}

Screenshot: configuration-step-3



Screenshot: configuration-step-3-diagram



{"diagramSpecification": {"type": "fischer", "top": "CH₃", "bottom": "CH₂CH₃", "left": "OH", "right": "H", "caption": "Mirror Fischer projection with C1 at the top; the horizontal groups are exchanged."}}

alkene | E/Z configuration and achirality

Rank each end separately

alkene:rule | Core Rules

E/Z is defined only if each double-bond carbon has two different substituents. Rank each end by CIP: Z puts the higher-priority groups on the same side; E puts them on opposite sides. E/Z geometry alone does not determine whether a whole molecule is chiral.

Schematic summary: Rank each end separately

Eligibility — Two different groups on each alkene carbon

Rank each end — Apply CIP separately

Compare the winners — Same side Z; opposite sides E

alkene:cue | Memory Cues

Z: higher-priority groups together. E: higher-priority groups opposite. Identify E/Z first, then assess the symmetry of the whole molecule separately.

Schematic summary: Choose the case that applies

Higher-priority groups together — Z

Higher-priority groups opposite — E

Identical groups at either end — E/Z is undefined

alkene:trap | Common Mistakes / MCAT Traps

Cis/trans cannot replace E/Z in every alkene. Both E- and Z-but-2-ene are achiral. Zero optical rotation can arise from an achiral compound or an equal mixture of enantiomers; it does not identify which situation applies.

Schematic summary: Shortcut versus a better check

Avoid — Assuming zero rotation proves a racemate

Check — An achiral molecule can also give zero rotation

alkene:app | MCAT Applications

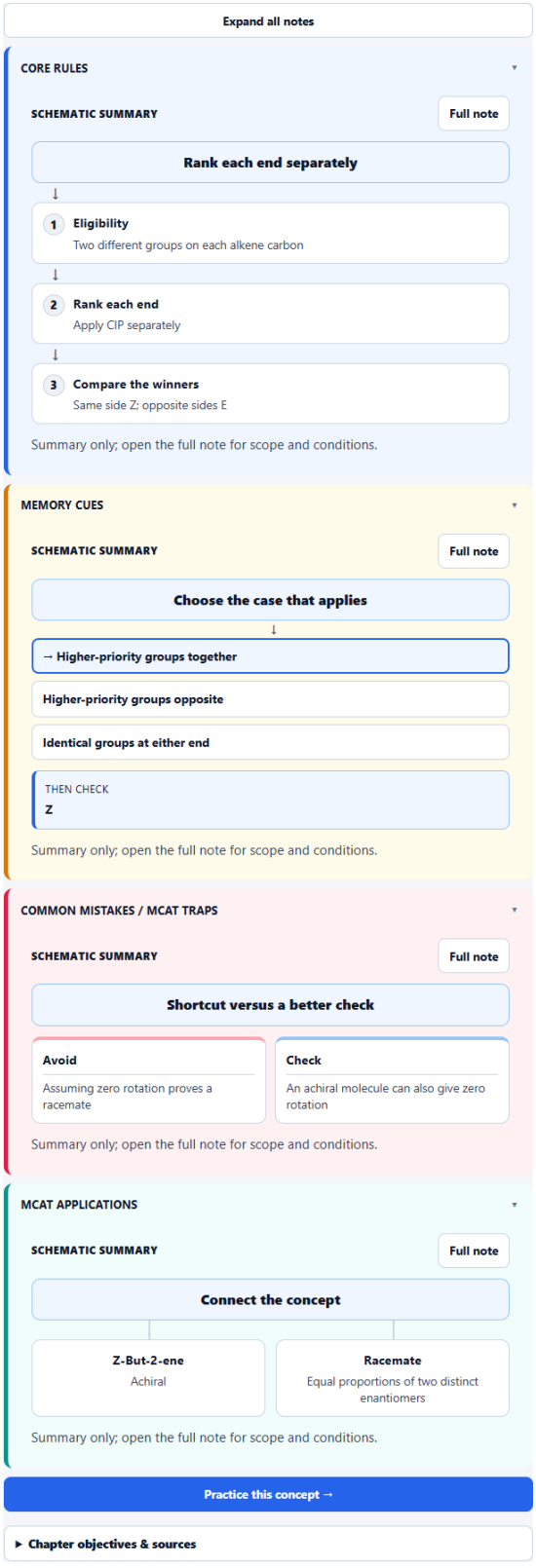
The pure Z isomer of but-2-ene is achiral and shows no ordinary optical activity. It is not a racemate. A racemate requires two distinct enantiomers in equal proportions.

Schematic summary: Connect the concept

Z-But-2-ene — Achiral

Racemate — Equal proportions of two distinct enantiomers

Screenshot: alkene-visuals



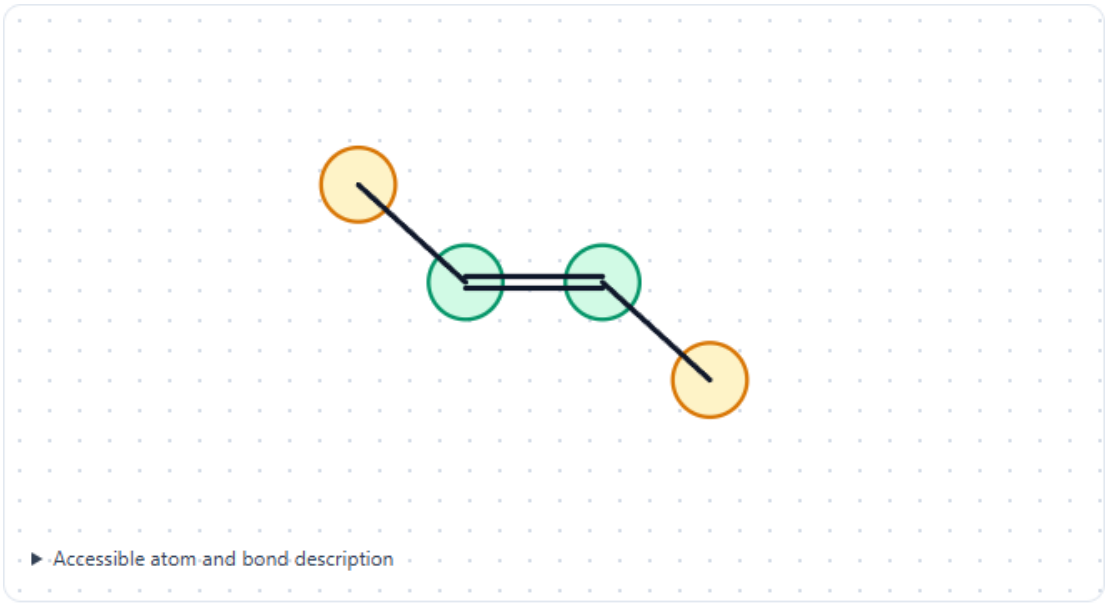
alkene:step-1 | 1. Compare the two ends

Each alkene carbon bears methyl and H. Methyl outranks H, and the methyl groups lie on opposite sides: E.

Displayed molecule ID: e-butene

{"highlights": {"prin": [1, 2], "sub": [0, 3]}, "revealName": true}

Screenshot: alkene-step-1



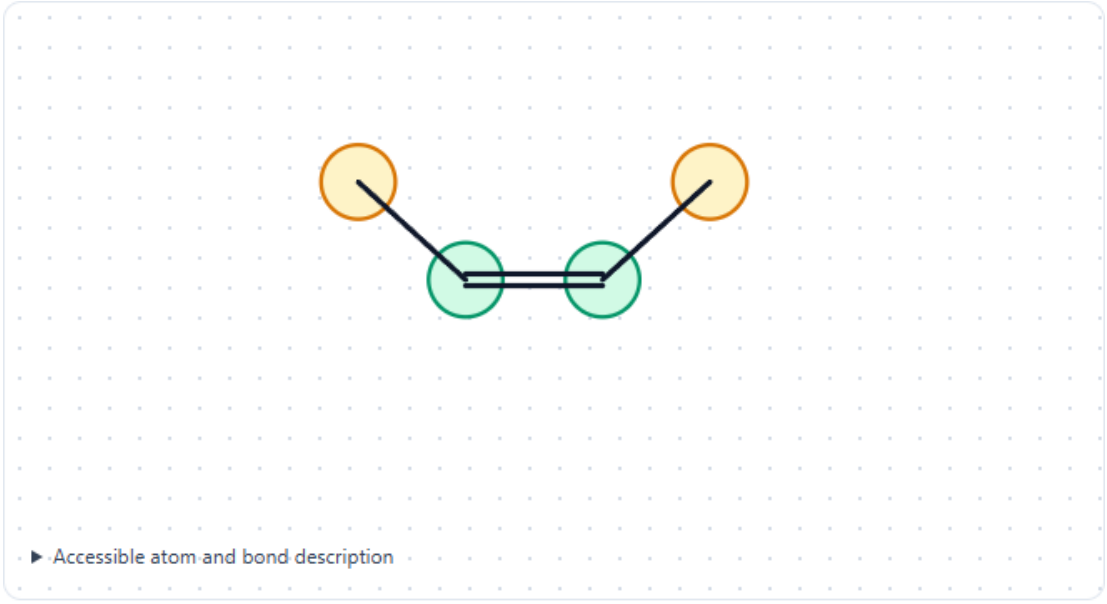
alkene:step-2 | 2. Same side

Both higher-priority methyl groups now lie on the same side: Z. Rotation about C=C would disrupt pi overlap, so these are configurational isomers.

Displayed molecule ID: z-butene

{"highlights": {"prin": [1, 2], "sub": [0, 3]}, "revealName": true}

Screenshot: alkene-step-2



optical | Optical activity, specific rotation, and resolution

Measure rotation; normalize units; keep configuration labels separate

optical:rule | Core Rules

Observed optical rotation αobs depends on concentration, path length, wavelength, temperature and solvent. In the common solution convention, [α] = αobs/(l × c), with l in dm and c in g/mL. If αobs is in degrees, the corresponding specific-rotation units are °·mL·g^{-1}·dm^{-1}. State wavelength, temperature and solvent when comparing measurements.

Schematic summary: Measure rotation; normalize units; keep configuration labels separate

Measure — Record wavelength, temperature and solvent

Convert — Path in dm; concentration in g/mL here

Normalize — [α] = αobs / (l × c)

optical:cue | Memory Cues

1 dm = 10 cm. Positive (+), dextrorotatory, means clockwise/rightward rotation as viewed by an observer looking toward the light source through the instrument; negative (−), levorotatory, is counterclockwise/leftward. Older texts use lowercase d/l for these signs. Uppercase D/L and R/S describe configuration and do not predict the rotation sign.

Schematic summary: Choose the case that applies

Path in cm — Divide by 10 to obtain dm

Concentration in g per 100 mL — Divide by 100 to obtain g/mL

R/S label — Does not predict +/− rotation

optical:trap | Common Mistakes / MCAT Traps

Do not put a length in cm directly into a formula requiring dm. Concentration in g per 100 mL needs conversion to g/mL in this convention. A racemate gives zero net rotation under matched ordinary conditions, but zero rotation alone does not prove a racemate. Specific rotation is not just an unnormalized angle.

Schematic summary: Shortcut versus a better check

Avoid — Using cm directly in a dm convention

Check — Units and measurement conditions before calculating

optical:app | MCAT Applications

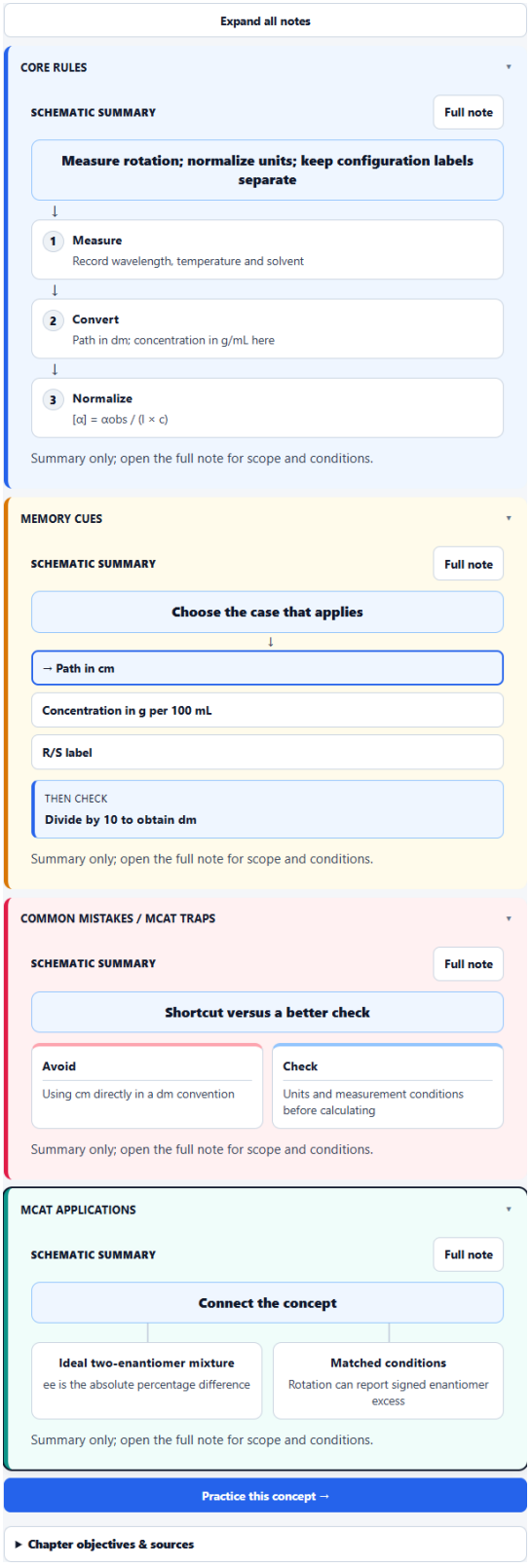
For an ideal mixture of only two enantiomers under matched conditions, ee = |%major − %minor| and observed rotation is the signed enantiomer excess times the pure-enantiomer observed rotation at the same total concentration and path length. A 75:25 mixture has 50% ee; if the major pure enantiomer gives +20°, the mixture gives +10°. These are illustrative practice data.

Schematic summary: Connect the concept

Ideal two-enantiomer mixture — ee is the absolute percentage difference

Matched conditions — Rotation can report signed enantiomer excess

Screenshot: optical-visuals



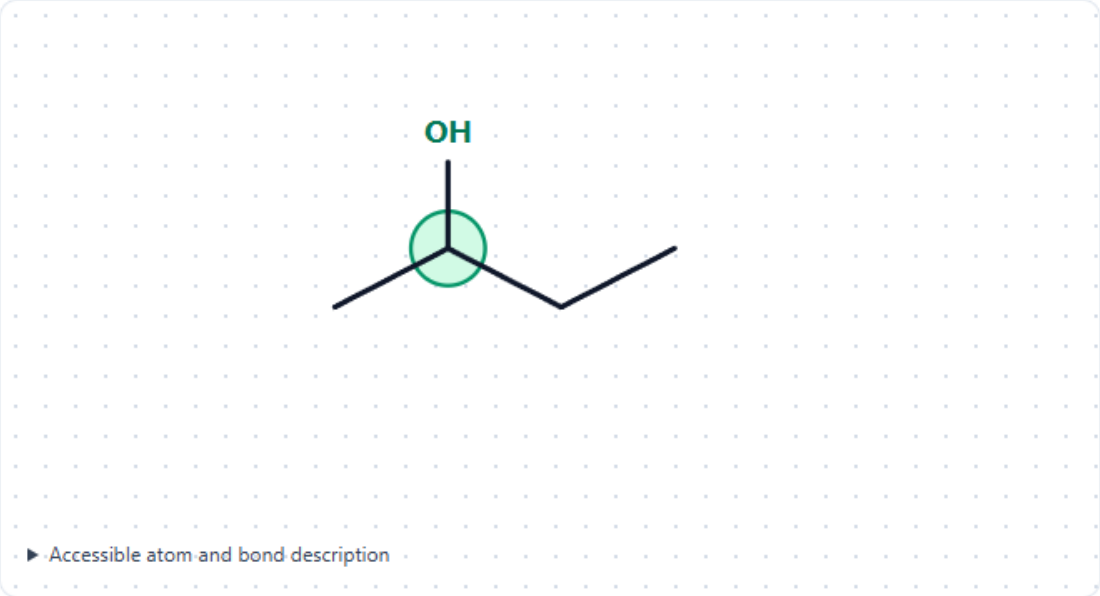
optical:step-1 | 1. Geometry is not optical sign

Butan-2-ol can exist as enantiomers. Its connectivity alone does not specify which enantiomer is present, and an R/S descriptor does not predict (+)/(−). Measure the rotation experimentally.

Displayed molecule ID: butanol2

{"highlights": {"prin": [1]}, "revealName": true}

Screenshot: optical-step-1



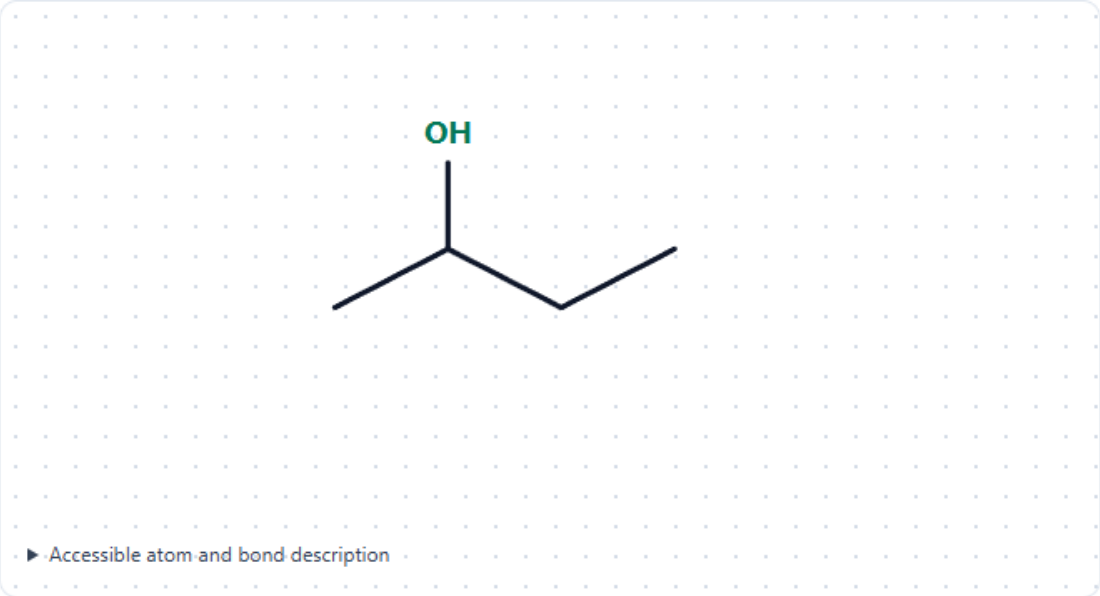
optical:step-2 | 2. Normalize a measurement

Illustrative data: αobs = +1.2°, c = 0.040 g/mL and l = 10 cm = 1.0 dm. Then [α] = +1.2/(0.040 × 1.0) = +30 °·mL·g^{-1}·dm^{-1}. These are practice numbers, not a tabulated value for the displayed connectivity.

Displayed molecule ID: butanol2

{"highlights": {}, "revealName": false}

Screenshot: optical-step-2



optical:step-3 | 3. Resolve a racemate

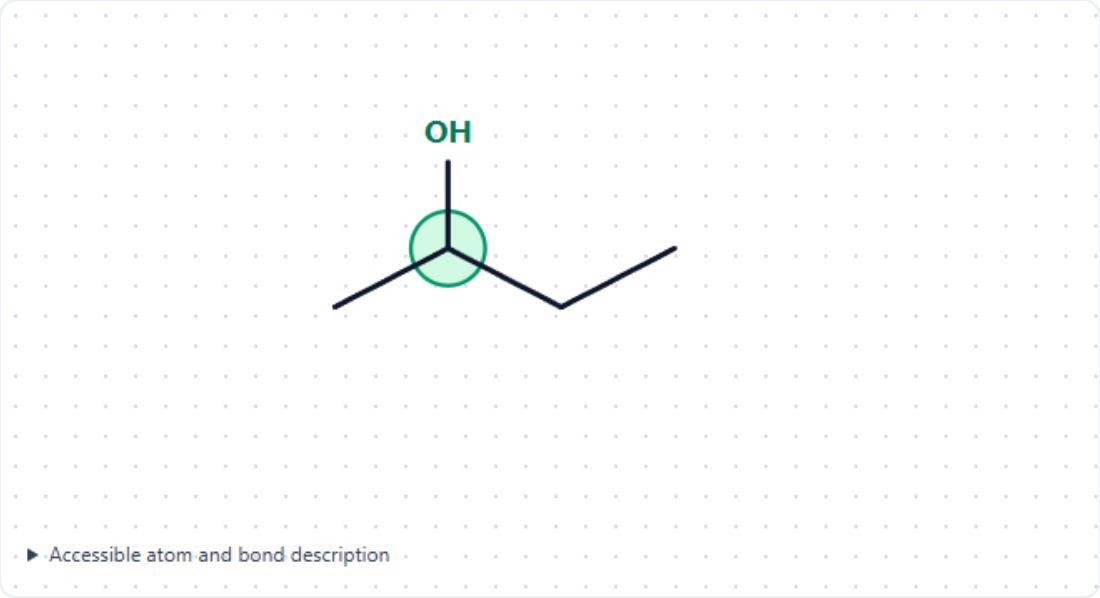
For a suitable substrate, use a single-enantiomer resolving agent to make diastereomeric salts or derivatives. Separate them by a suitable property difference, then regenerate the starting enantiomers. Ordinary achiral distillation does not generally

separate enantiomers, which share the same boiling point under achiral conditions.

Displayed molecule ID: butanol2

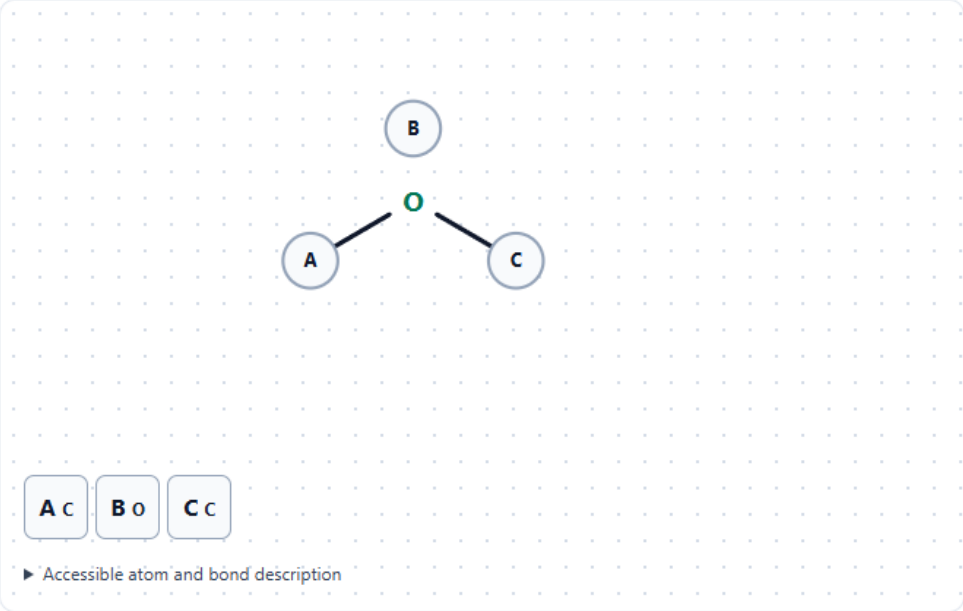
{"highlights": {"prin": [1]}, "revealName": false}

Screenshot: optical-step-3



isomer-pair | Classify an isomer pair

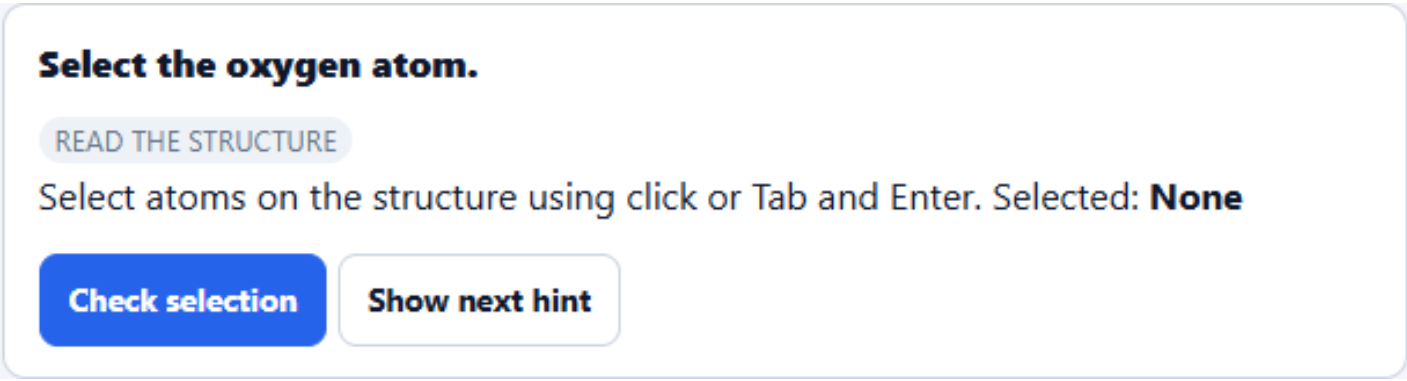
Screenshot: isomer-pair-stage-1-molecule



isomer-pair:stage-1 | Select the oxygen atom.

Answer type: atoms

Screenshot: isomer-pair-stage-1-question



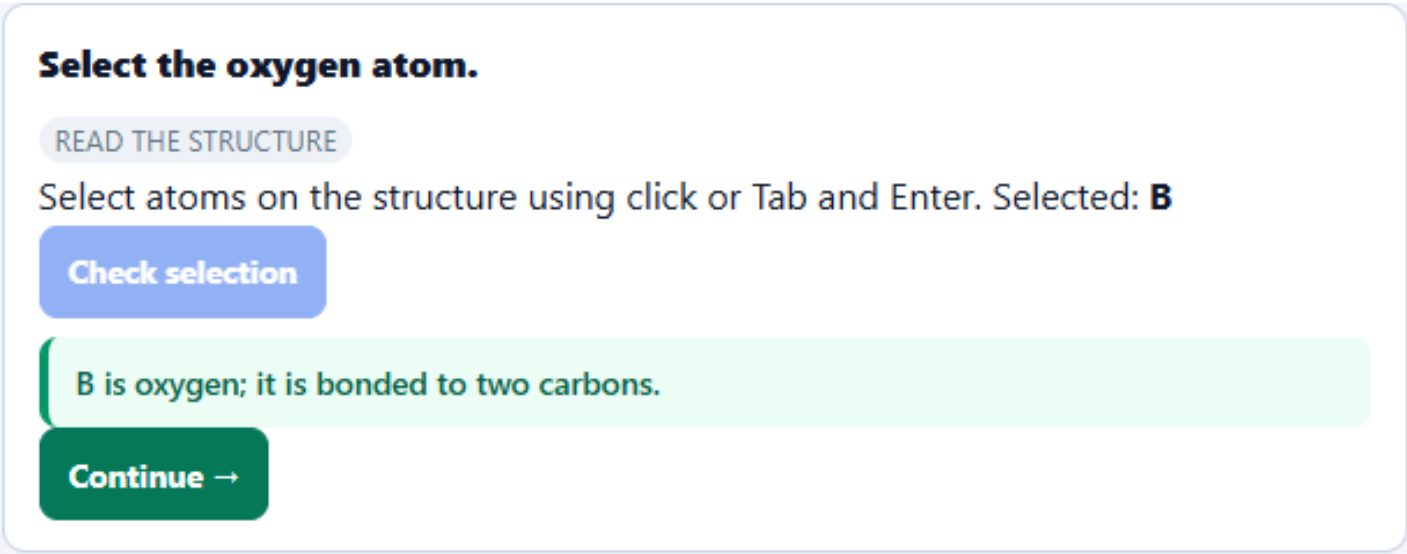
{"authorAnswer": [1], "tolerance": null, "highlightsAfterCorrectAnswer": {"prin": [1]}}

Hint 1: Read the element label.

Hint 2: The oxygen is between the carbon endpoints.

Explanation: B is oxygen; it is bonded to two carbons.

Screenshot: isomer-pair-stage-1-feedback



isomer-pair:stage-2 | Compared with ethanol (C2H6O), this compound is a...

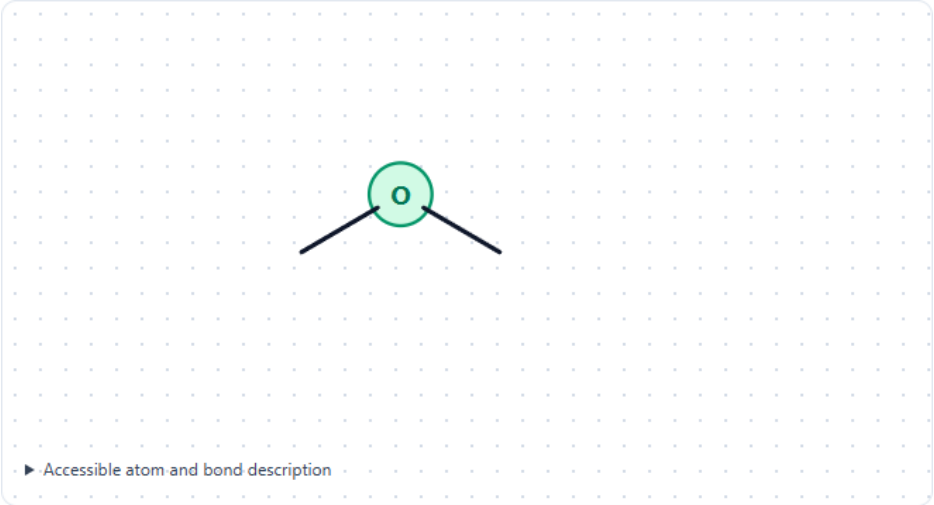
Answer type: choice

Option 0 (A): Constitutional isomer

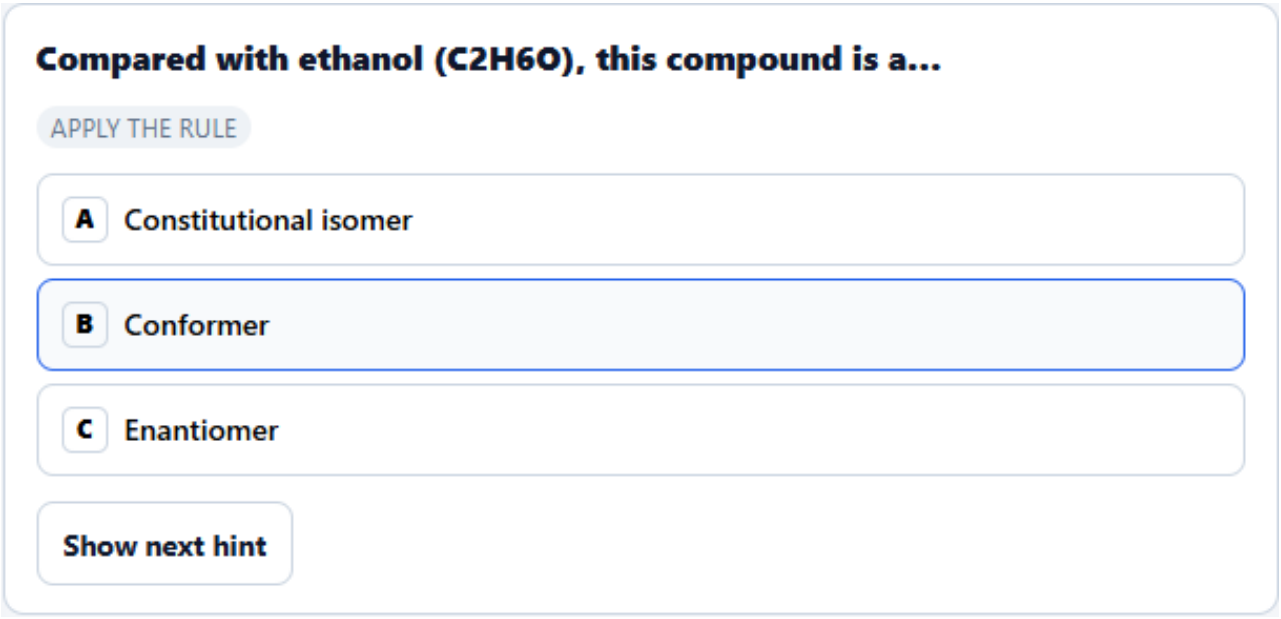
Option 1 (B): Conformer

Option 2 (C): Enantiomer

Screenshot: isomer-pair-stage-2-molecule



Screenshot: isomer-pair-stage-2-question



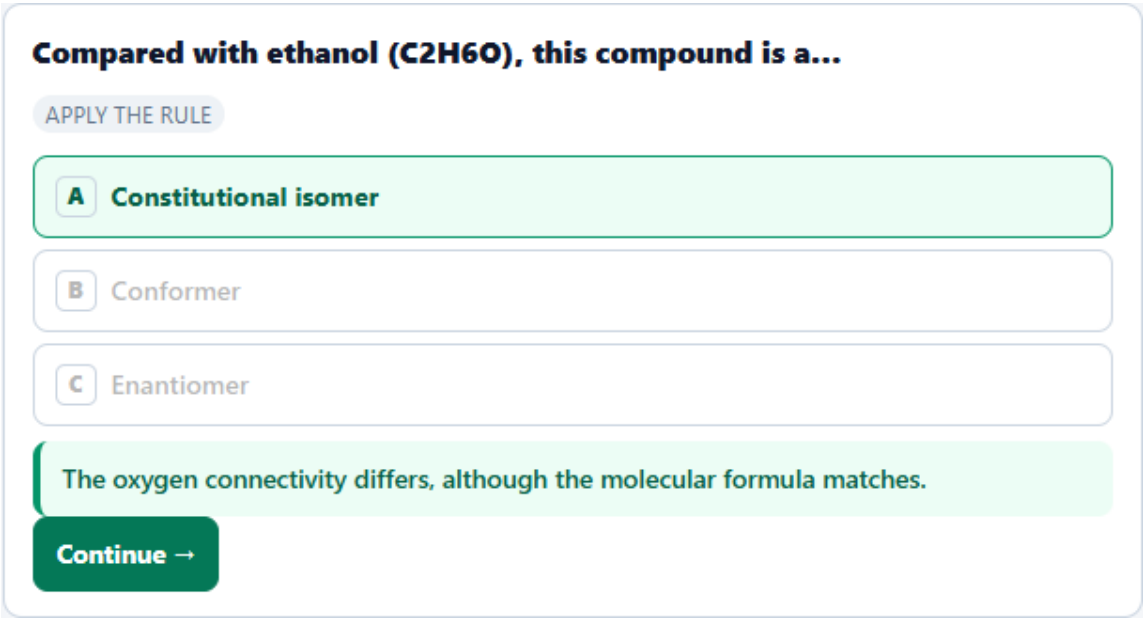
{"authorAnswer": 0, "tolerance": null, "highlightsAfterCorrectAnswer": {}}

Hint 1: Compare bonds before spatial arrangements.

Hint 2: Ethanol contains an O–H bond.

Explanation: The oxygen connectivity differs, although the molecular formula matches.

Screenshot: isomer-pair-stage-2-feedback



isomer-pair:stage-3 | Which can donate an O–H hydrogen bond?

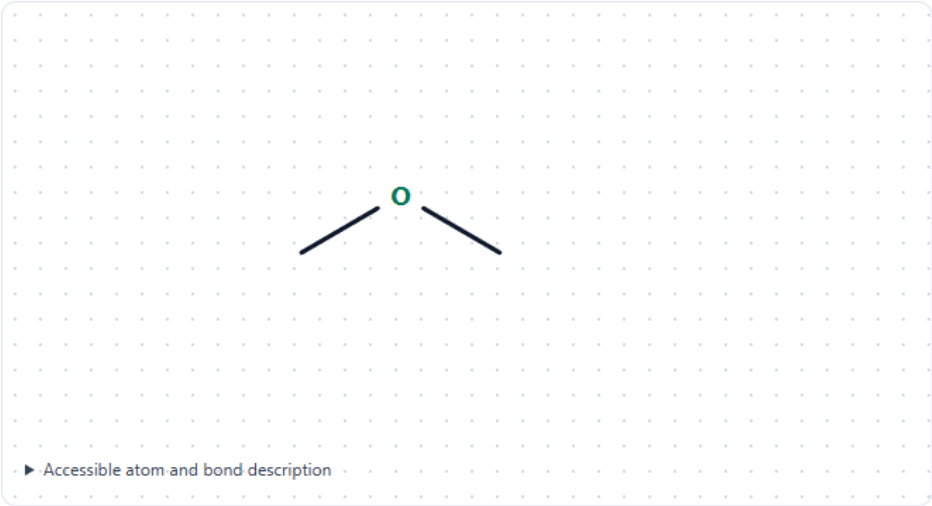
Answer type: choice

Option 0 (A): Both

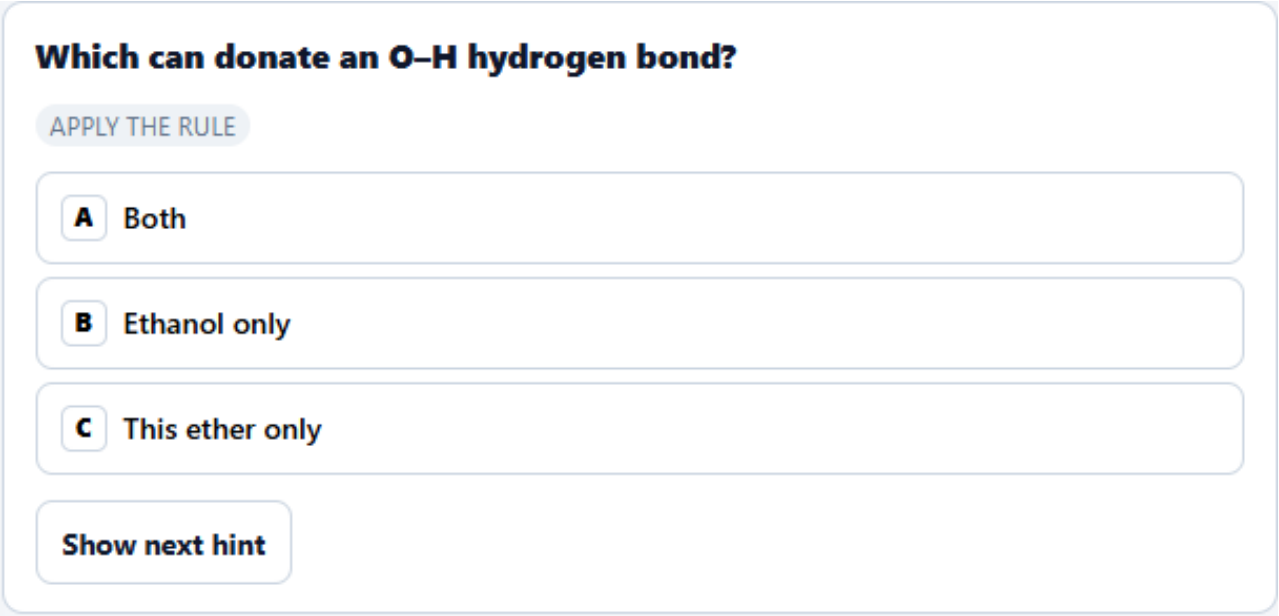
Option 1 (B): Ethanol only

Option 2 (C): This ether only

Screenshot: isomer-pair-stage-3-molecule



Screenshot: isomer-pair-stage-3-question



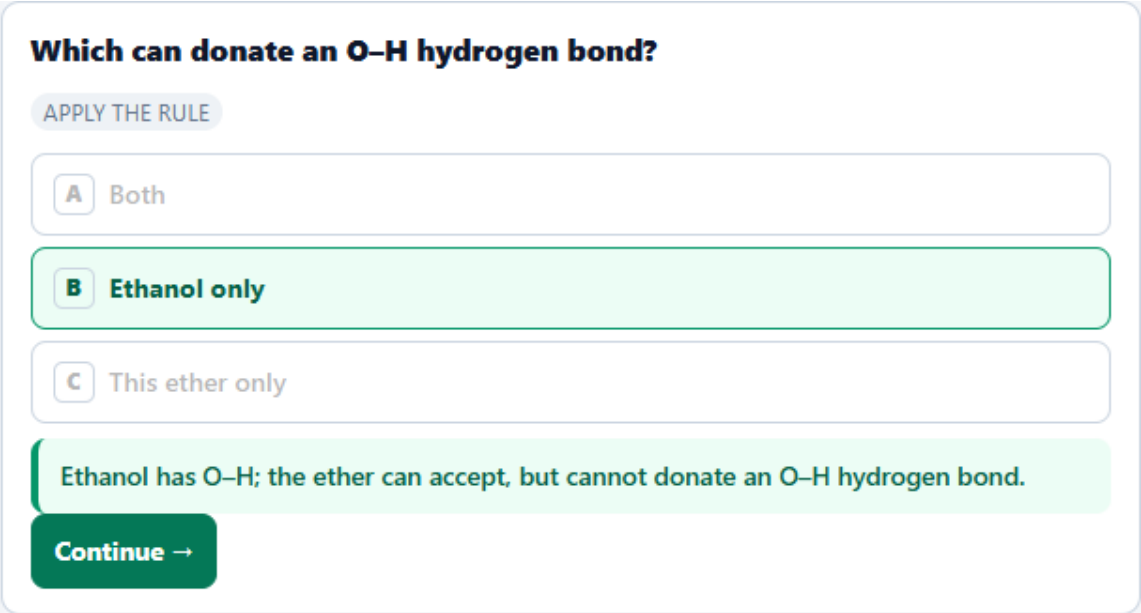
{"authorAnswer": 1, "tolerance": null, "highlightsAfterCorrectAnswer": {}}

Hint 1: Donation requires H bonded to an electronegative atom.

Hint 2: Look for an O–H bond.

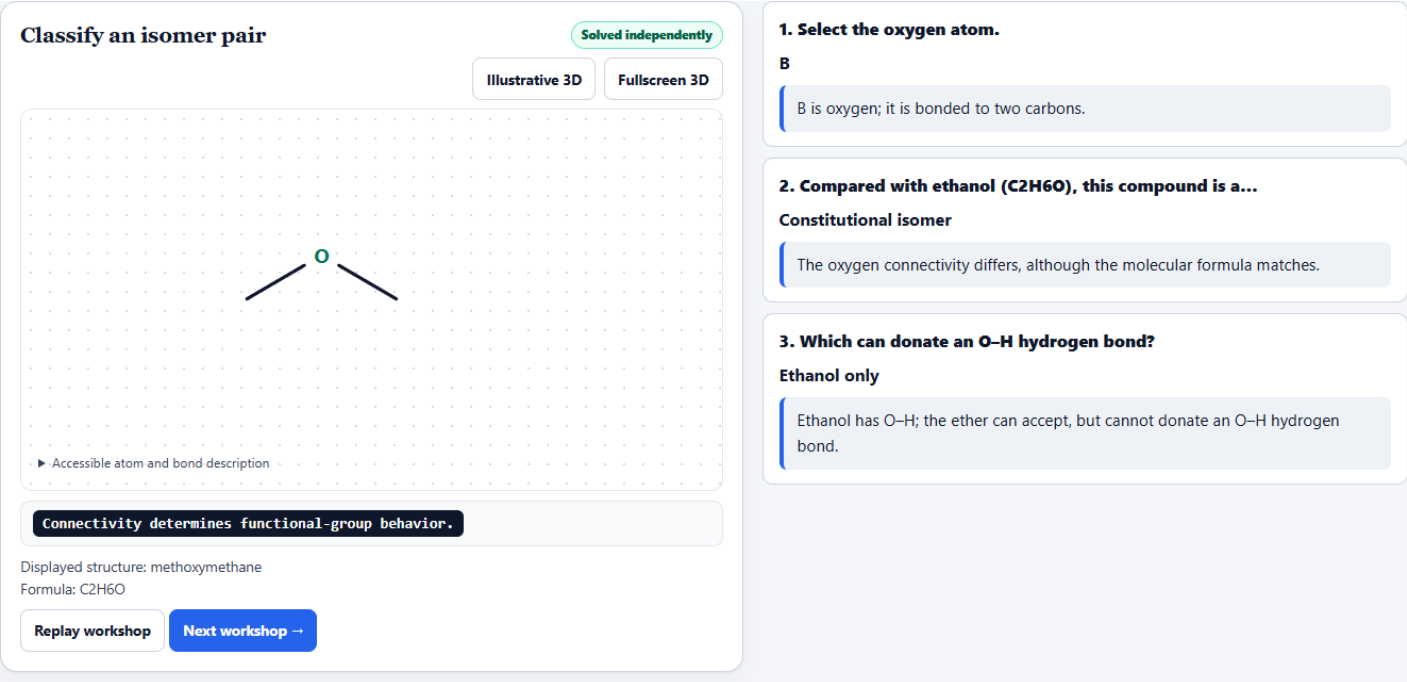
Explanation: Ethanol has O–H; the ether can accept, but cannot donate an O–H hydrogen bond.

Screenshot: isomer-pair-stage-3-feedback



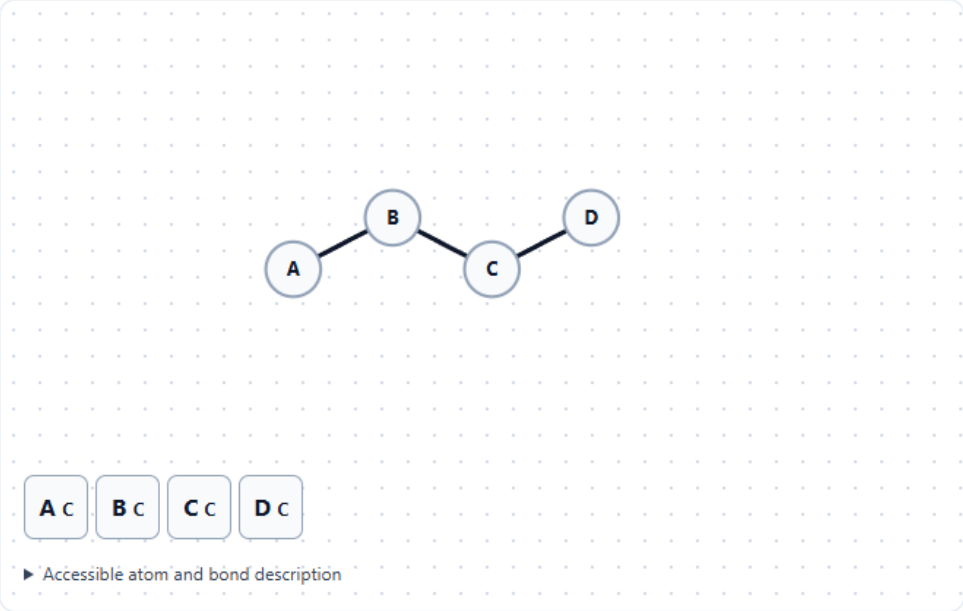
Completion summary: Connectivity determines functional-group behavior.

Screenshot: isomer-pair-complete

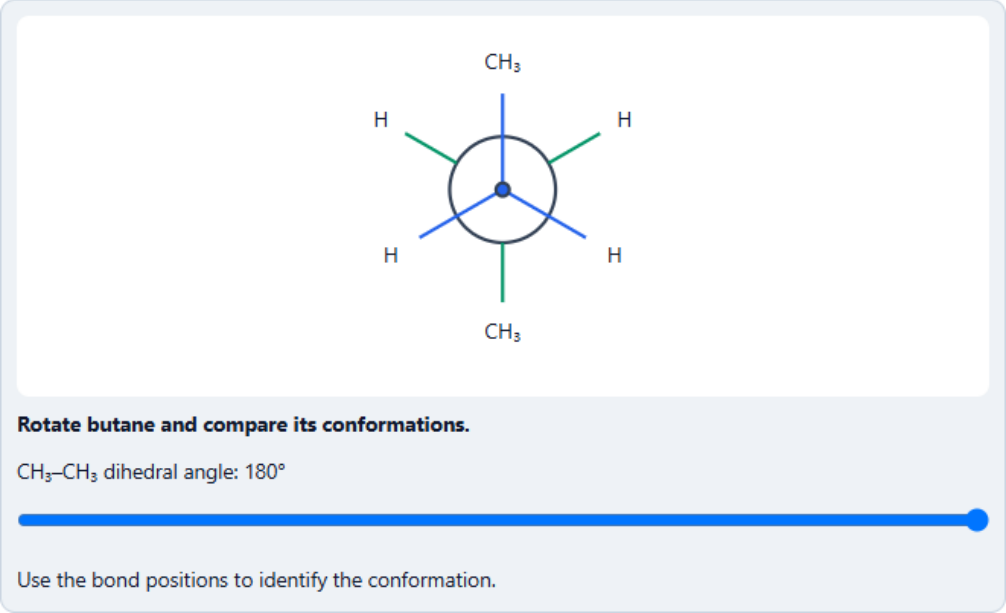


butane-rotation | Reason about conformational energy

Screenshot: butane-rotation-stage-1-molecule



Screenshot: butane-rotation-diagram

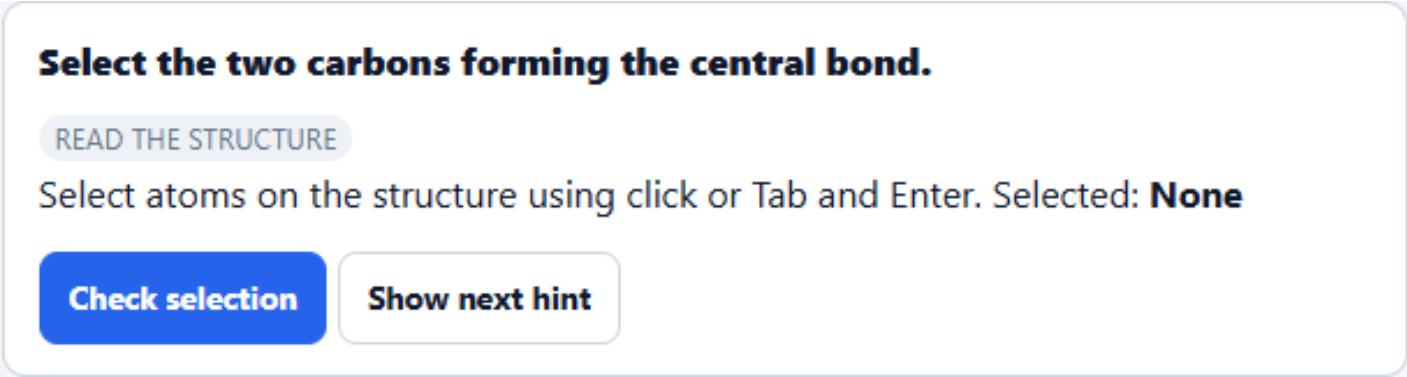


{"diagramSpecification": {"type": "newman", "caption": "Rotate butane and compare its conformations."}}

butane-rotation:stage-1 | Select the two carbons forming the central bond.

Answer type: atoms

Screenshot: butane-rotation-stage-1-question



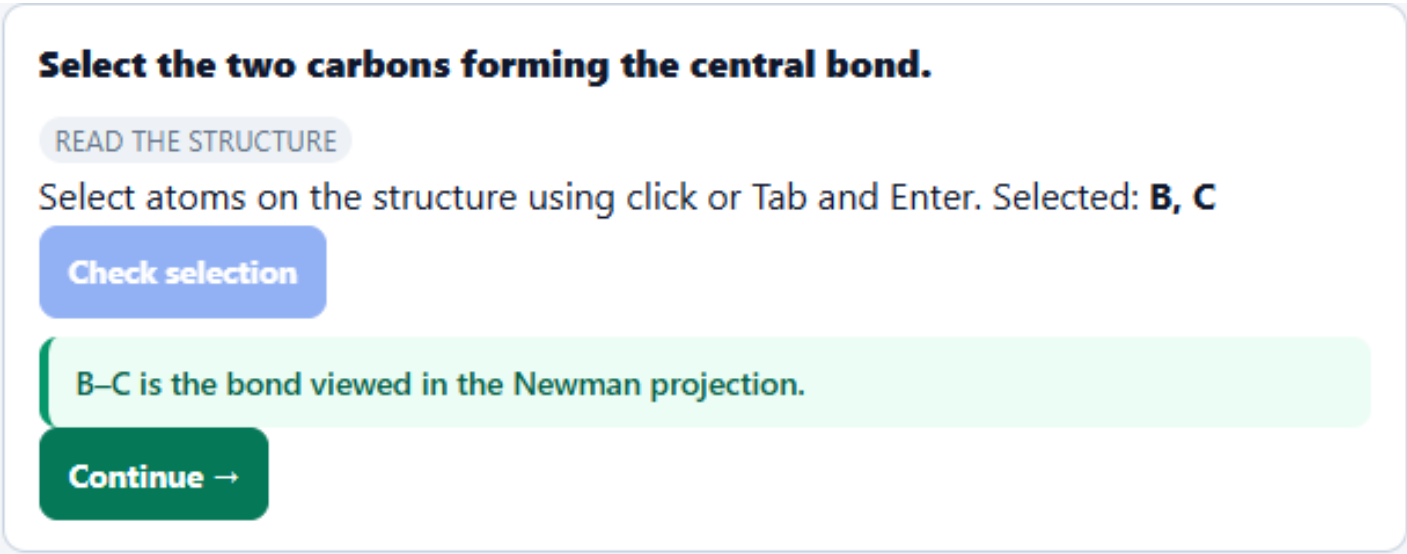
{"authorAnswer": [1, 2], "tolerance": null, "highlightsAfterCorrectAnswer": {"prin": [1, 2]}}

Hint 1: Find the four-carbon chain.

Hint 2: Choose the two internal atoms.

Explanation: B–C is the bond viewed in the Newman projection.

Screenshot: butane-rotation-stage-1-feedback



butane-rotation:stage-2 | At a 180° methyl–methyl dihedral, the conformation is…

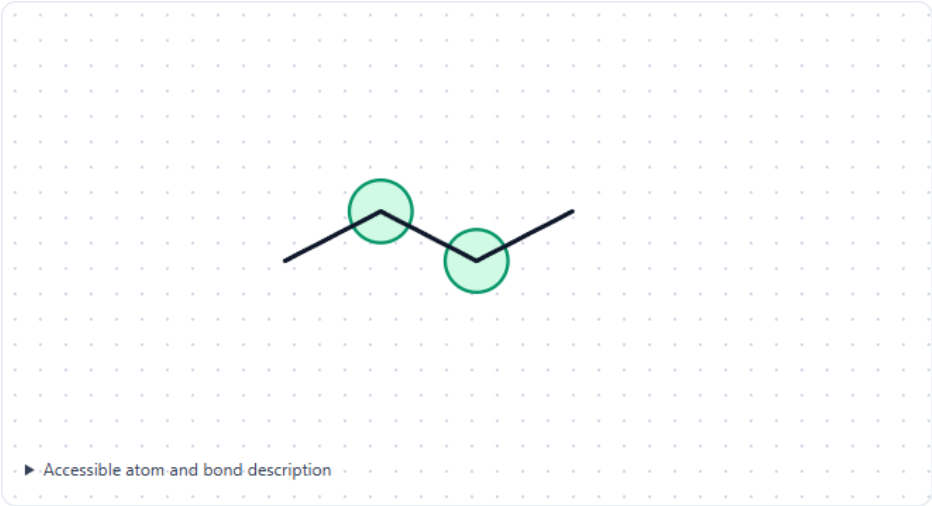
Answer type: choice

Option 0 (A): Anti

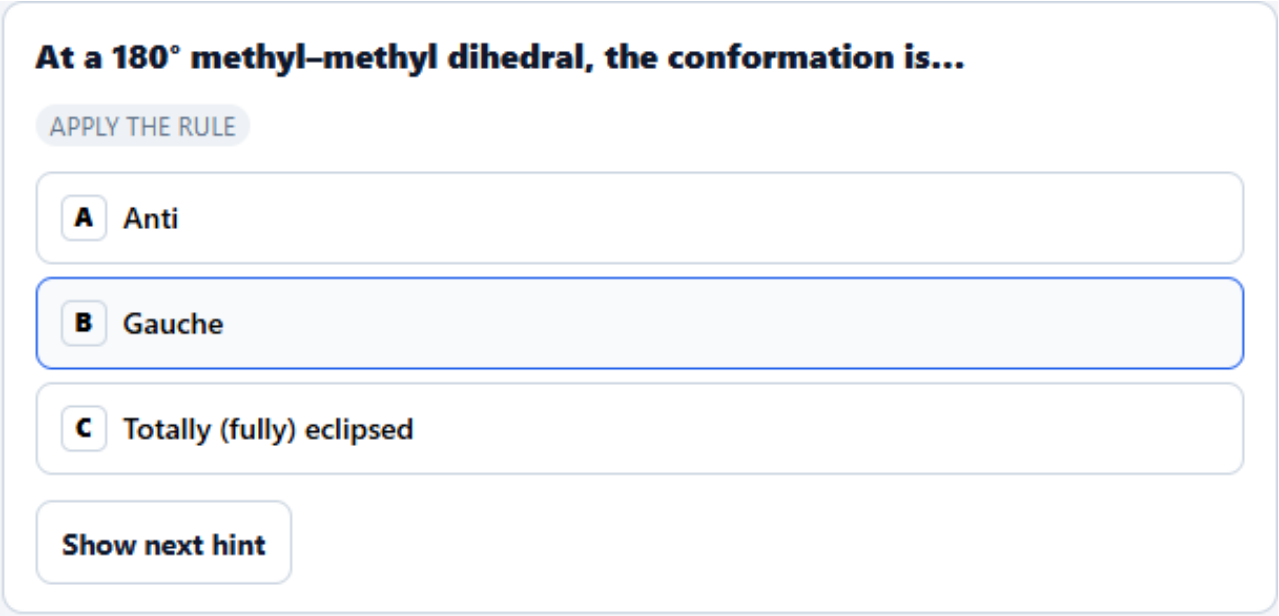
Option 1 (B): Gauche

Option 2 (C): Totally (fully) eclipsed

Screenshot: butane-rotation-stage-2-molecule



Screenshot: butane-rotation-stage-2-question



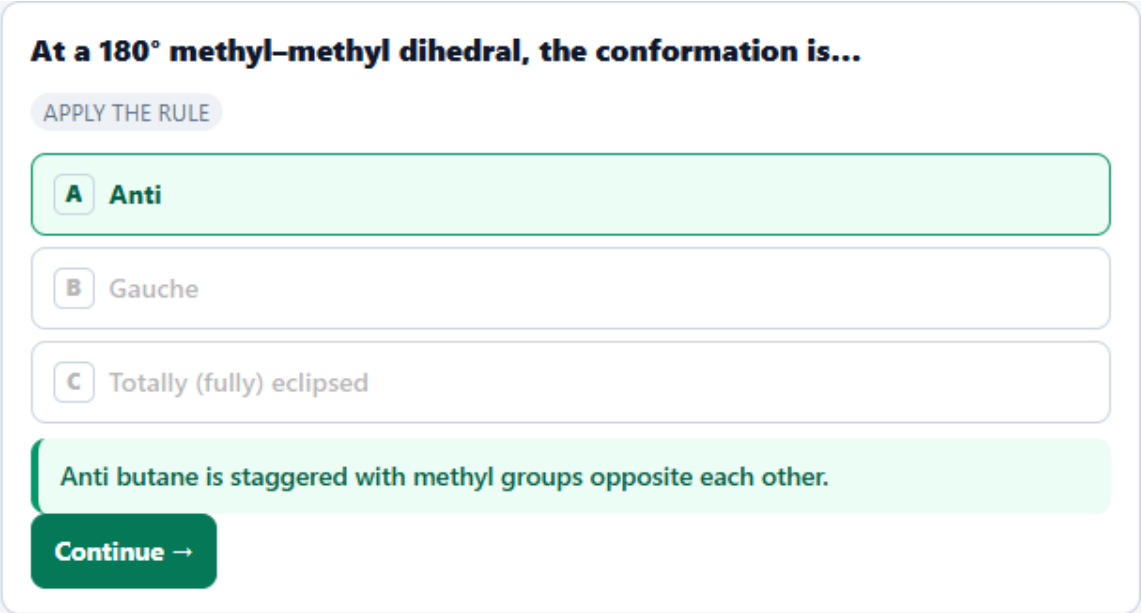
{"authorAnswer": 0, "tolerance": null, "highlightsAfterCorrectAnswer": {}}

Hint 1: Distinguish staggered from eclipsed.

Hint 2: Opposite groups are separated by half a turn.

Explanation: Anti butane is staggered with methyl groups opposite each other.

Screenshot: butane-rotation-stage-2-feedback



This stage uses a different molecule (connectivity shown).

butane-rotation:stage-3 | For cyclohexane (connectivity shown), a chair flip changes...

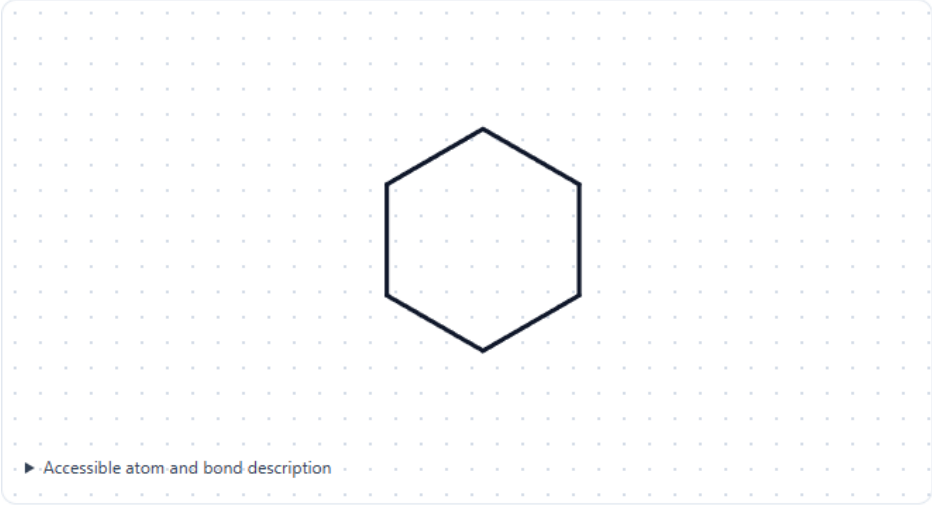
Answer type: choice

Option 0 (A): Up into down

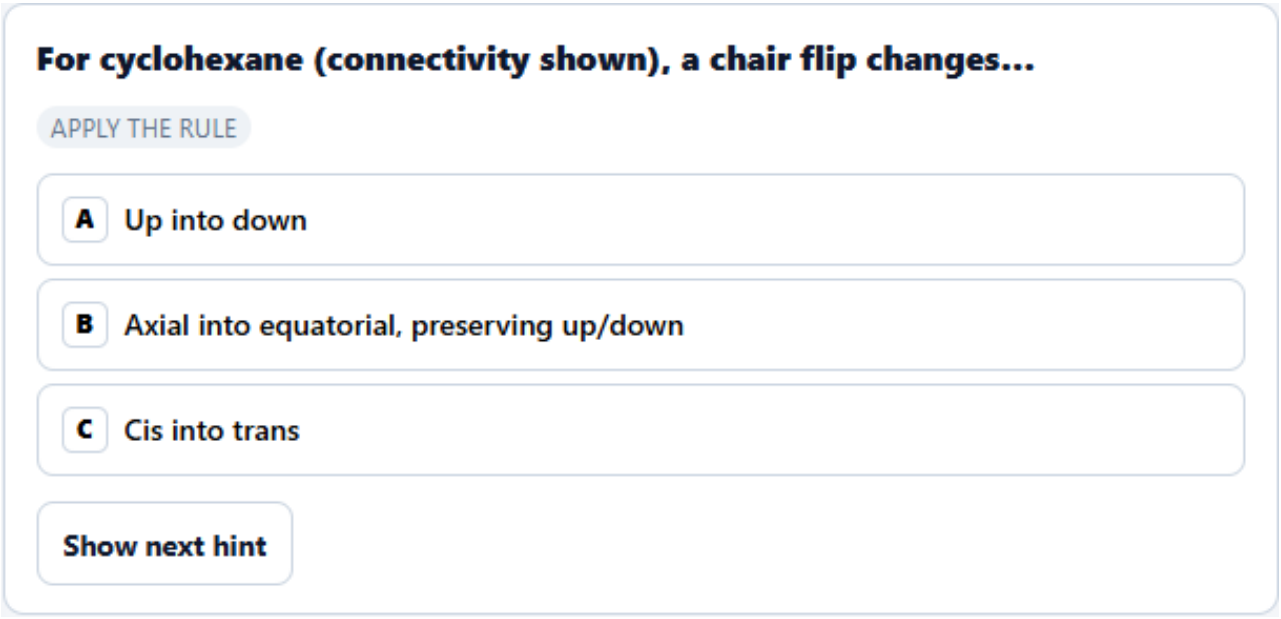
Option 1 (B): Axial into equatorial, preserving up/down

Option 2 (C): Cis into trans

Screenshot: butane-rotation-stage-3-molecule



Screenshot: butane-rotation-stage-3-question



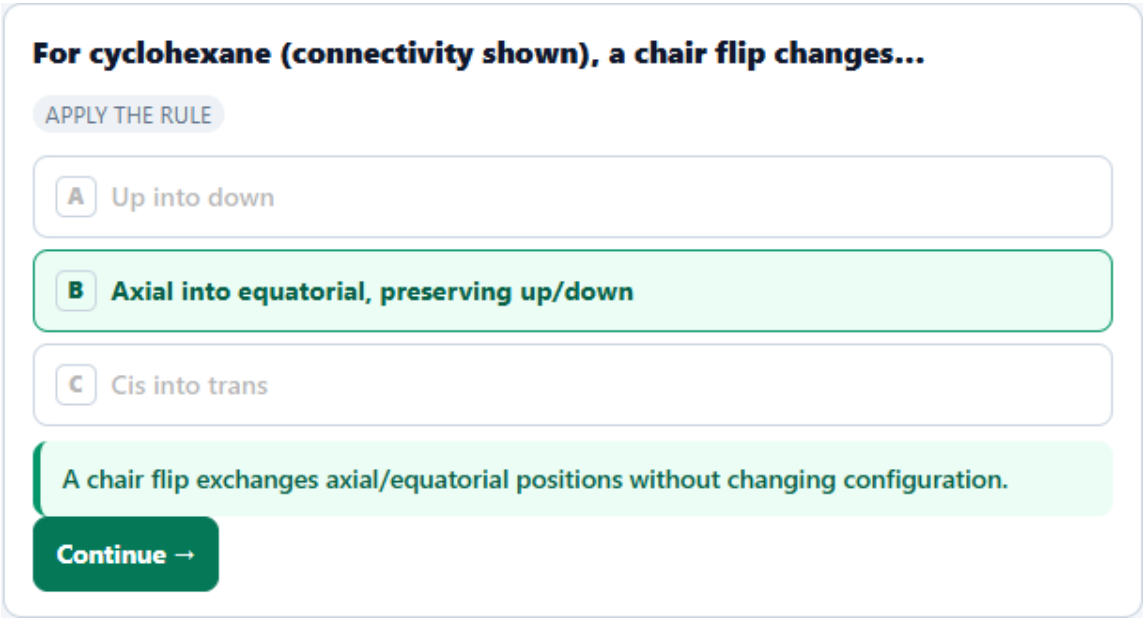
{"authorAnswer": 1, "tolerance": null, "highlightsAfterCorrectAnswer": {}}

Hint 1: Track the bond orientation relative to the ring.

Hint 2: Up/down and axial/equatorial are different labels.

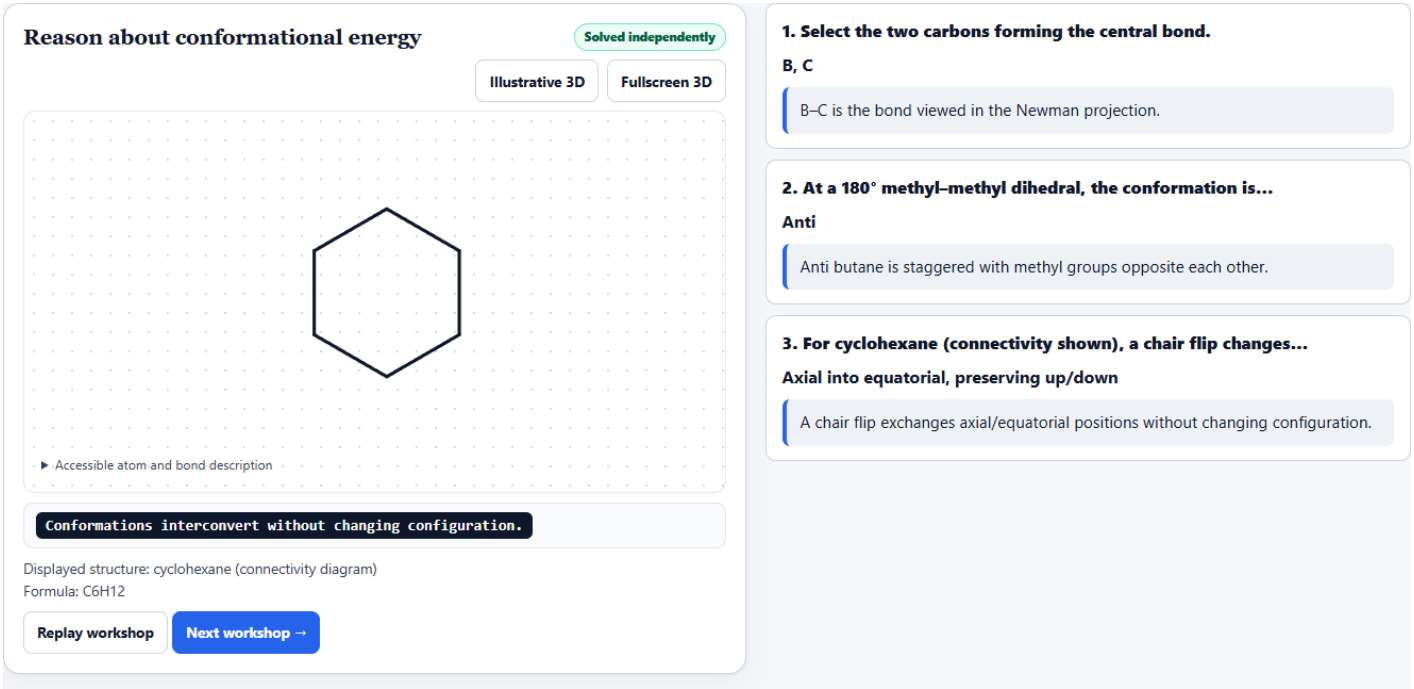
Explanation: A chair flip exchanges axial/equatorial positions without changing configuration.

Screenshot: butane-rotation-stage-3-feedback



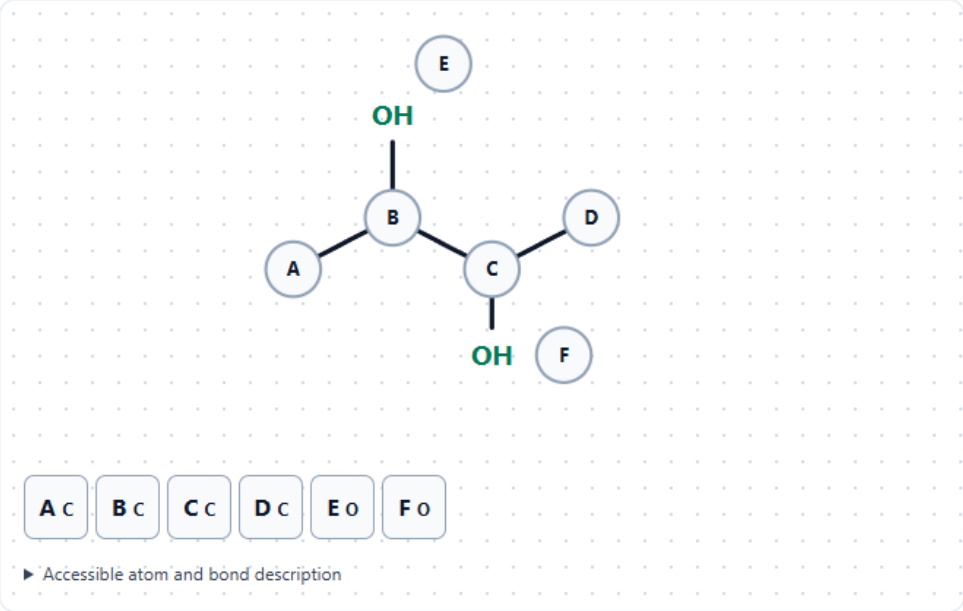
Completion summary: Conformations interconvert without changing configuration.

Screenshot: butane-rotation-complete



stereocenters | Count stereoisomers with symmetry

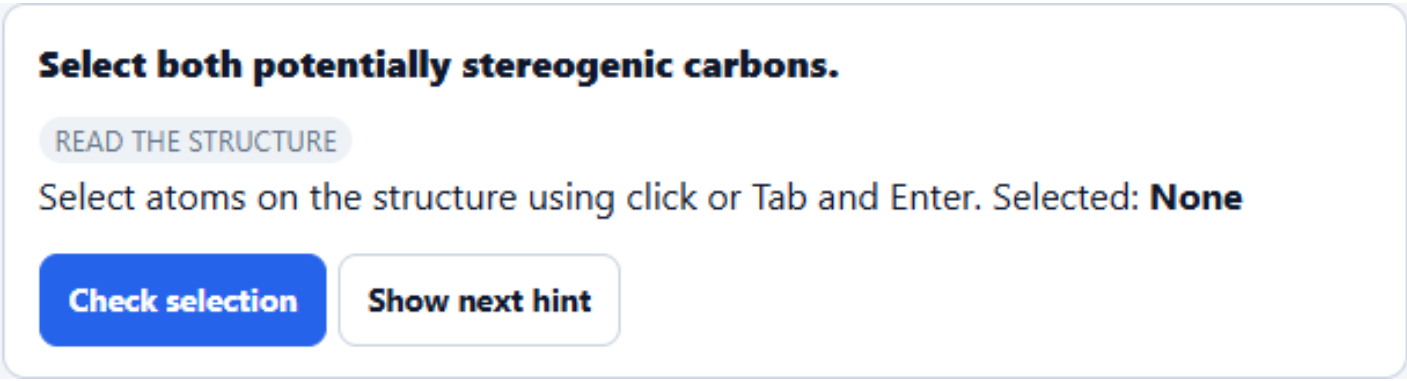
Screenshot: stereocenters-stage-1-molecule



stereocenters:stage-1 | Select both potentially stereogenic carbons.

Answer type: atoms

Screenshot: stereocenters-stage-1-question



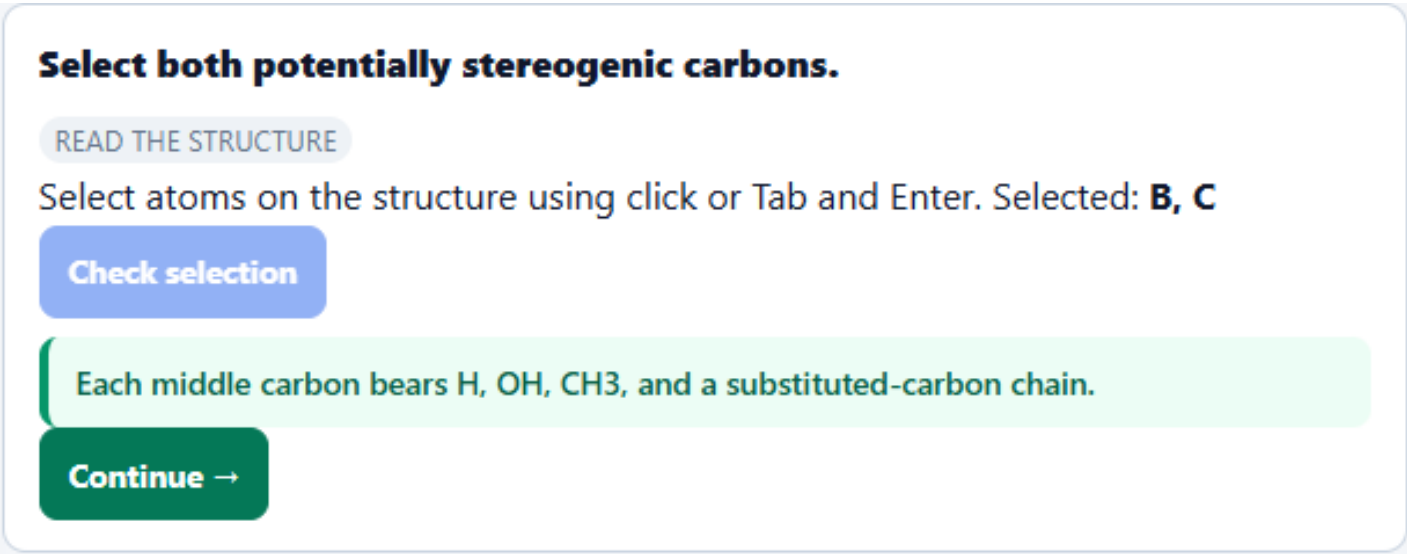
{"authorAnswer": [1, 2], "tolerance": null, "highlightsAfterCorrectAnswer": {"prin": [1, 2]}}

Hint 1: Look for tetrahedral atoms with four different groups.

Hint 2: The OH-bearing carbons are the candidates.

Explanation: Each middle carbon bears H, OH, CH3, and a substituted-carbon chain.

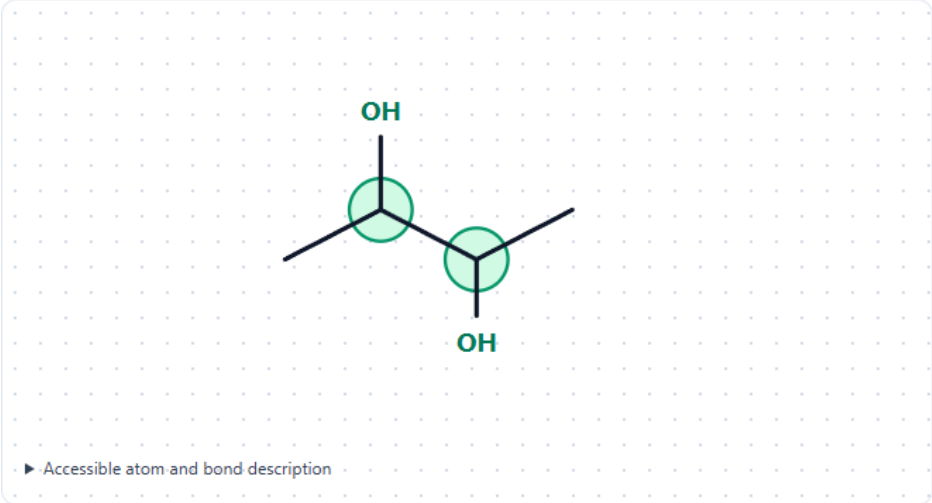
Screenshot: stereocenters-stage-1-feedback



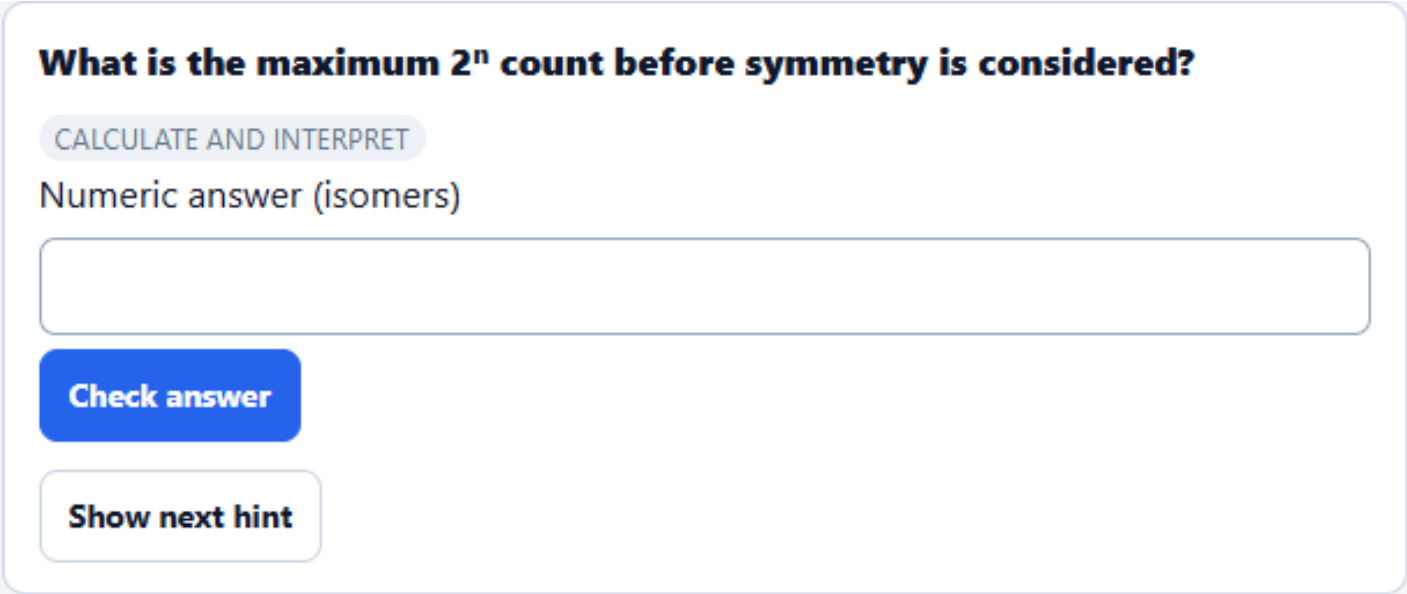
stereocenters:stage-2 | What is the maximum 2^n count before symmetry is considered?

Answer type: number | unit: isomers

Screenshot: stereocenters-stage-2-molecule



Screenshot: stereocenters-stage-2-question



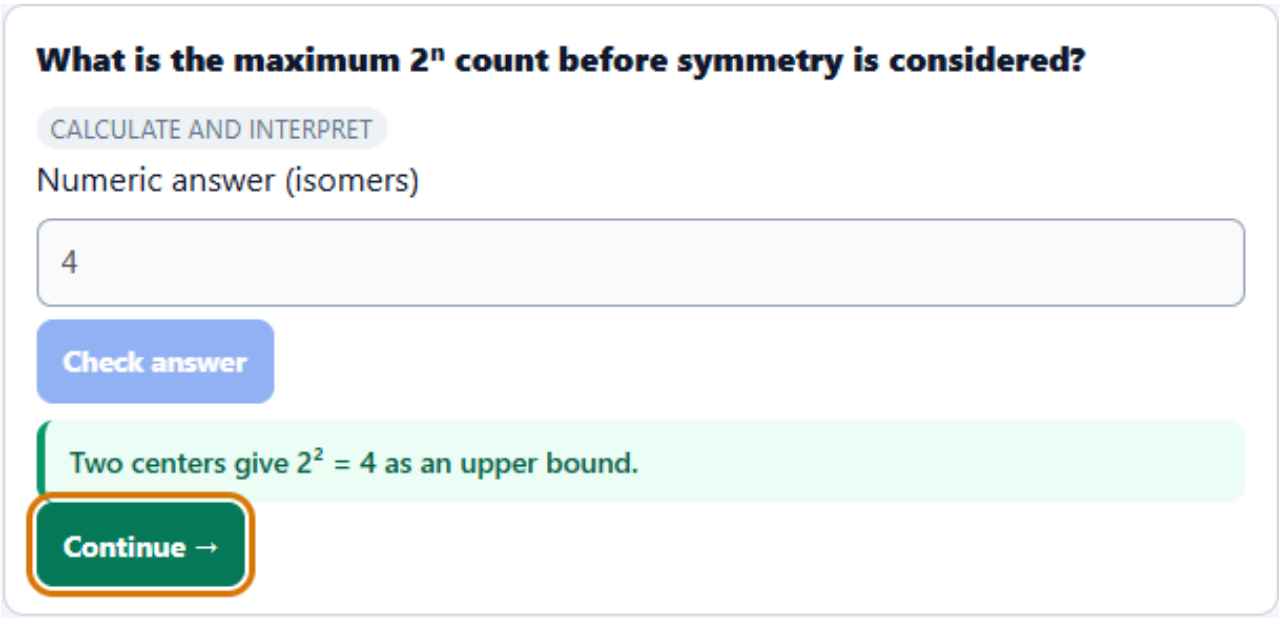
{"authorAnswer": 4, "tolerance": 0, "highlightsAfterCorrectAnswer": {}}

Hint 1: n is the number of independent stereocenters.

Hint 2: Raise two to the power of two.

Explanation: Two centers give $2^2 = 4$ as an upper bound.

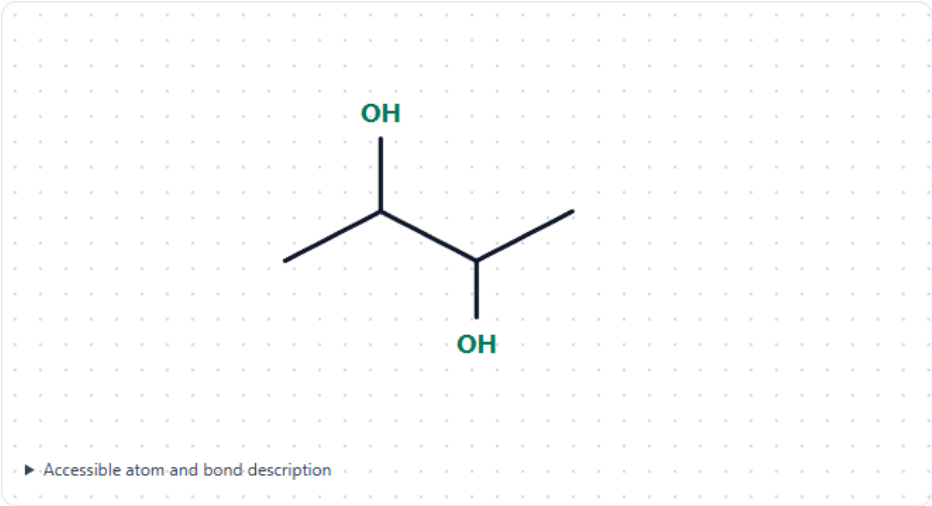
Screenshot: stereocenters-stage-2-feedback



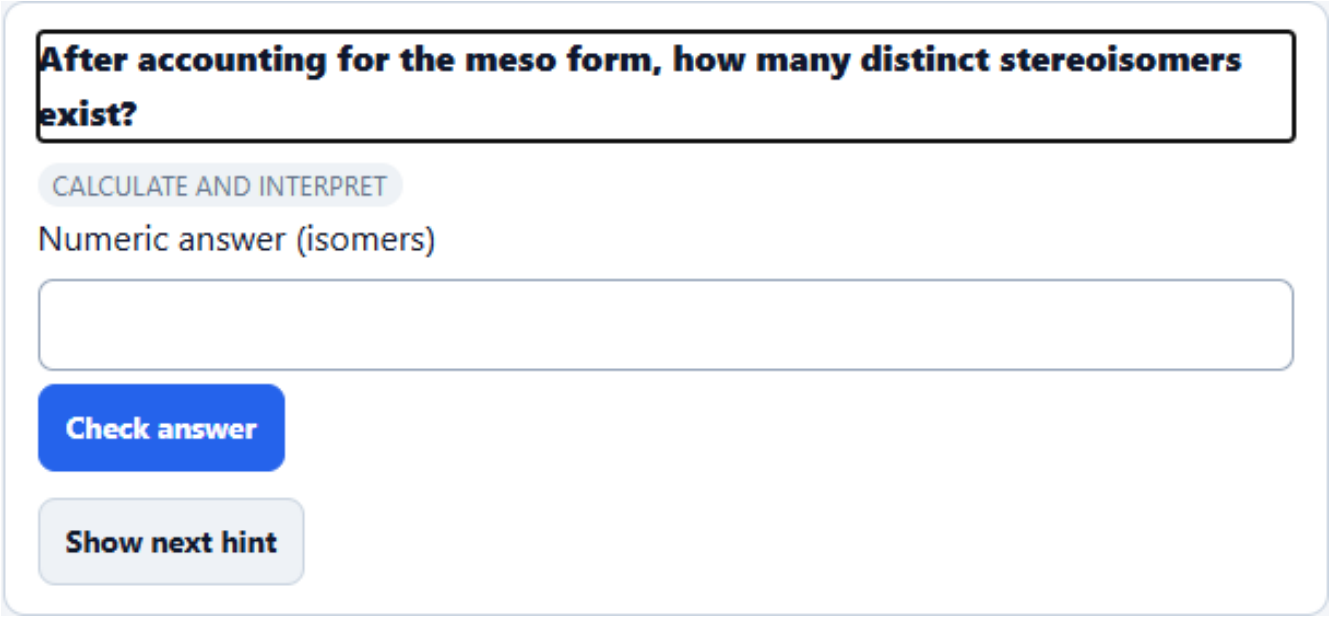
stereocenters:stage-3 | After accounting for the meso form, how many distinct stereoisomers exist?

Answer type: number | unit: isomers

Screenshot: stereocenters-stage-3-molecule



Screenshot: stereocenters-stage-3-question



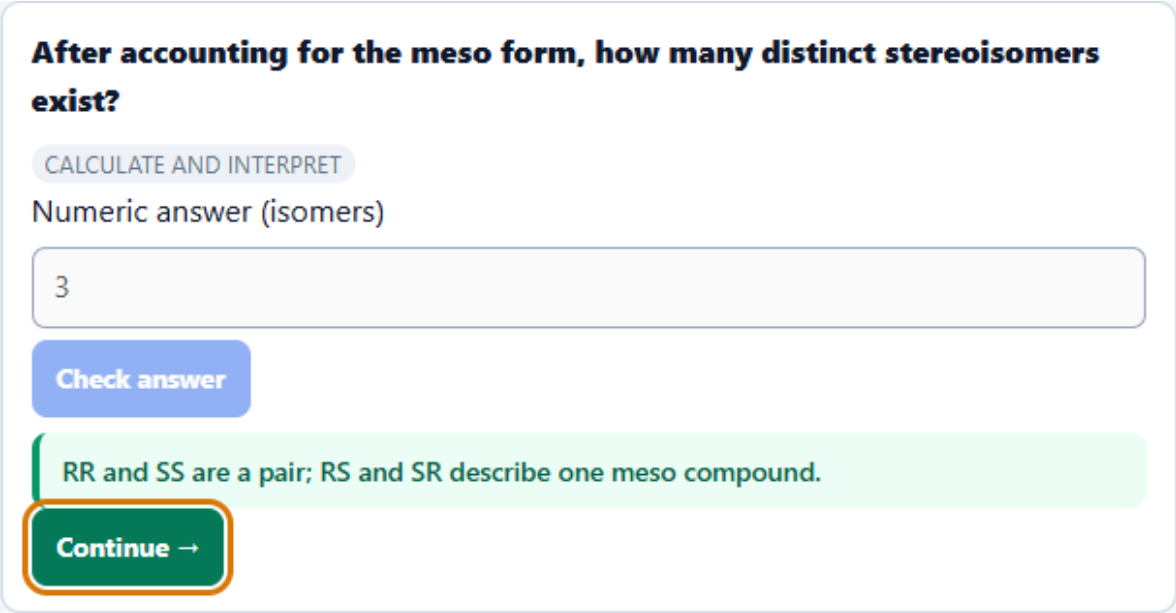
{"authorAnswer": 3, "tolerance": 0, "highlightsAfterCorrectAnswer": {}}

Hint 1: Check whether exchanging the equivalent ends duplicates a structure.

Hint 2: There is one achiral meso form plus one mirror-image pair.

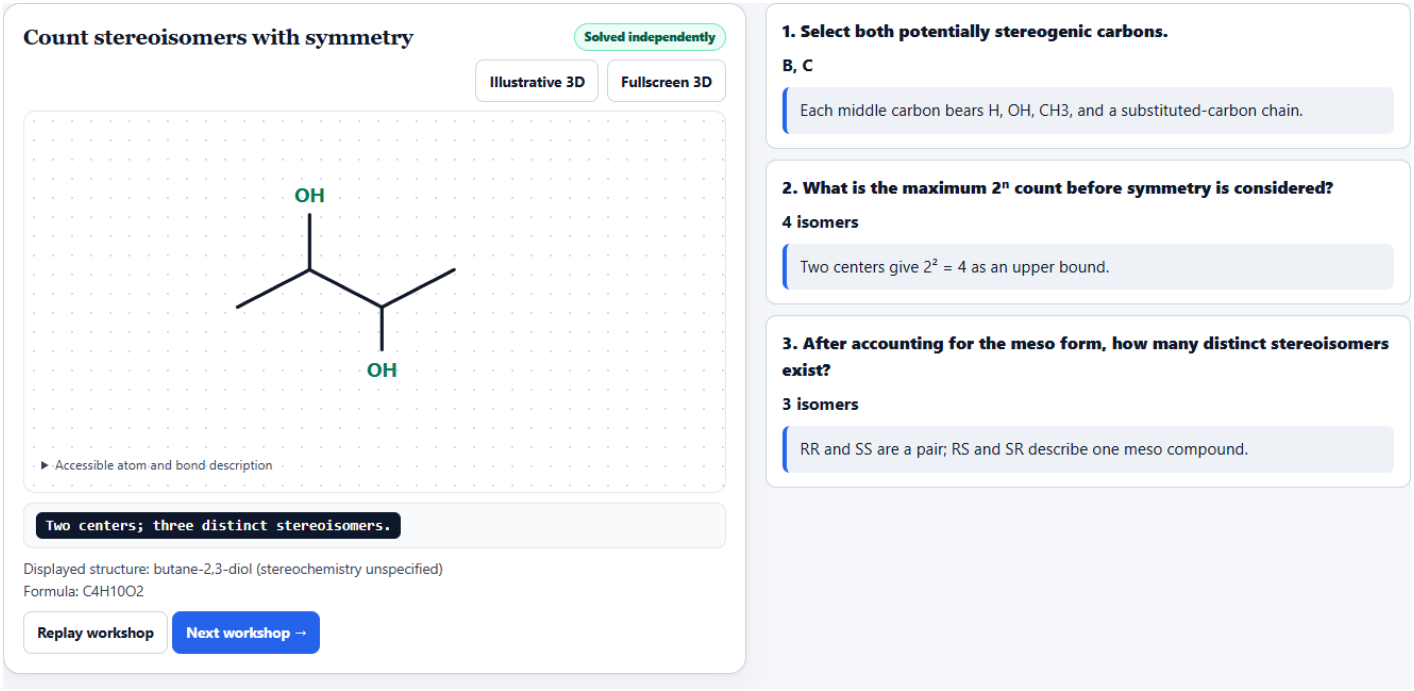
Explanation: RR and SS are a pair; RS and SR describe one meso compound.

Screenshot: stereocenters-stage-3-feedback



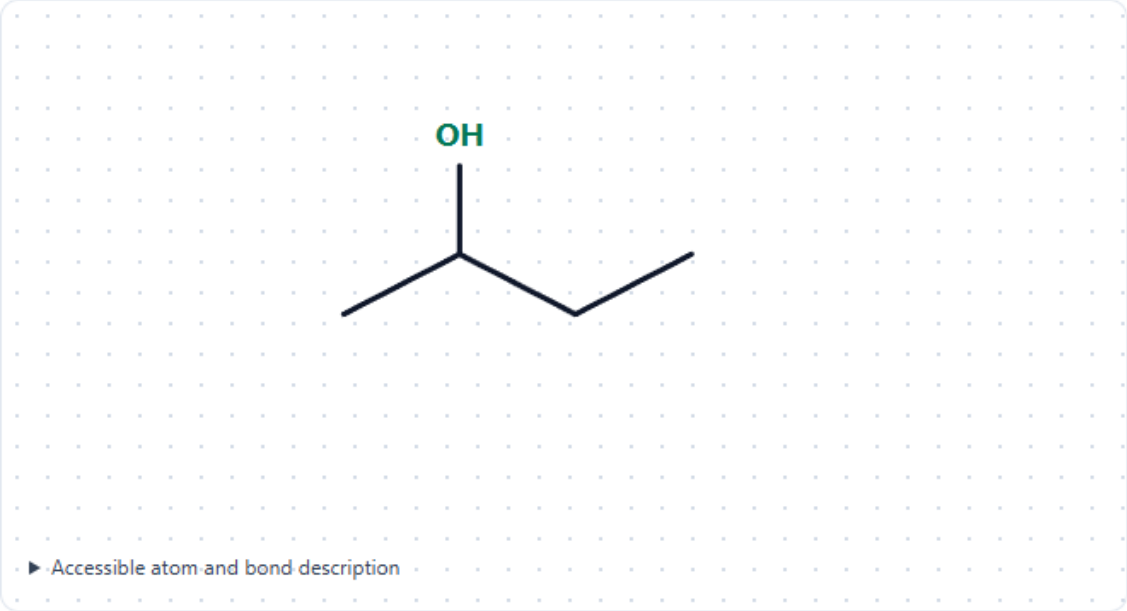
Completion summary: Two centers; three distinct stereoisomers.

Screenshot: stereocenters-complete

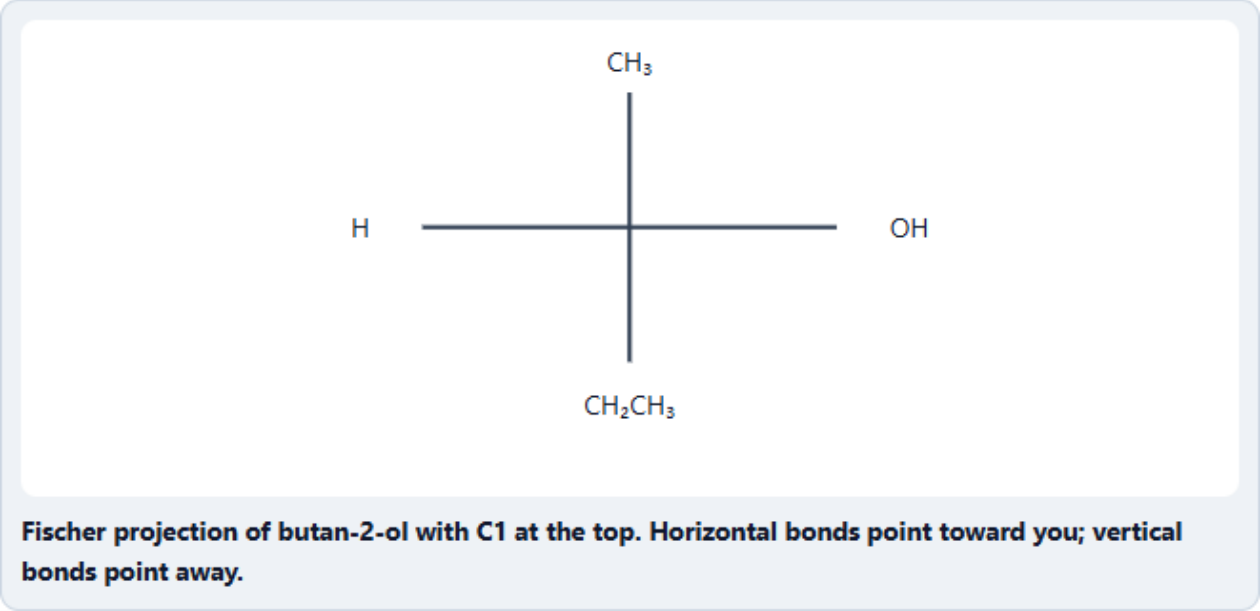


fischer-cip | Assign a Fischer configuration

Screenshot: fischer-cip-stage-1-molecule



Screenshot: fischer-cip-diagram



Fischer projection of butan-2-ol with C1 at the top. Horizontal bonds point toward you; vertical bonds point away.

{"diagramSpecification": {"type": "fischer", "top": "CH₃", "bottom": "CH₂CH₃", "left": "H", "right": "OH", "caption": "Fischer projection of butan-2-ol with C1 at the top. Horizontal bonds point toward you; vertical bonds point away."}}

fischer-cip:stage-1 | Which substituent has highest CIP priority?

Answer type: choice

Option 0 (A): OH

Option 1 (B): Ethyl

Option 2 (C): Methyl

Option 3 (D): H

Screenshot: fischer-cip-stage-1-question



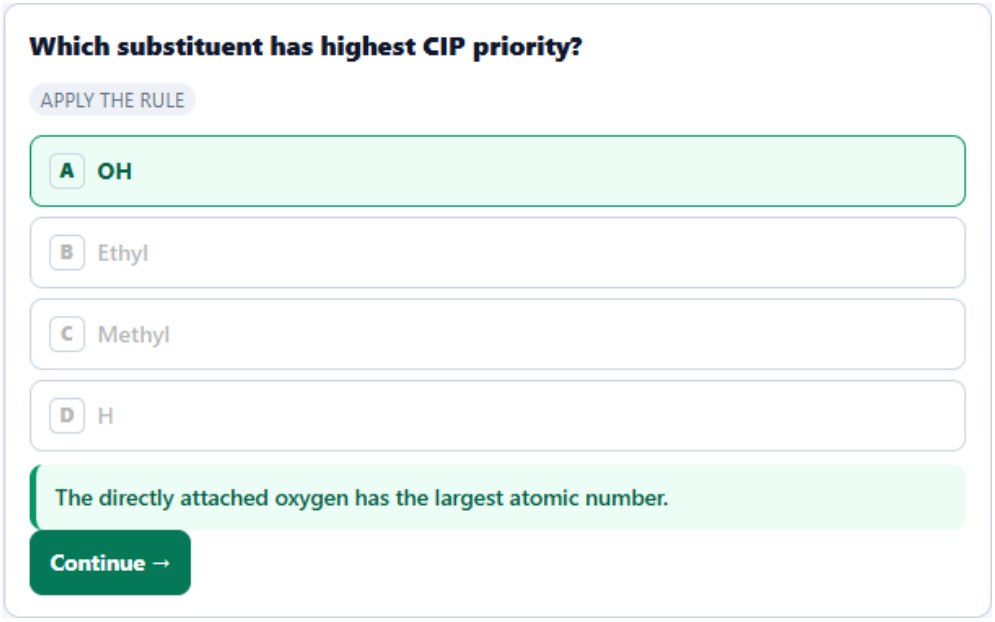
{"authorAnswer": 0, "tolerance": null, "highlightsAfterCorrectAnswer": {}}

Hint 1: Start at directly attached atoms.

Hint 2: O has atomic number 8; C is 6.

Explanation: The directly attached oxygen has the largest atomic number.

Screenshot: fischer-cip-stage-1-feedback



fischer-cip:stage-2 | In this projection, the lowest-priority H points…

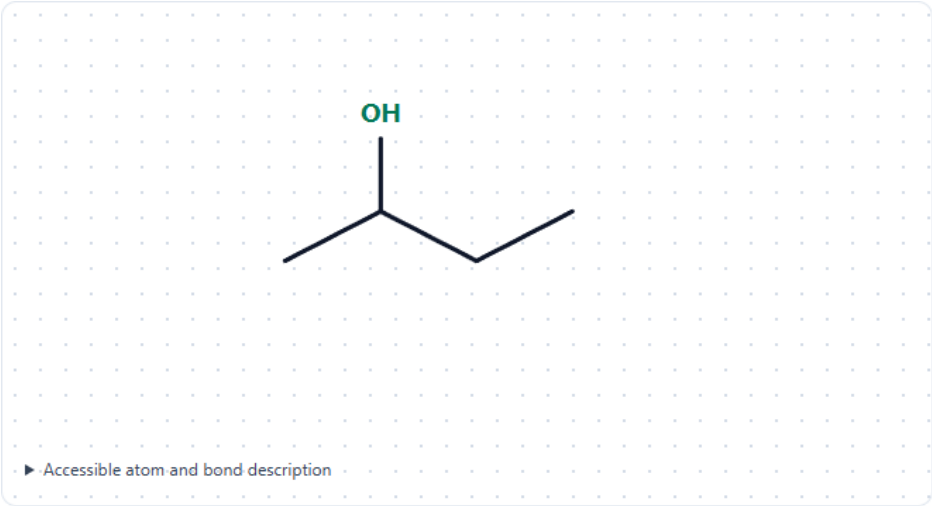
Answer type: choice

Option 0 (A): Away

Option 1 (B): Toward you

Option 2 (C): In the paper plane

Screenshot: fischer-cip-stage-2-molecule



Screenshot: fischer-cip-stage-2-question



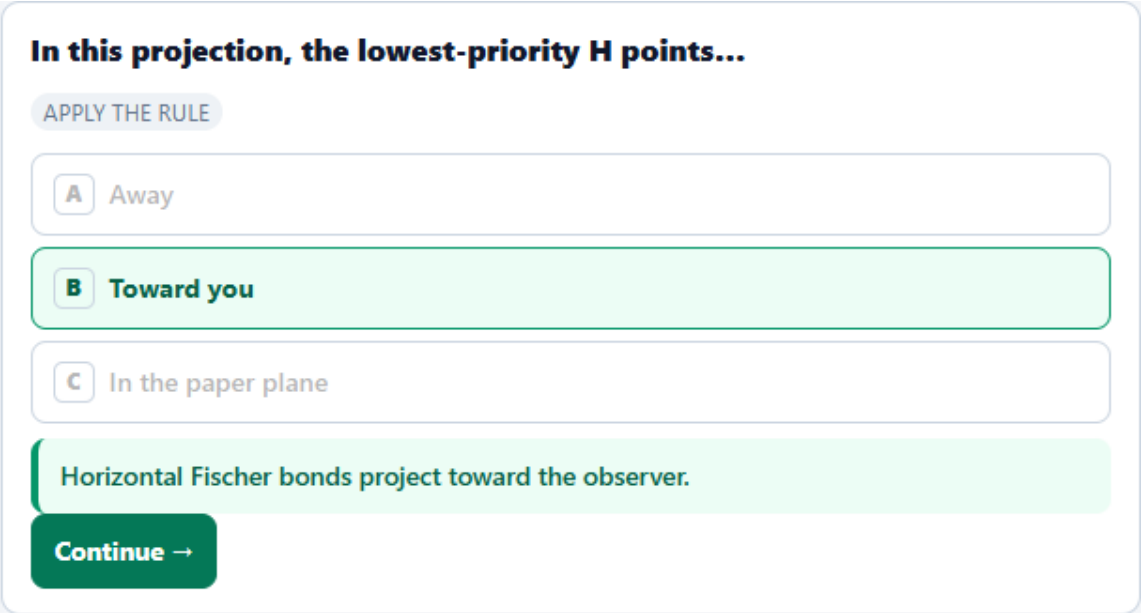
{"authorAnswer": 1, "tolerance": null, "highlightsAfterCorrectAnswer": {}}

Hint 1: Use the Fischer convention.

Hint 2: Vertical and horizontal bonds have opposite depth directions.

Explanation: Horizontal Fischer bonds project toward the observer.

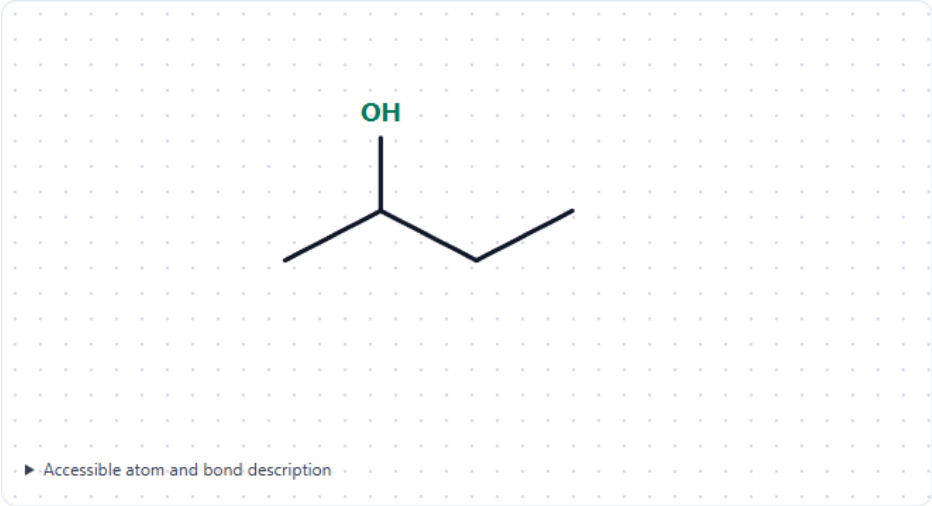
Screenshot: fischer-cip-stage-2-feedback



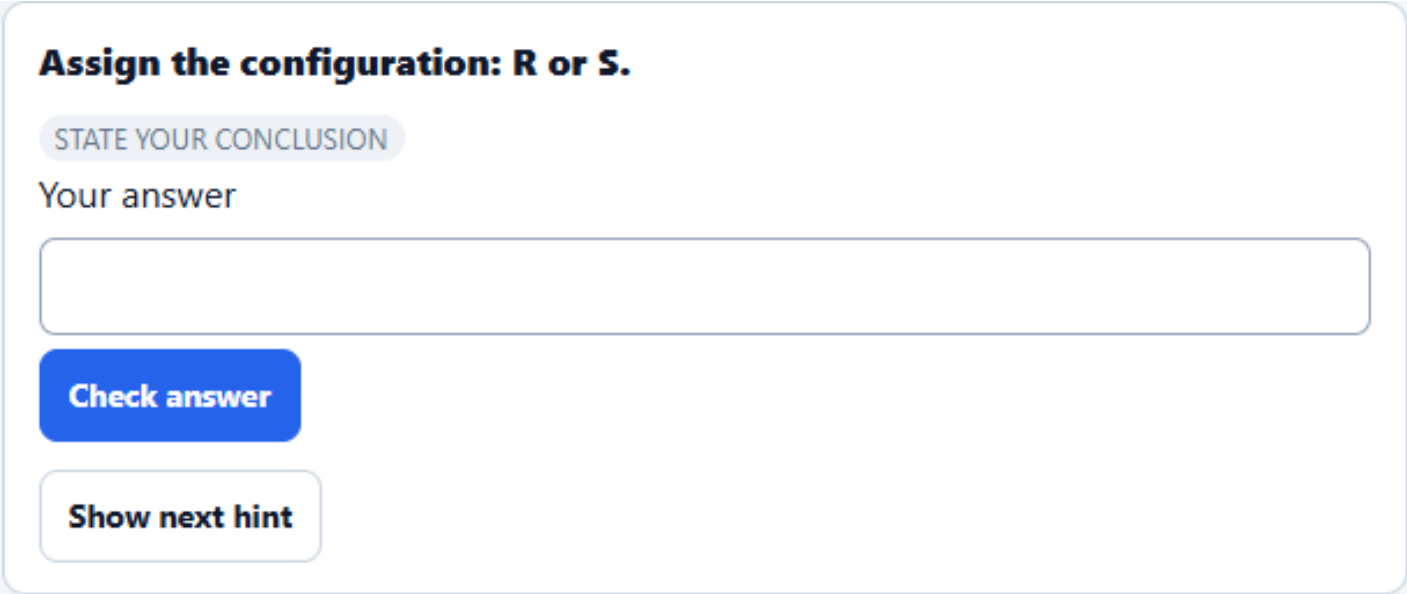
fischer-cip:stage-3 | Assign the configuration: R or S.

Answer type: text

Screenshot: fischer-cip-stage-3-molecule



Screenshot: fischer-cip-stage-3-question



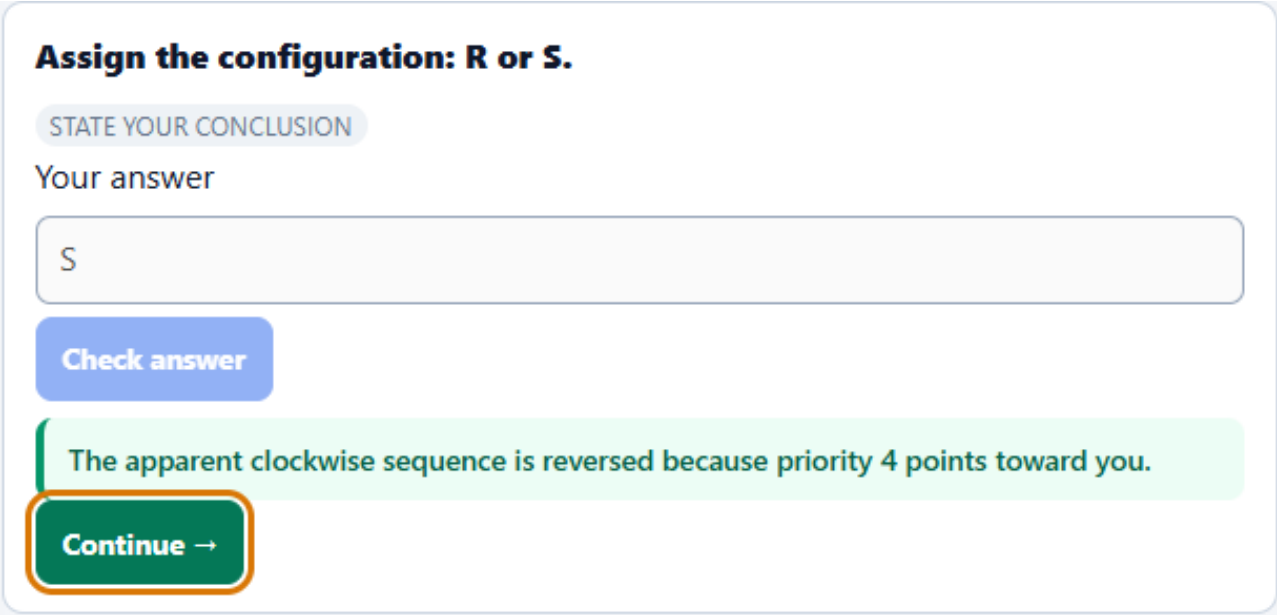
{"authorAnswer": ["S", "(S)", "2S", "(2S)", "(S)-butan-2-ol", "(2S)-butan-2-ol", "(S)-2-butanol", "(2S)-2-butanol"], "tolerance": null, "highlightsAfterCorrectAnswer": {}}

Hint 1: Rank OH > ethyl > methyl > H.

Hint 2: Trace right → bottom → top, then reverse because H is horizontal.

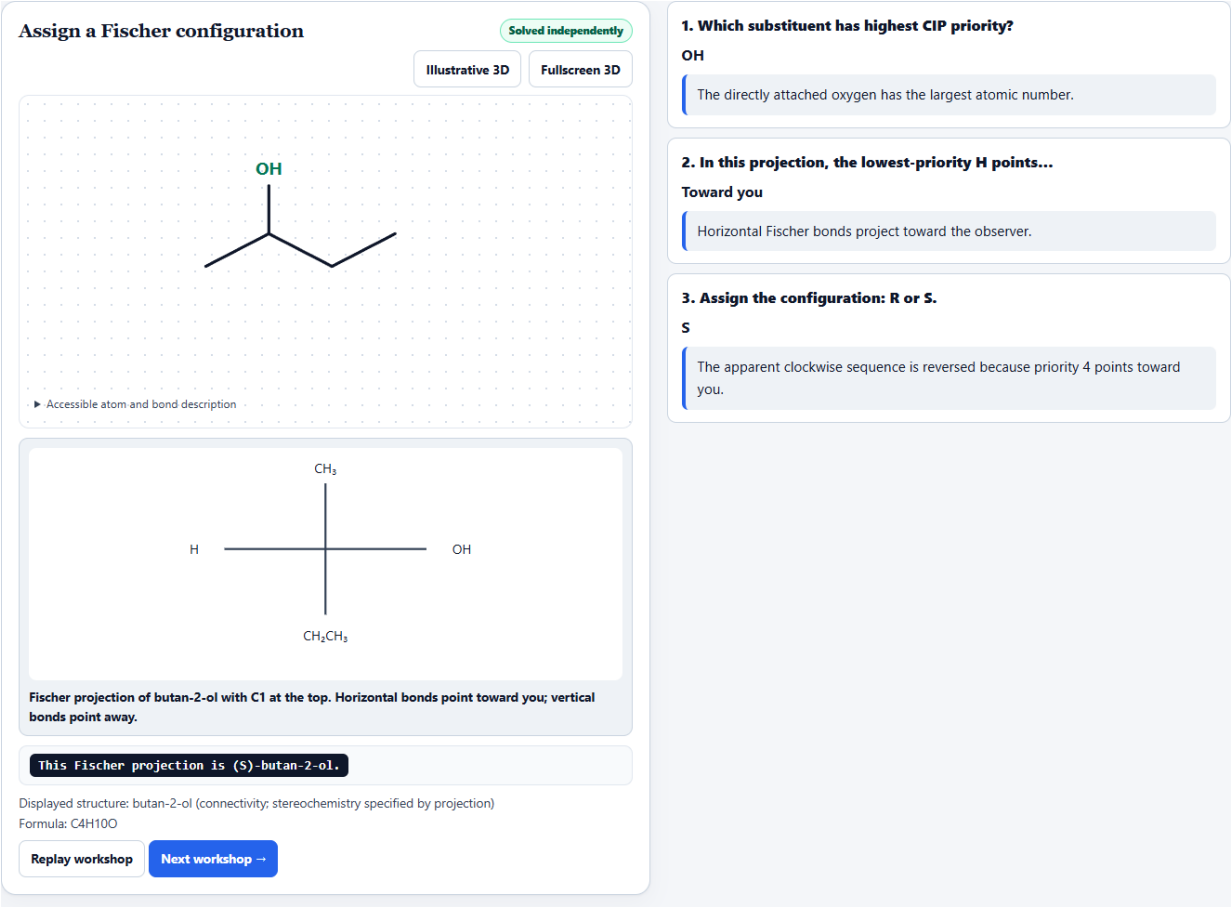
Explanation: The apparent clockwise sequence is reversed because priority 4 points toward you.

Screenshot: fischer-cip-stage-3-feedback



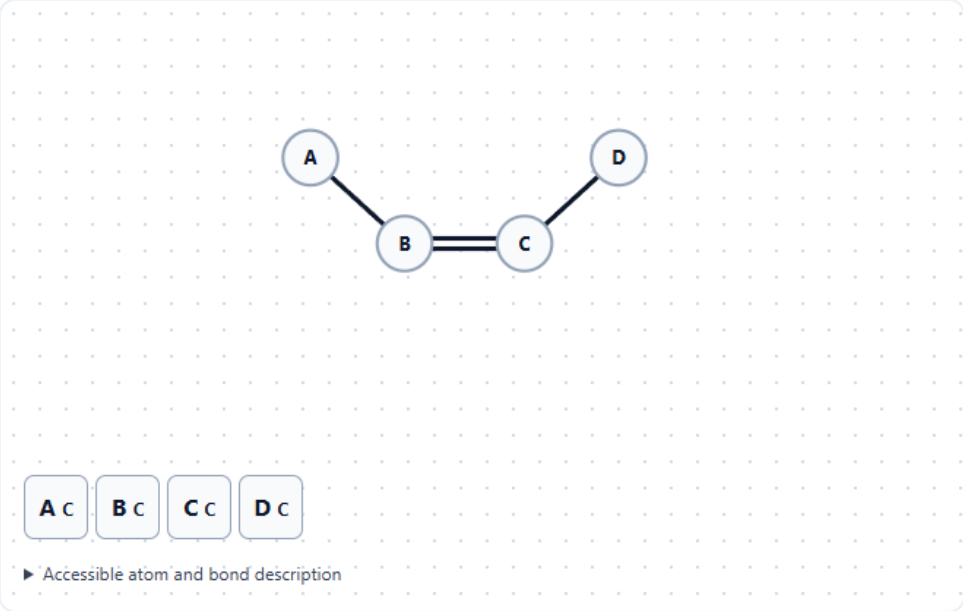
Completion summary: This Fischer projection is (S)-butan-2-ol.

Screenshot: fischer-cip-complete



ez-chirality | Separate E/Z geometry from chirality

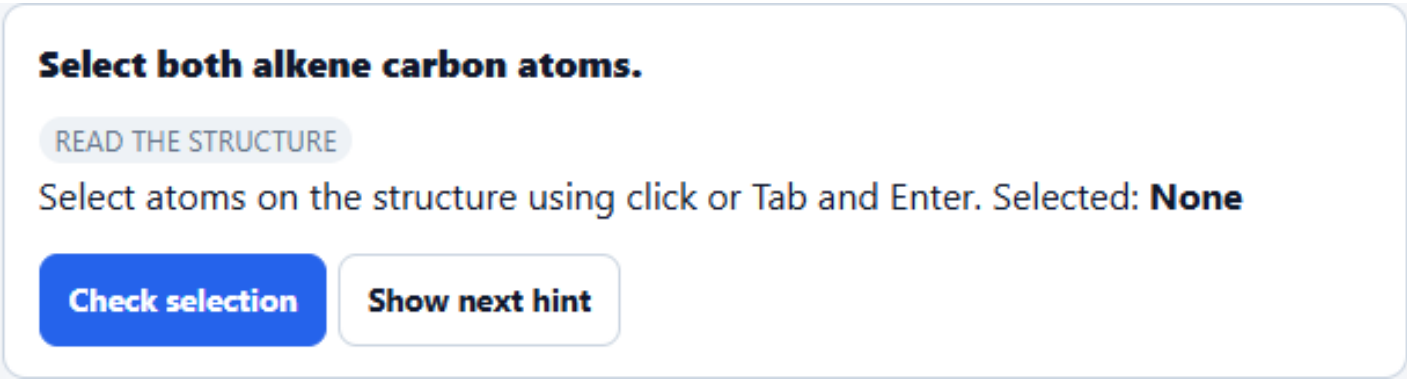
Screenshot: ez-chirality-stage-1-molecule



ez-chirality:stage-1 | Select both alkene carbon atoms.

Answer type: atoms

Screenshot: ez-chirality-stage-1-question



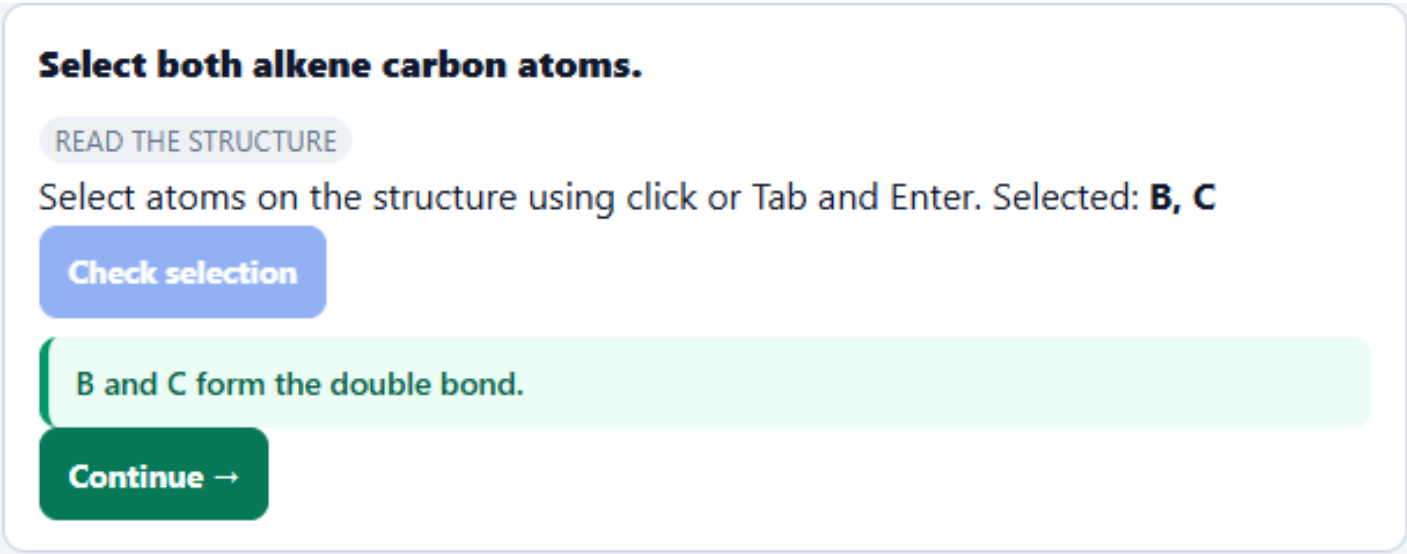
{"authorAnswer": [1, 2], "tolerance": null, "highlightsAfterCorrectAnswer": {"prin": [1, 2]}}

Hint 1: Look for two parallel bond lines.

Hint 2: Choose both endpoints of that bond.

Explanation: B and C form the double bond.

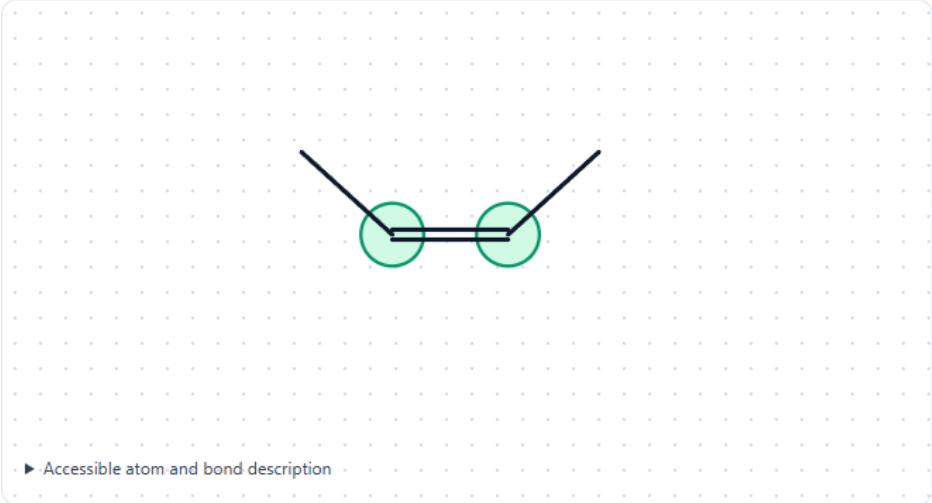
Screenshot: ez-chirality-stage-1-feedback



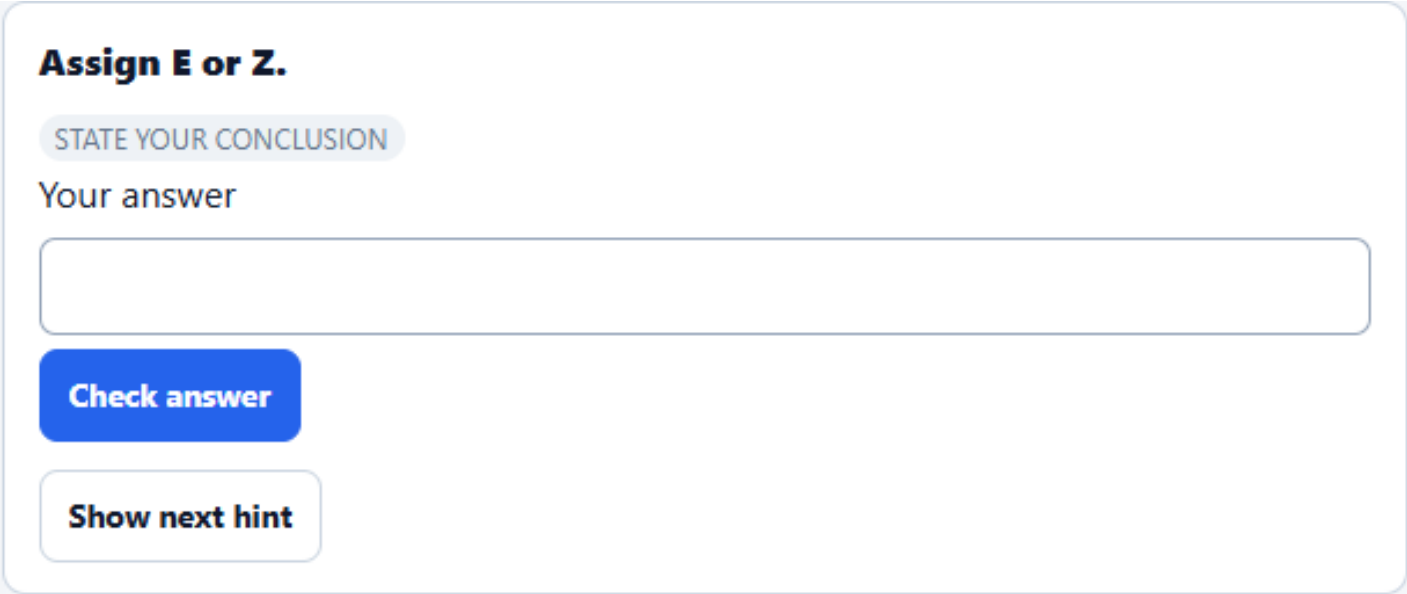
ez-chirality:stage-2 | Assign E or Z.

Answer type: text

Screenshot: ez-chirality-stage-2-molecule



Screenshot: ez-chirality-stage-2-question



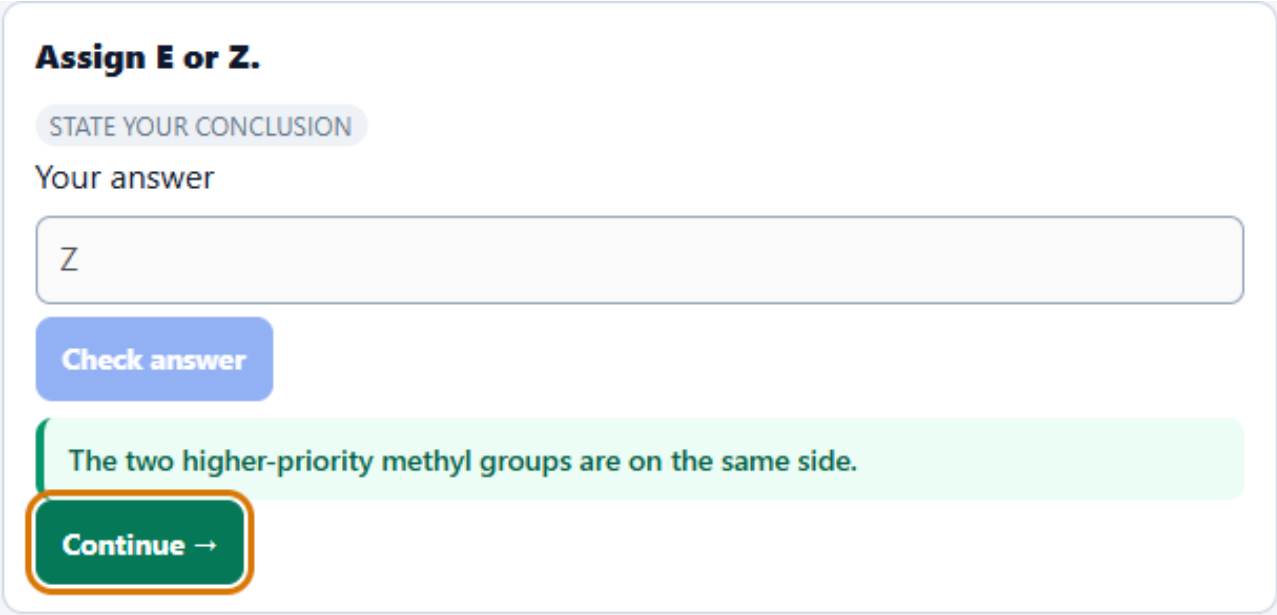
{"authorAnswer": ["Z", "(Z)", "2Z", "(2Z)", "(Z)-but-2-ene", "(2Z)-but-2-ene", "(Z)-2-butene", "(2Z)-2-butene"], "tolerance": null, "highlightsAfterCorrectAnswer": {}}

Hint 1: Rank methyl versus H on each end.

Hint 2: Same-side higher-priority groups determine the descriptor.

Explanation: The two higher-priority methyl groups are on the same side.

Screenshot: ez-chirality-stage-2-feedback



ez-chirality:stage-3 | Why does a pure sample of this alkene isomer show no ordinary optical rotation?

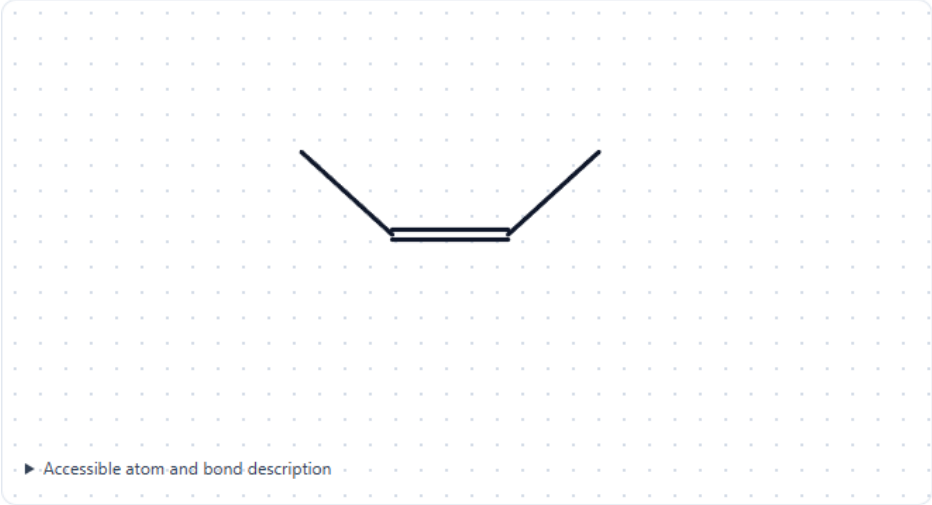
Answer type: choice

Option 0 (A): It is achiral

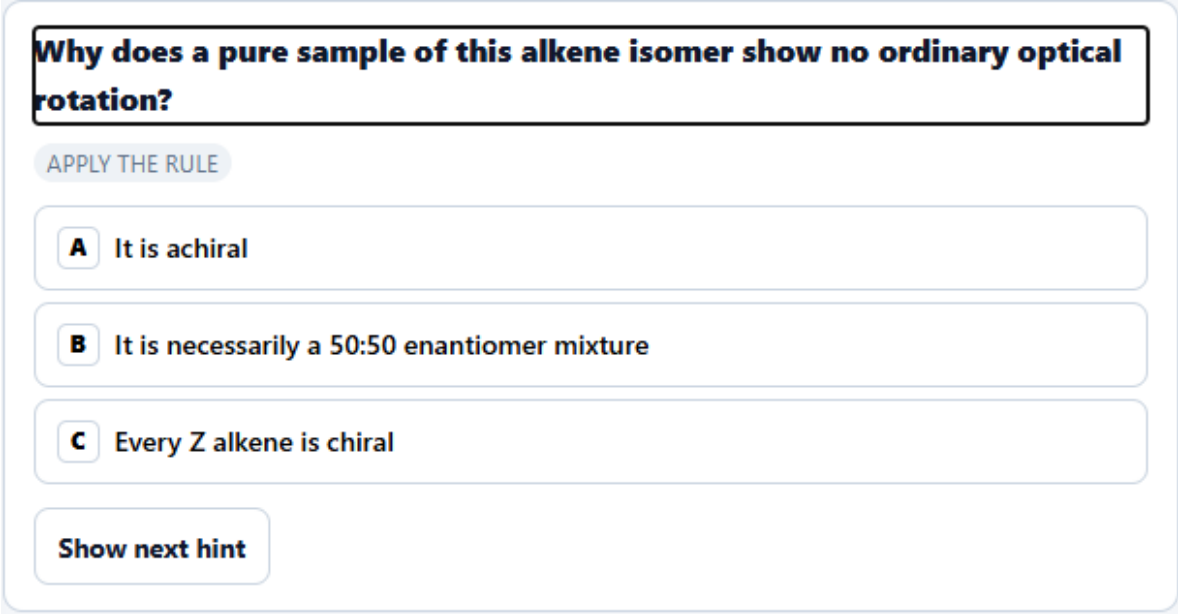
Option 1 (B): It is necessarily a 50:50 enantiomer mixture

Option 2 (C): Every Z alkene is chiral

Screenshot: ez-chirality-stage-3-molecule



Screenshot: ez-chirality-stage-3-question



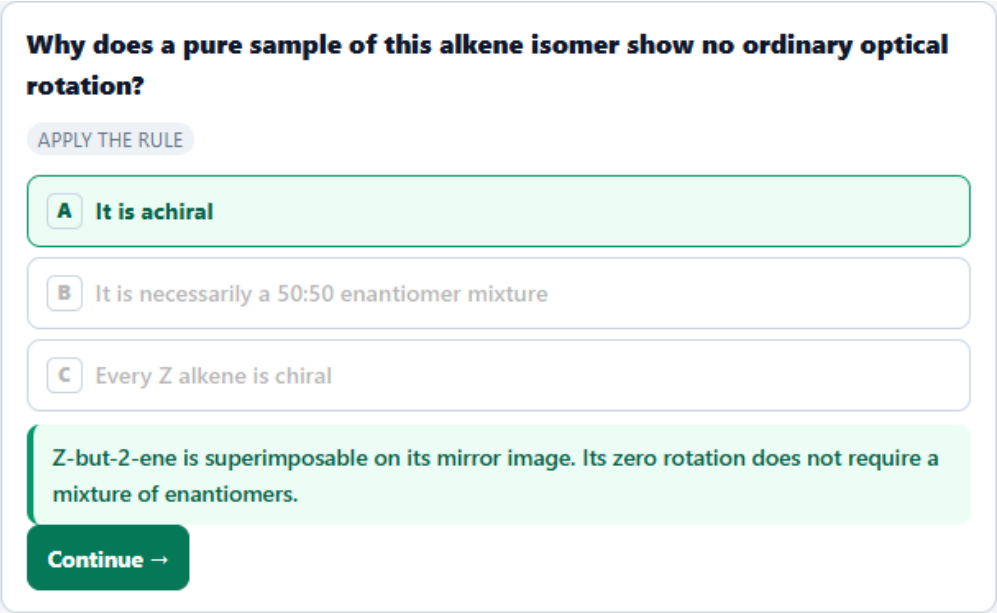
{"authorAnswer": 0, "tolerance": null, "highlightsAfterCorrectAnswer": {}}

Hint 1: Distinguish a property of one molecule from cancellation in a mixture.

Hint 2: Inspect the symmetry of the entire planar but-2-ene structure.

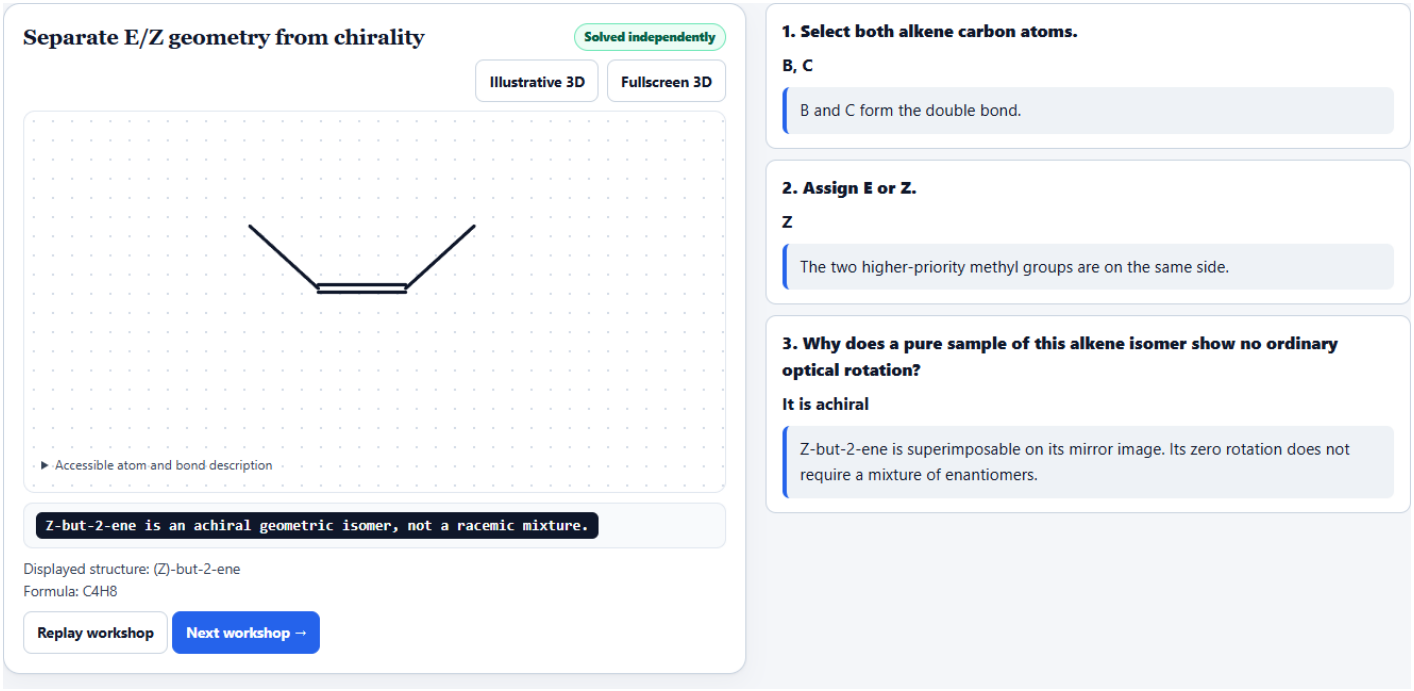
Explanation: Z-but-2-ene is superimposable on its mirror image. Its zero rotation does not require a mixture of enantiomers.

Screenshot: ez-chirality-stage-3-feedback



Completion summary: Z-but-2-ene is an achiral geometric isomer, not a racemic mixture.

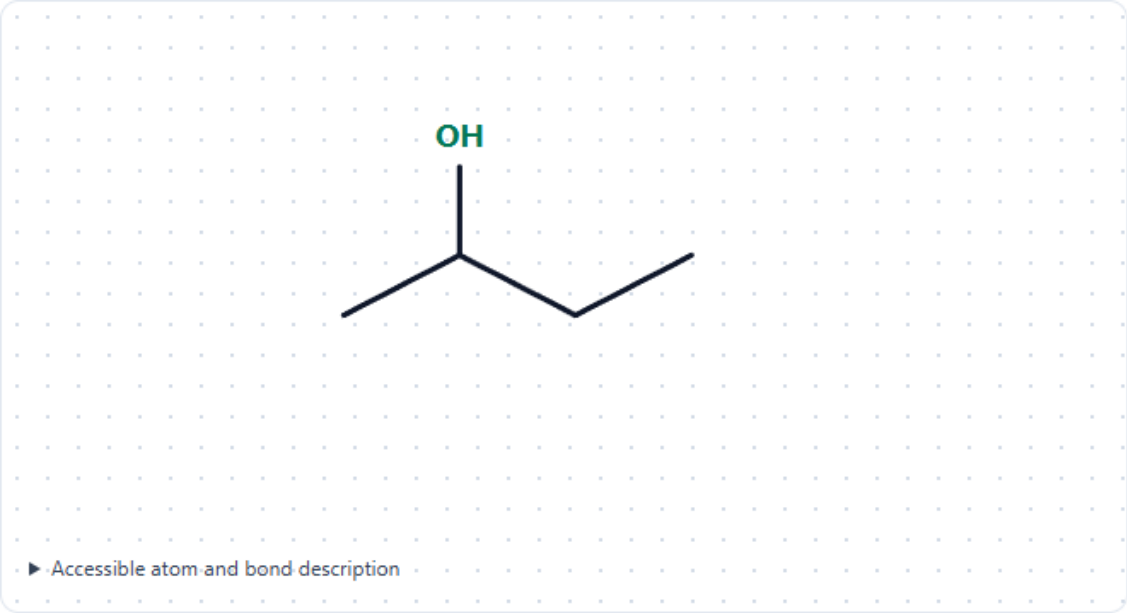
Screenshot: ez-chirality-complete



specific-rotation | Normalize a polarimeter measurement

Hypothetical polarimetry data for an enantiomerically pure chiral sample with the illustrated connectivity: αobs = +2.4°, c = 0.080 g/mL, path length = 5.0 cm. These are practice data, not literature measurements for butan-2-ol. Do not infer the sign from the flat molecular drawing.

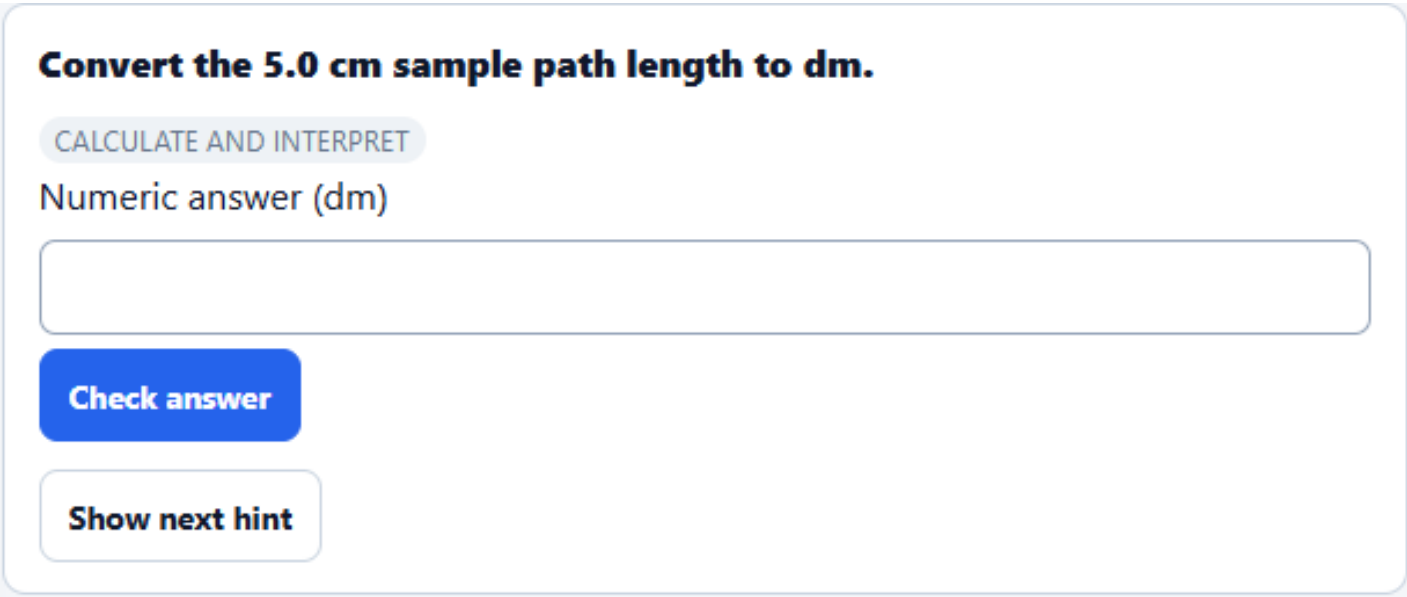
Screenshot: specific-rotation-stage-1-molecule



specific-rotation:stage-1 | Convert the 5.0 cm sample path length to dm.

Answer type: number | unit: dm

Screenshot: specific-rotation-stage-1-question



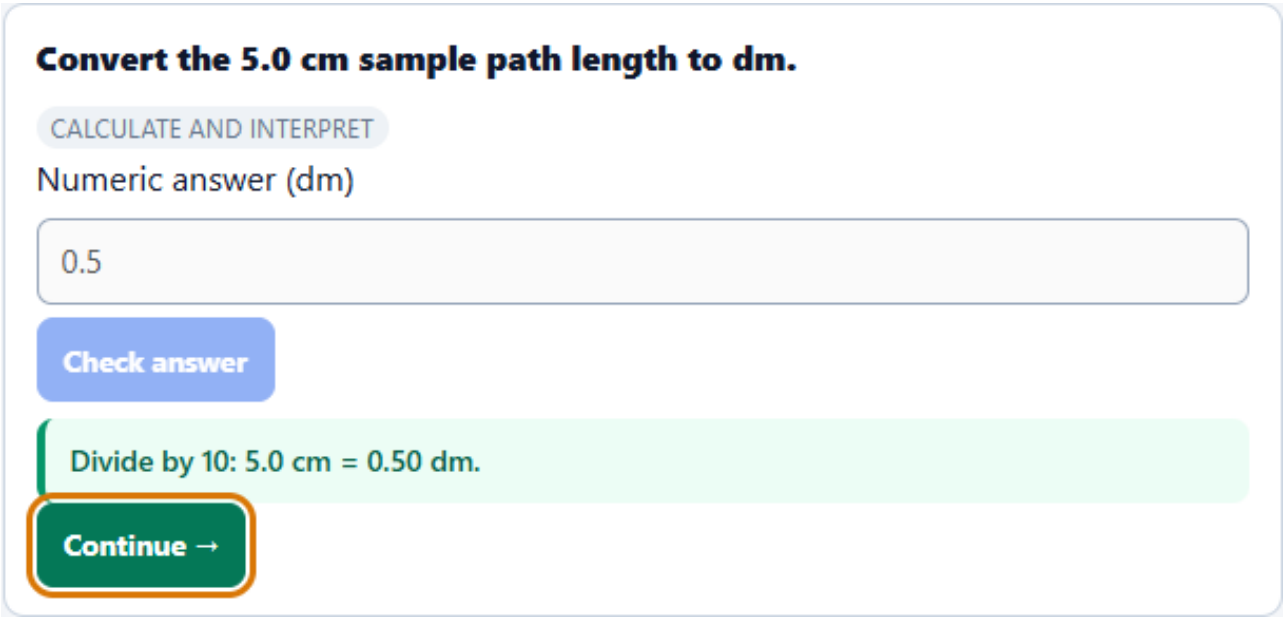
{"authorAnswer": 0.5, "tolerance": 0.0001, "highlightsAfterCorrectAnswer": {}}

Hint 1: The specific-rotation formula here requires dm.

Hint 2: One decimeter is ten centimeters.

Explanation: Divide by 10: 5.0 cm = 0.50 dm.

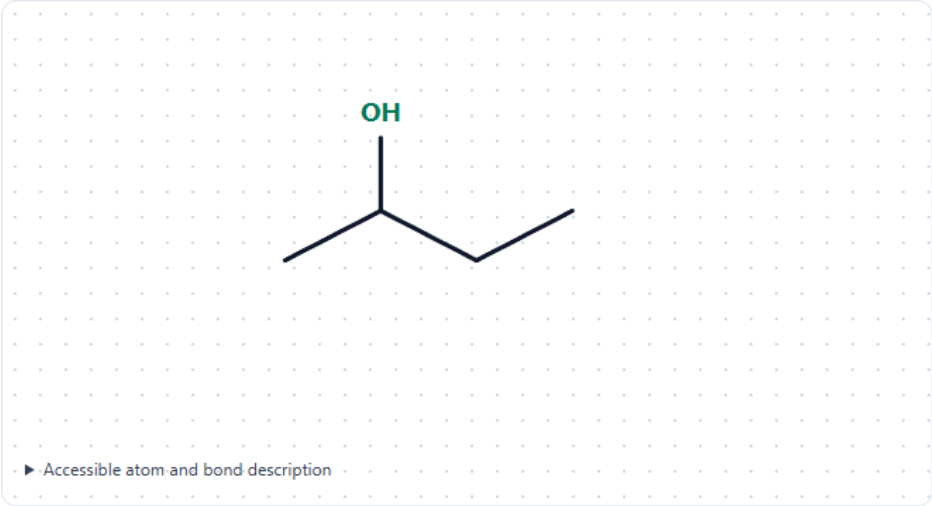
Screenshot: specific-rotation-stage-1-feedback



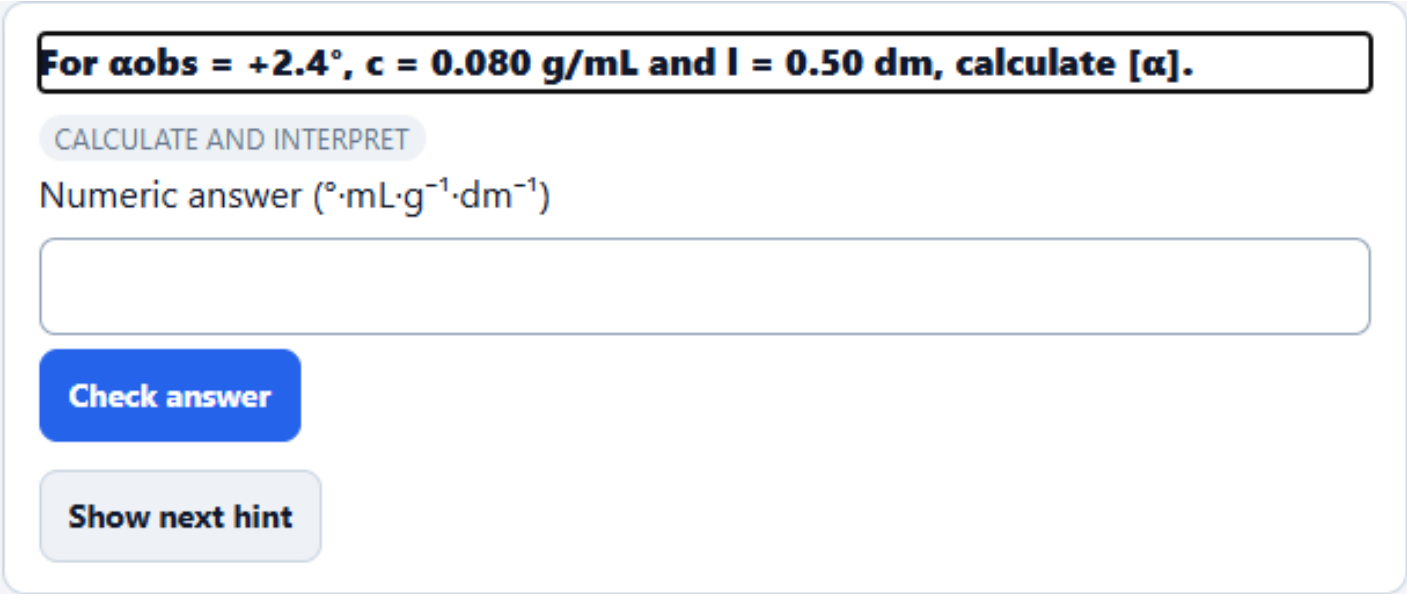
specific-rotation:stage-2 | For αobs = +2.4°, c = 0.080 g/mL and l = 0.50 dm, calculate [α].

Answer type: number | unit: °·mL·g^{-1}·dm^{-1}

Screenshot: specific-rotation-stage-2-molecule



Screenshot: specific-rotation-stage-2-question



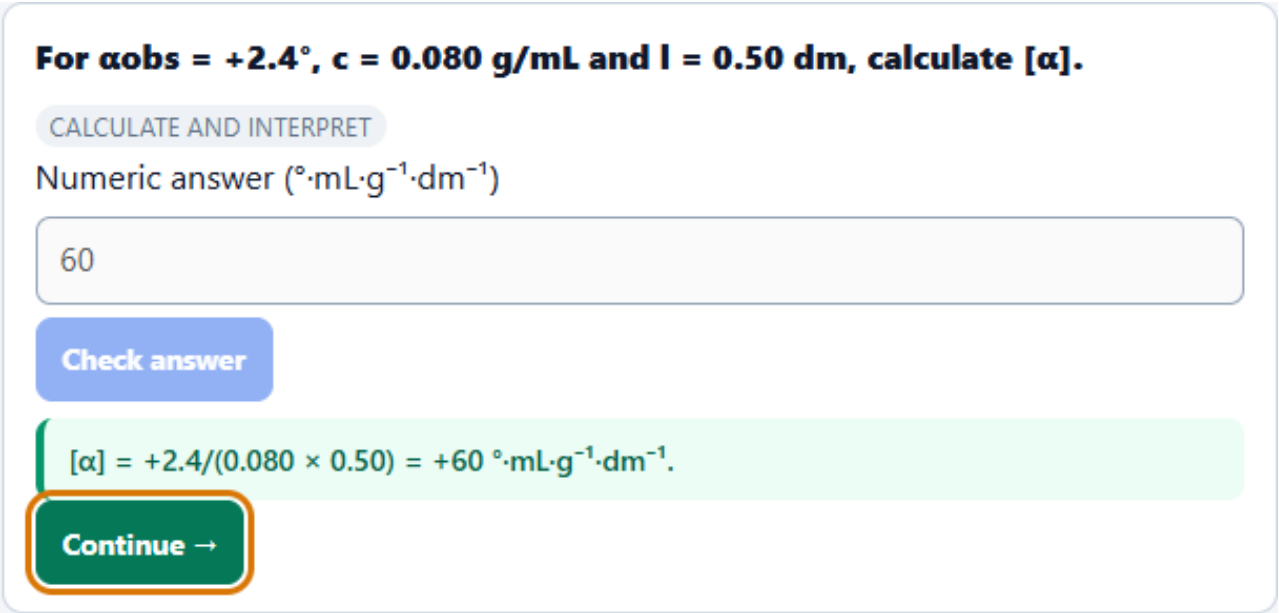
{"authorAnswer": 60, "tolerance": 0.01, "highlightsAfterCorrectAnswer": {}}

Hint 1: Divide observed rotation by concentration times length.

Hint 2: The denominator is 0.080 × 0.50 = 0.040.

Explanation: [α] = +2.4/(0.080 × 0.50) = +60 °·mL·g^{-1}·dm^{-1}.

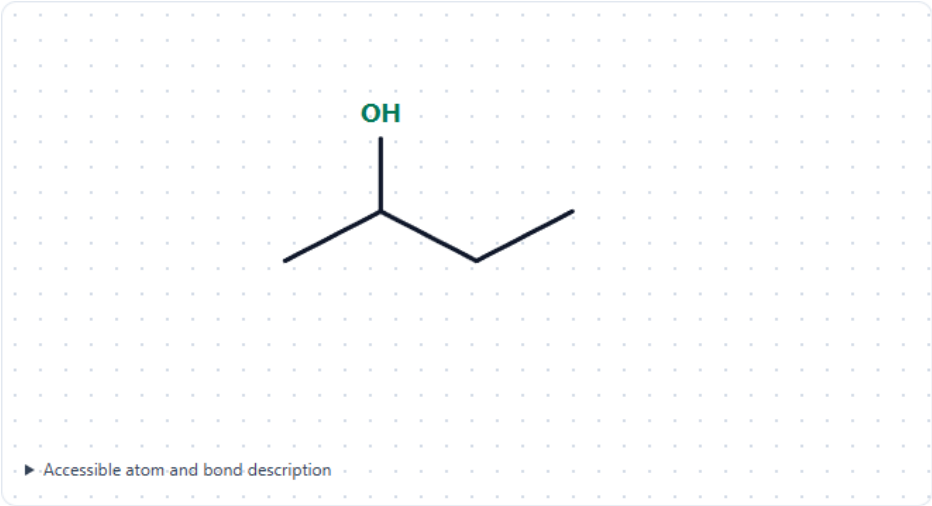
Screenshot: specific-rotation-stage-2-feedback



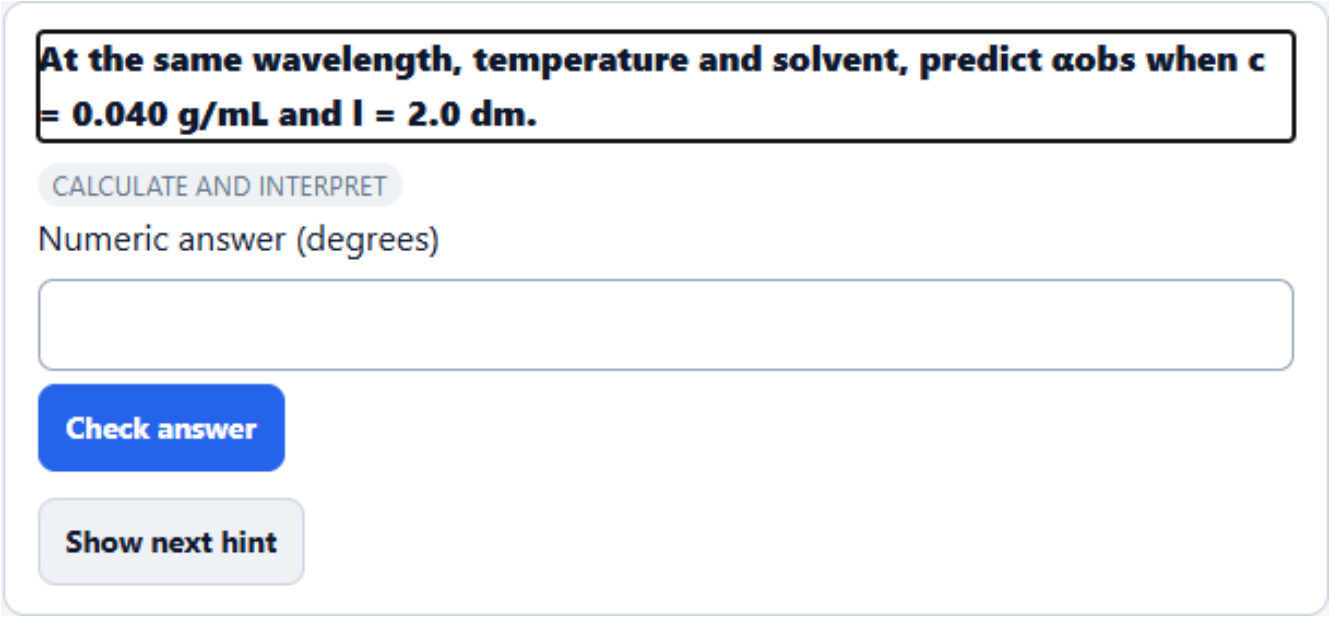
specific-rotation:stage-3 | At the same wavelength, temperature and solvent, predict αobs when c = 0.040 g/mL and l = 2.0 dm.

Answer type: number | unit: degrees

Screenshot: specific-rotation-stage-3-molecule



Screenshot: specific-rotation-stage-3-question



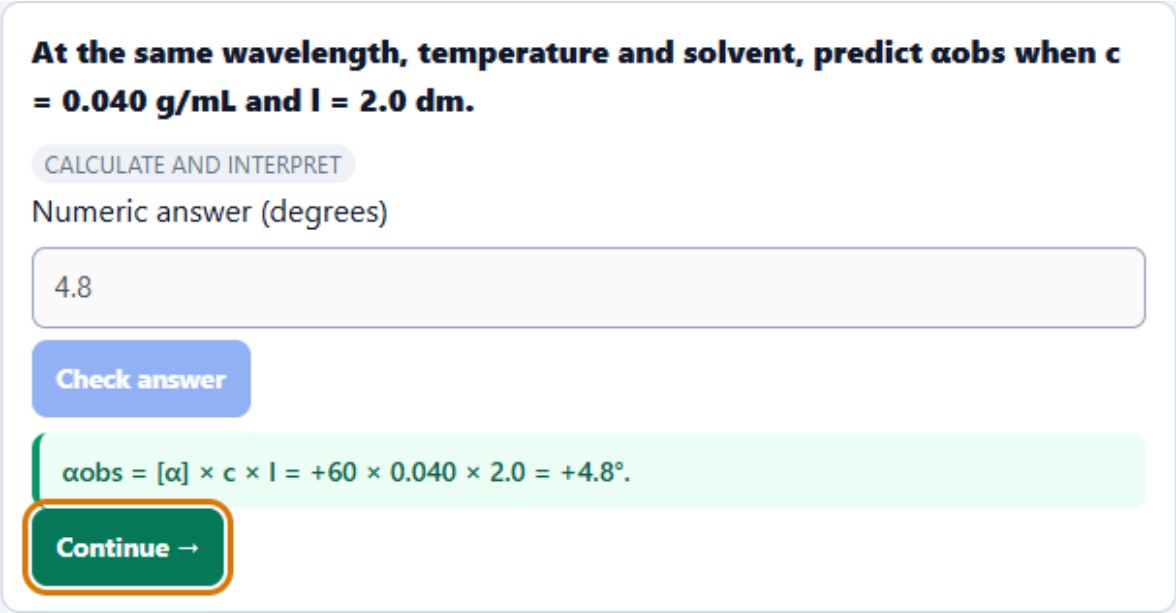
{"authorAnswer": 4.8, "tolerance": 0.001, "highlightsAfterCorrectAnswer": {}}

Hint 1: Rearrange the same normalization equation.

Hint 2: Use the previously calculated specific rotation with the new concentration and length.

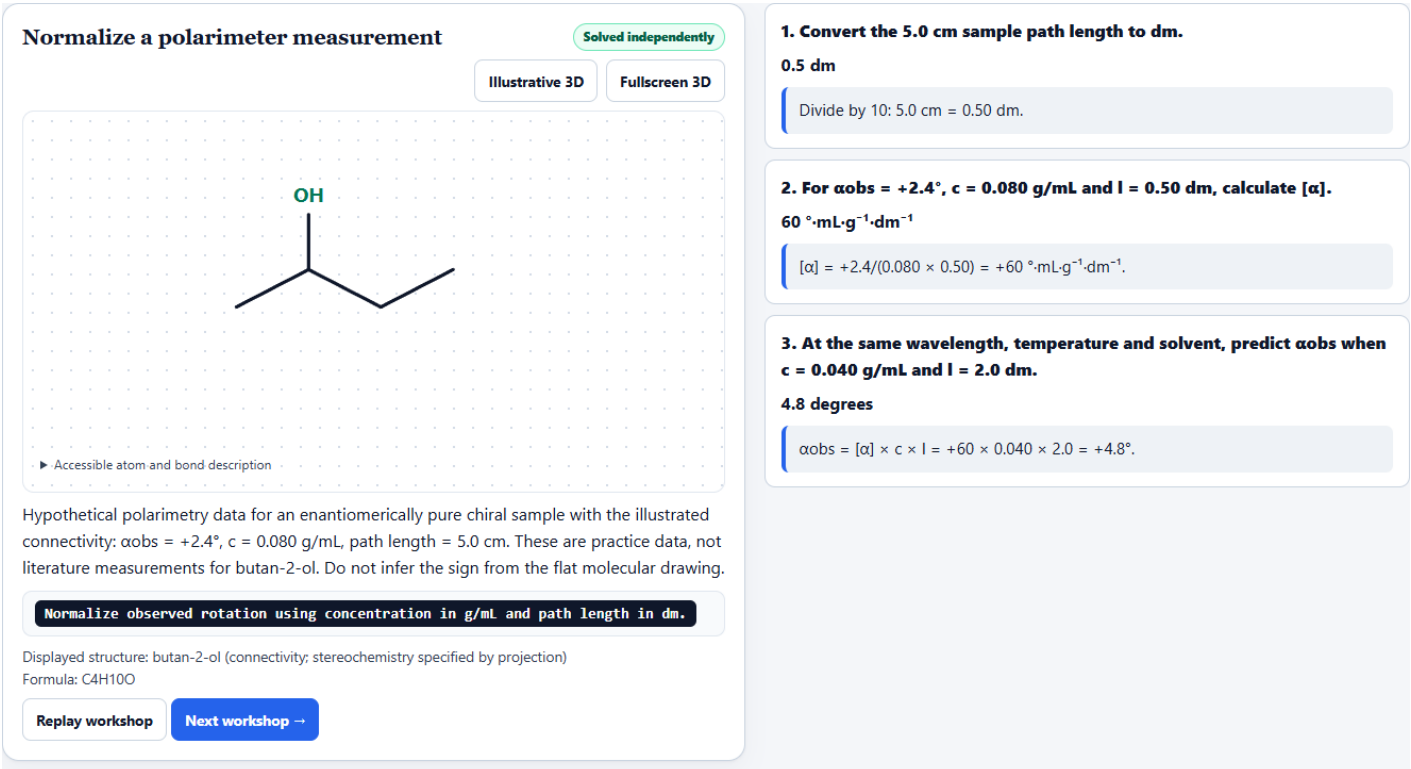
Explanation: αobs = [α] × c × l = +60 × 0.040 × 2.0 = +4.8°.

Screenshot: specific-rotation-stage-3-feedback



Completion summary: Normalize observed rotation using concentration in g/mL and path length in dm.

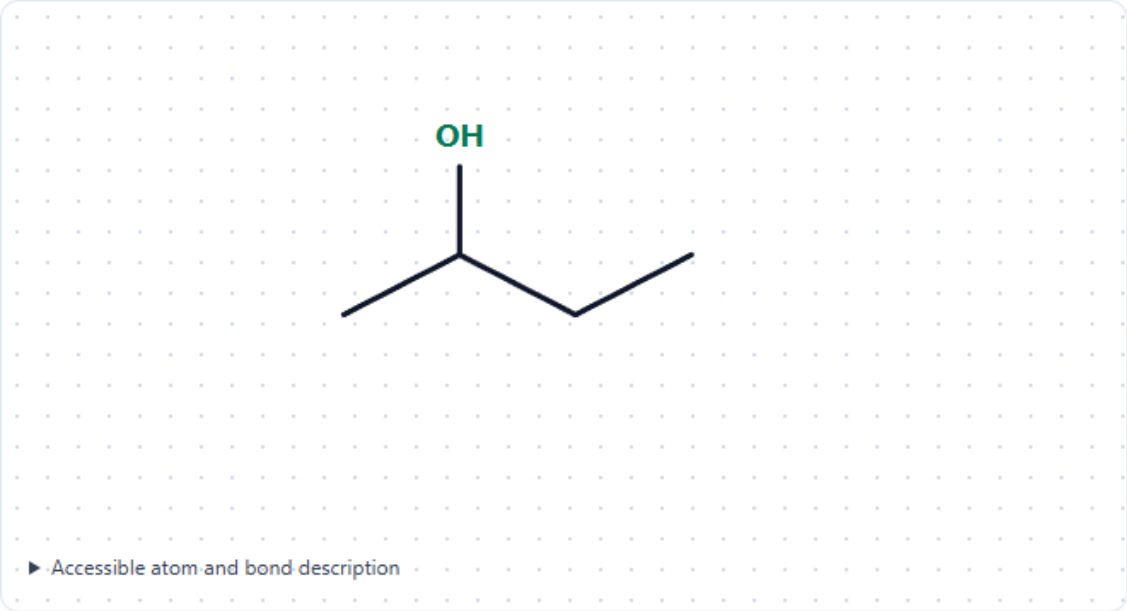
Screenshot: specific-rotation-complete



optical-mixture | Interpret an enantiomer mixture

The illustrated chiral connectivity represents a pair of enantiomers. The 85:15 composition and +40° reference rotation are hypothetical practice data; no (+)/(−) assignment is made from the drawing.

Screenshot: optical-mixture-stage-1-molecule



optical-mixture:stage-1 | The major enantiomer is dextrorotatory. Which sign describes its observed rotation?

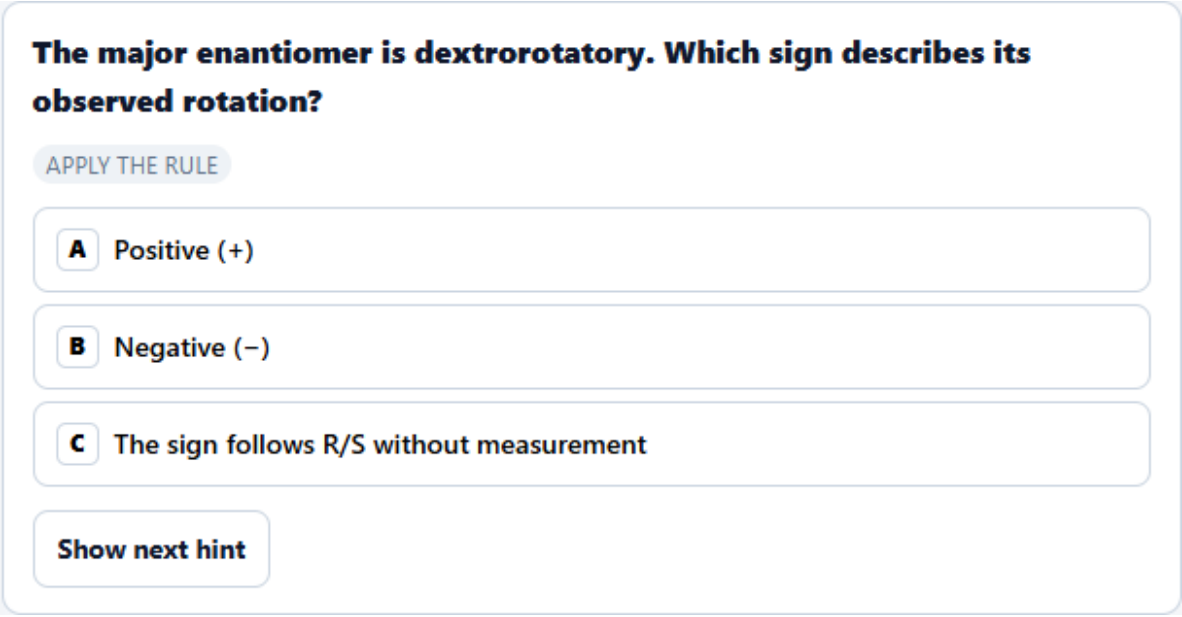
Answer type: choice

Option 0 (A): Positive (+)

Option 1 (B): Negative (−)

Option 2 (C): The sign follows R/S without measurement

Screenshot: optical-mixture-stage-1-question



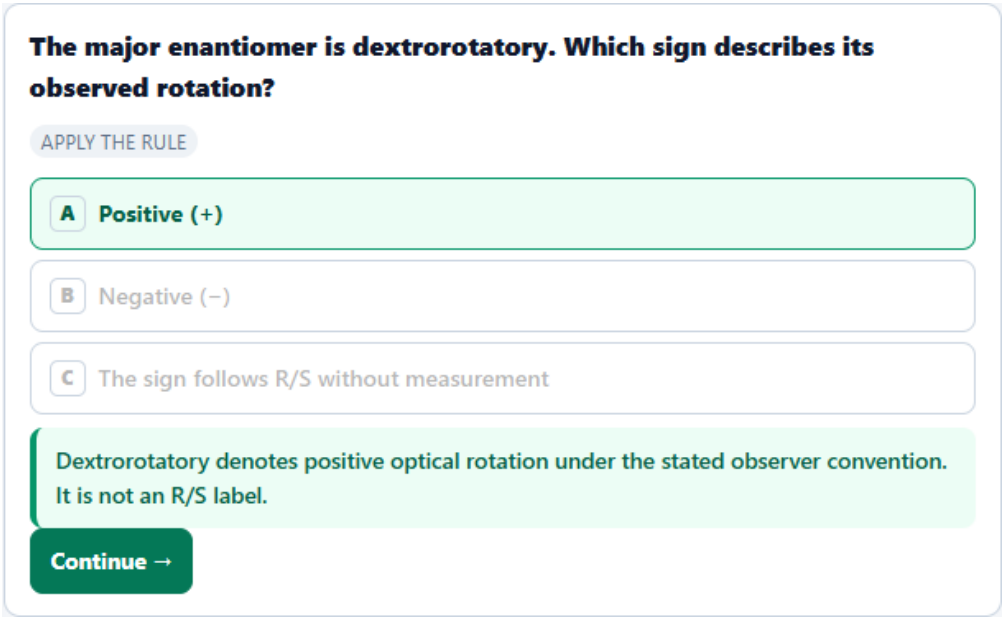
{"authorAnswer": 0, "tolerance": null, "highlightsAfterCorrectAnswer": {}}

Hint 1: Use the optical-rotation vocabulary.

Hint 2: Configuration descriptors do not determine this sign.

Explanation: Dextrorotatory denotes positive optical rotation under the stated observer convention. It is not an R/S label.

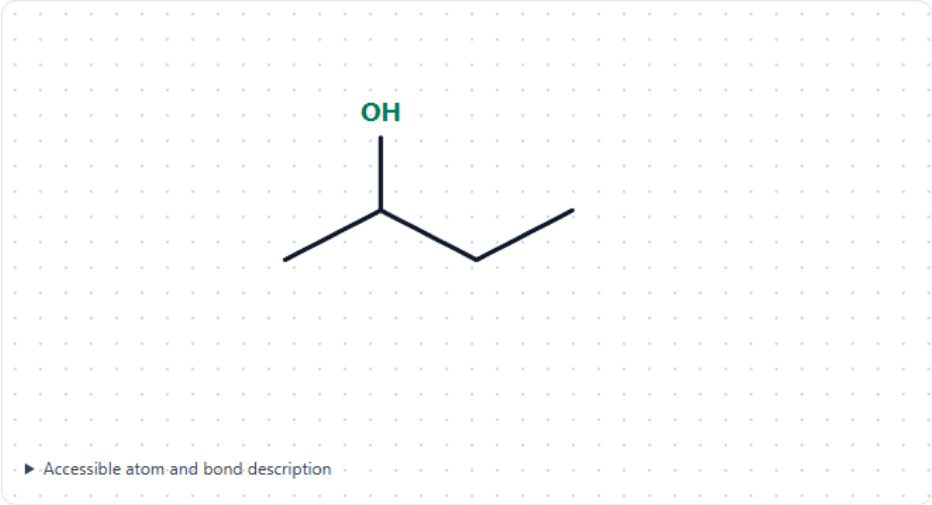
Screenshot: optical-mixture-stage-1-feedback



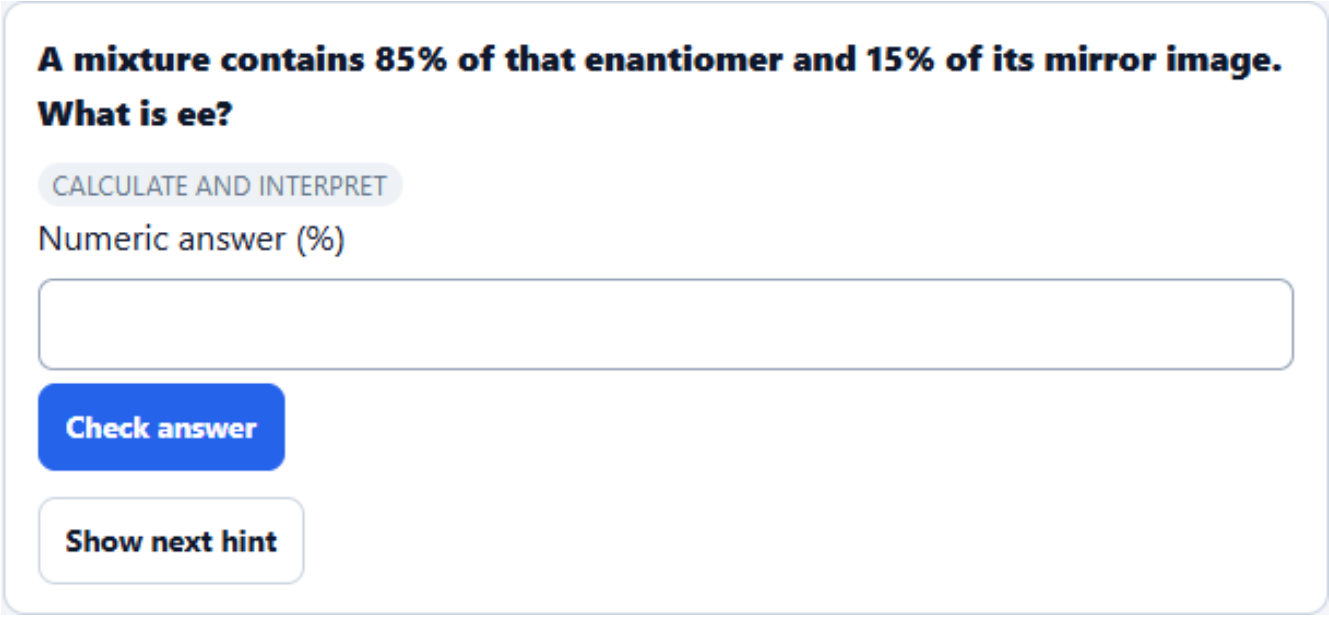
optical-mixture:stage-2 | A mixture contains 85% of that enantiomer and 15% of its mirror image. What is ee?

Answer type: number | unit: %

Screenshot: optical-mixture-stage-2-molecule



Screenshot: optical-mixture-stage-2-question



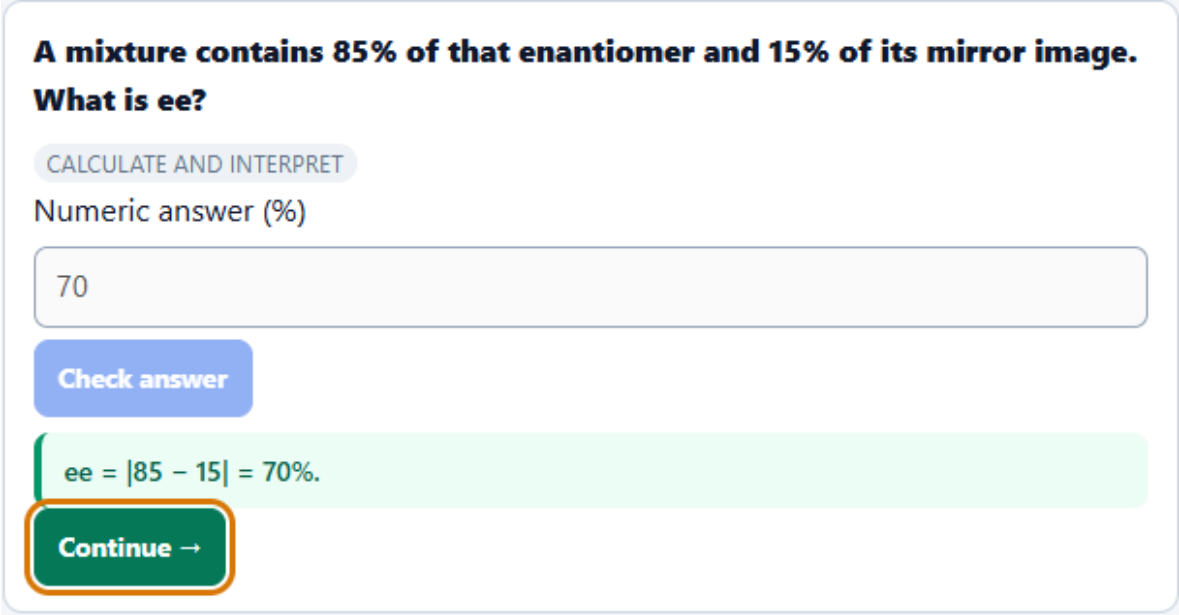
{"authorAnswer": 70, "tolerance": 0, "highlightsAfterCorrectAnswer": {}}

Hint 1: Subtract the smaller enantiomer percentage from the larger.

Hint 2: Do not use the 75:25 worked example without recalculating.

Explanation: ee = |85 − 15| = 70%.

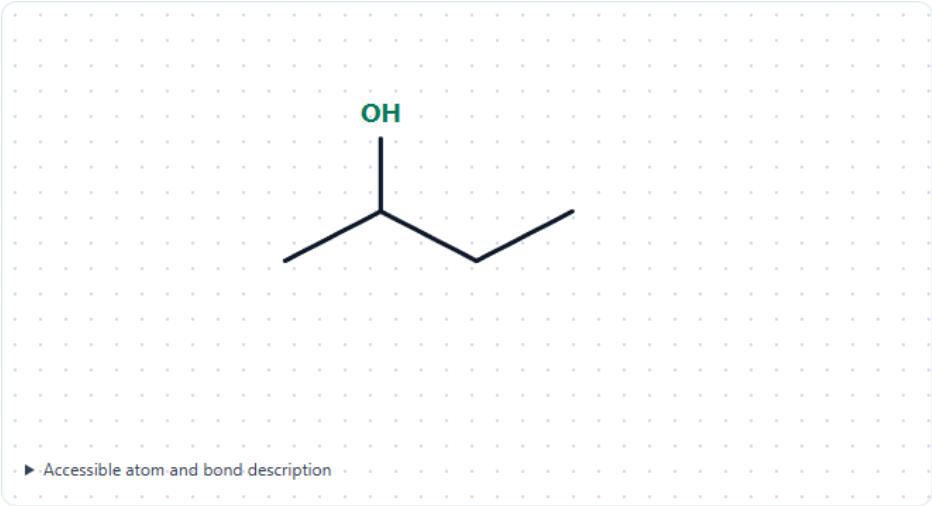
Screenshot: optical-mixture-stage-2-feedback



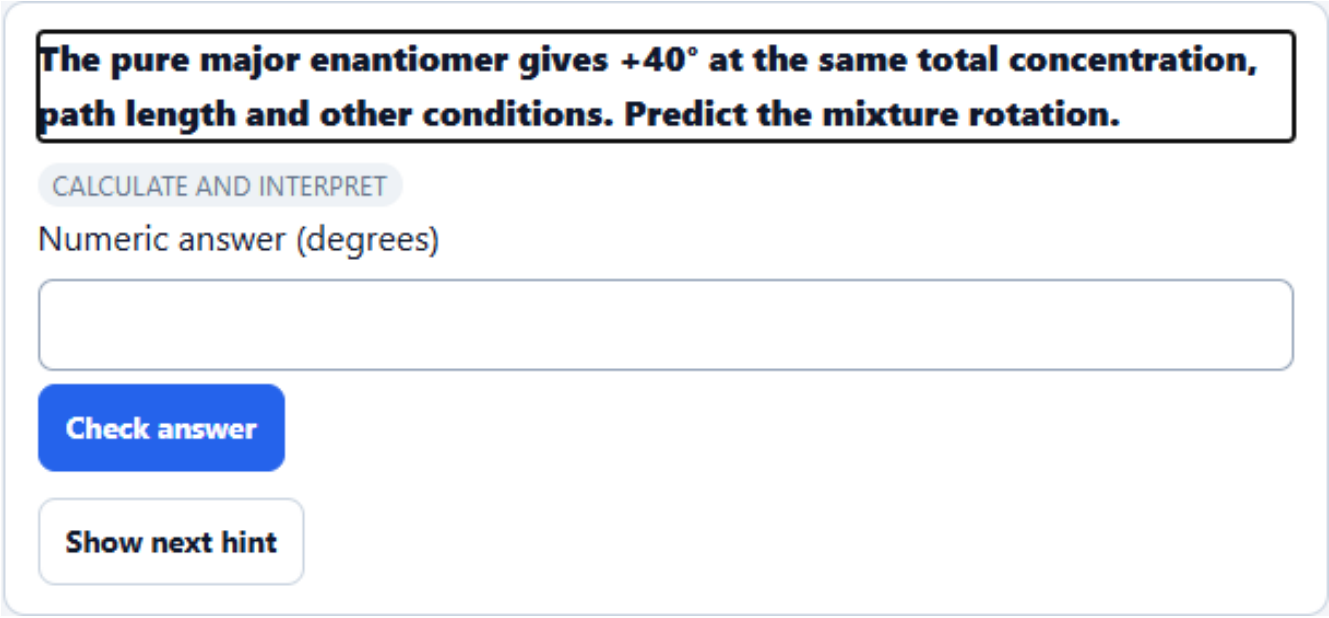
optical-mixture:stage-3 | The pure major enantiomer gives +40° at the same total concentration, path length and other conditions. Predict the mixture rotation.

Answer type: number | unit: degrees

Screenshot: optical-mixture-stage-3-molecule



Screenshot: optical-mixture-stage-3-question



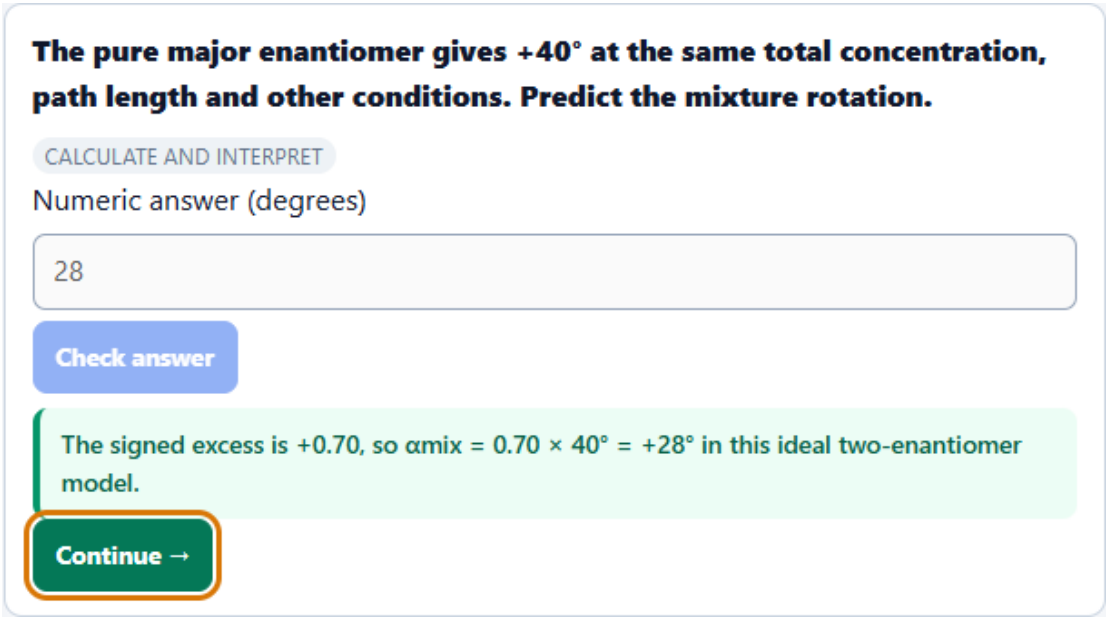
{"authorAnswer": 28, "tolerance": 0.001, "highlightsAfterCorrectAnswer": {}}

Hint 1: The two enantiomers rotate equally and oppositely.

Hint 2: Multiply the pure-enantiomer angle by the fractional ee.

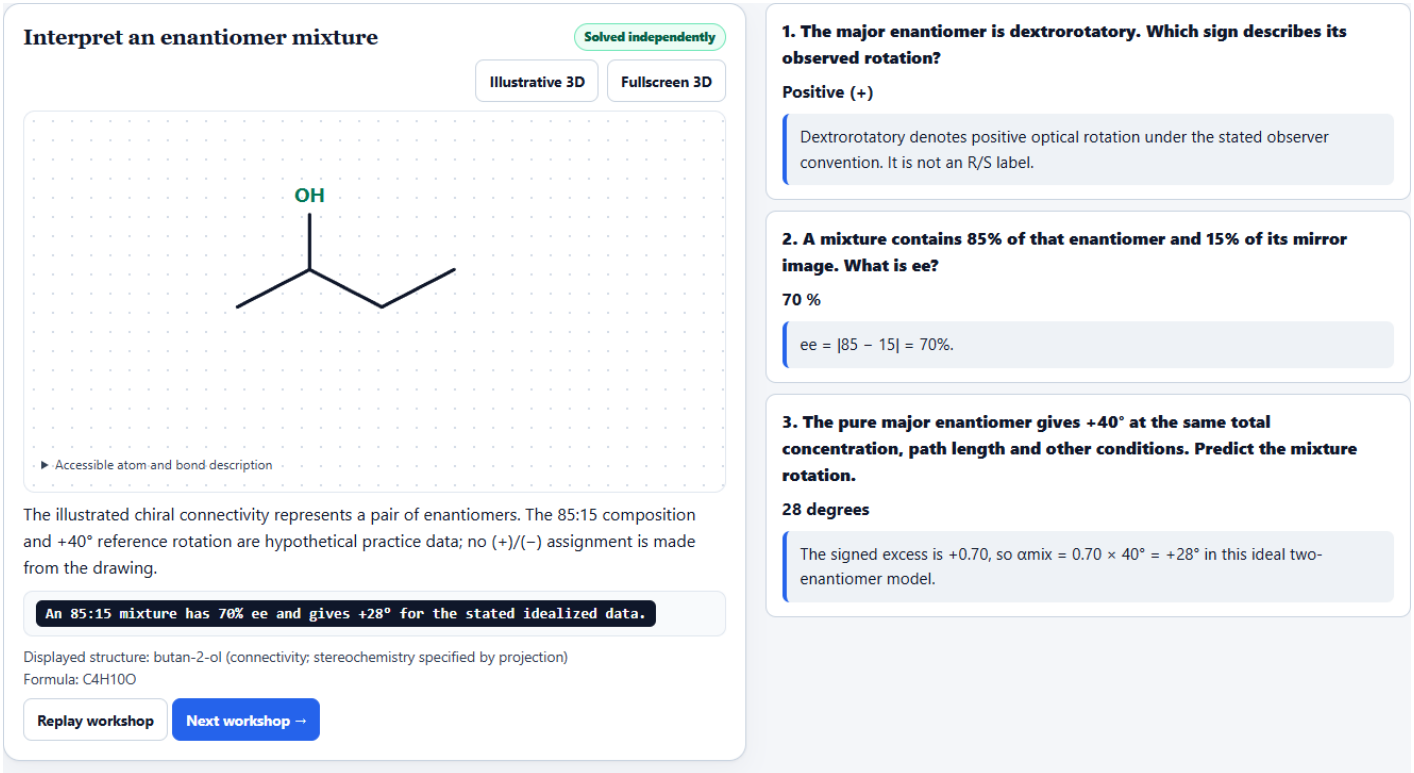
Explanation: The signed excess is +0.70, so αmix = 0.70 × 40° = +28° in this ideal two-enantiomer model.

Screenshot: optical-mixture-stage-3-feedback



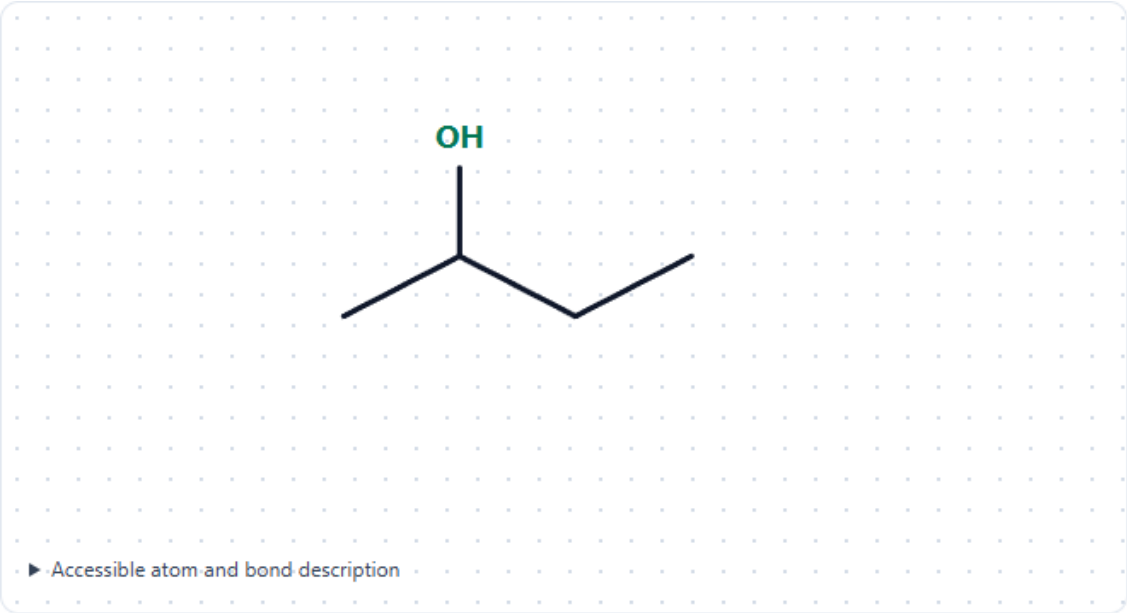
Completion summary: An 85:15 mixture has 70% ee and gives +28° for the stated idealized data.

Screenshot: optical-mixture-complete



chiral-resolution | Choose a chiral resolution strategy

Screenshot: chiral-resolution-stage-1-molecule



chiral-resolution:stage-1 | Why does ordinary achiral distillation generally fail to resolve two enantiomers?

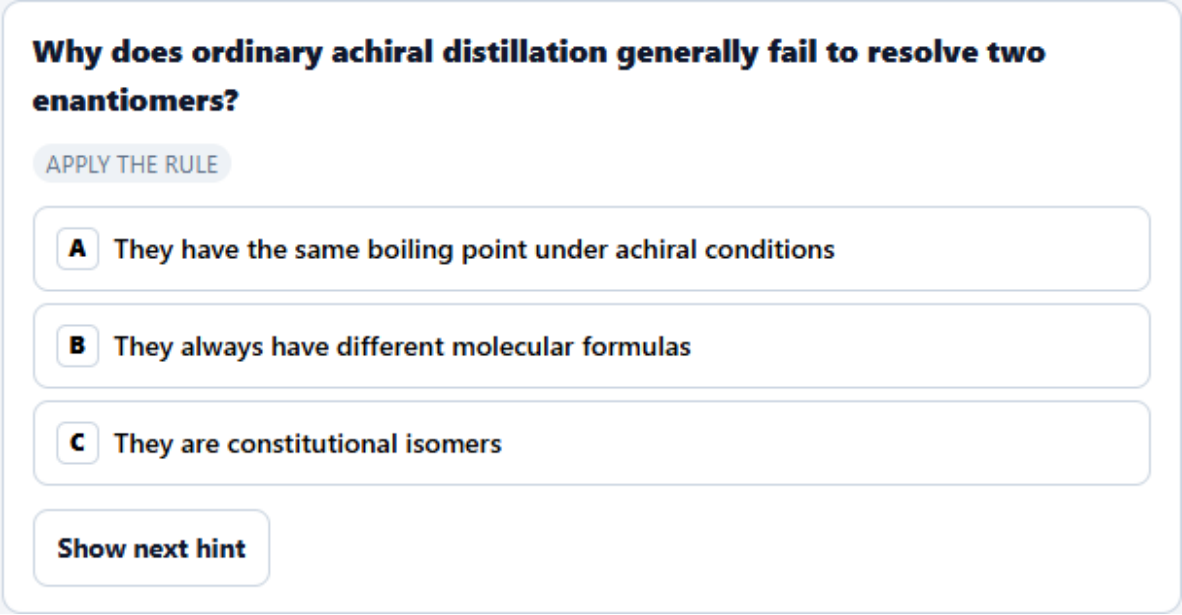
Answer type: choice

Option 0 (A): They have the same boiling point under achiral conditions

Option 1 (B): They always have different molecular formulas

Option 2 (C): They are constitutional isomers

Screenshot: chiral-resolution-stage-1-question



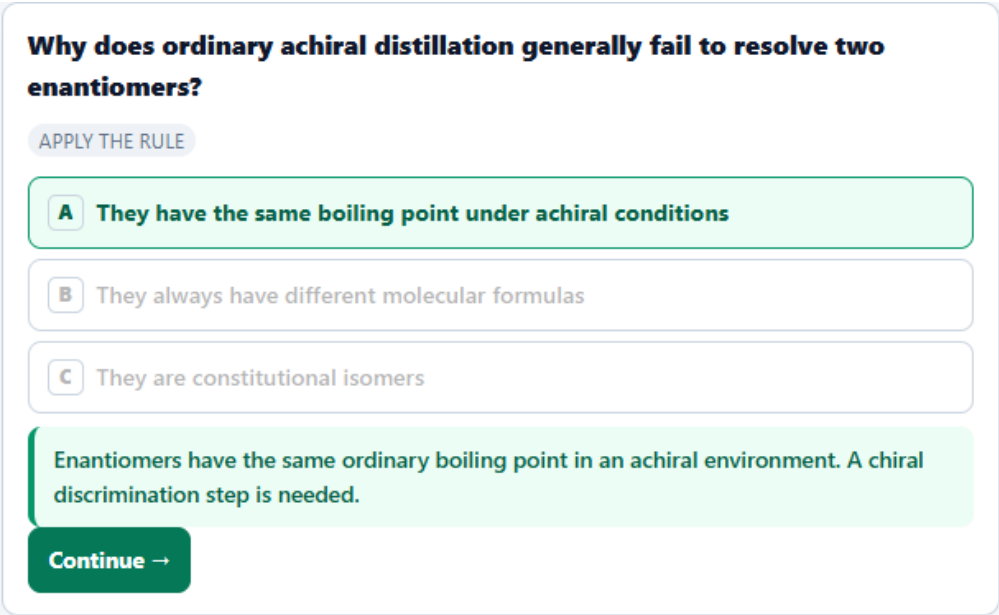
{"authorAnswer": 0, "tolerance": null, "highlightsAfterCorrectAnswer": {}}

Hint 1: Compare their properties in an achiral environment.

Hint 2: Opposite optical rotation does not imply different boiling points.

Explanation: Enantiomers have the same ordinary boiling point in an achiral environment. A chiral discrimination step is needed.

Screenshot: chiral-resolution-stage-1-feedback



chiral-resolution:stage-2 | For a suitable substrate, reaction with one enantiomer of a resolving agent can create…

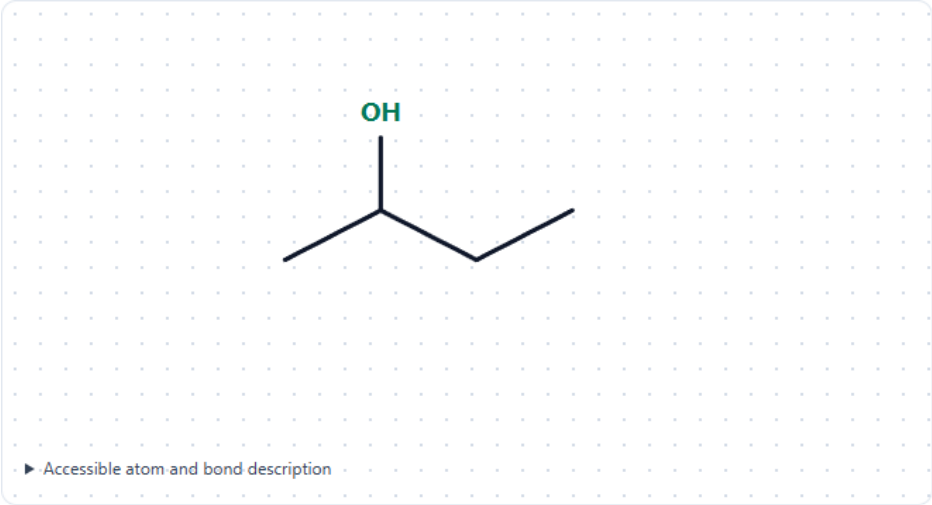
Answer type: choice

Option 0 (A): Separable diastereomeric derivatives or salts

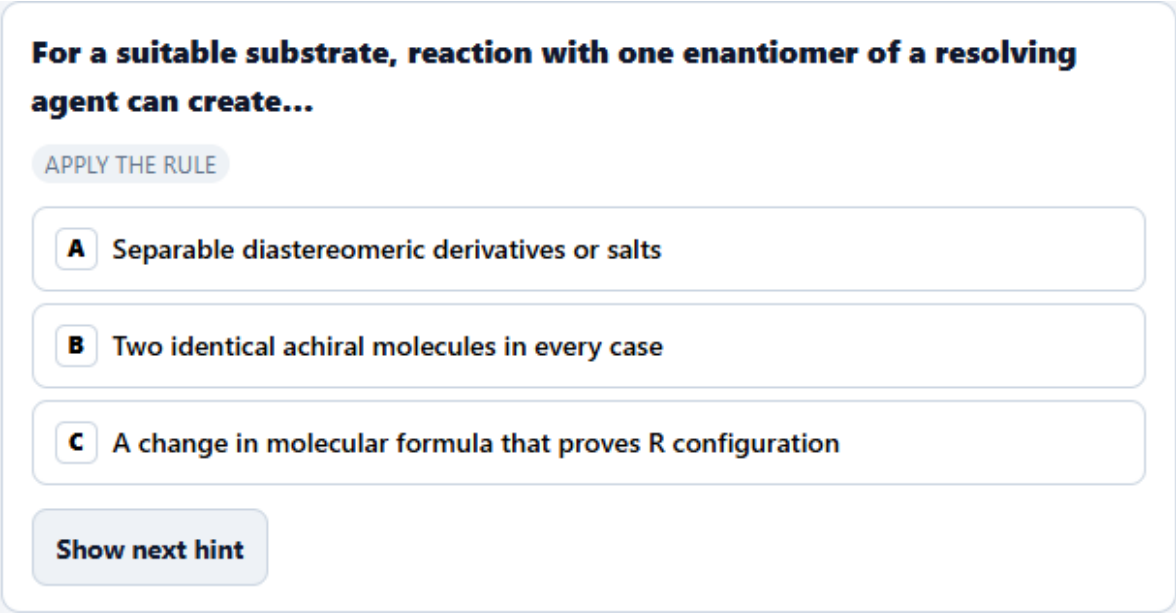
Option 1 (B): Two identical achiral molecules in every case

Option 2 (C): A change in molecular formula that proves R configuration

Screenshot: chiral-resolution-stage-2-molecule



Screenshot: chiral-resolution-stage-2-question



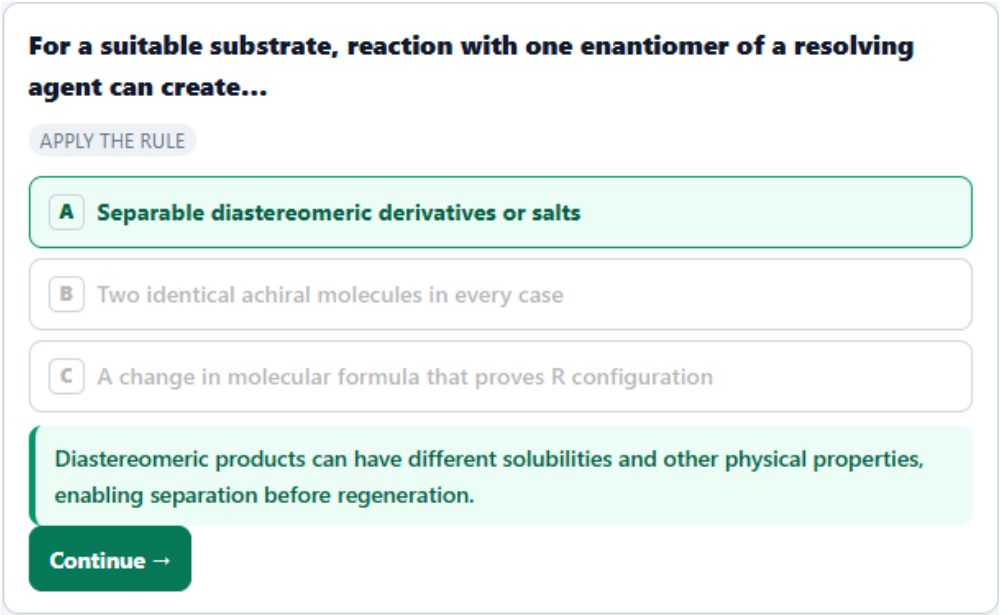
{"authorAnswer": 0, "tolerance": null, "highlightsAfterCorrectAnswer": {}}

Hint 1: Add one fixed chiral influence to both enantiomers.

Hint 2: The two resulting products are no longer mirror-image counterparts.

Explanation: Diastereomeric products can have different solubilities and other physical properties, enabling separation before regeneration.

Screenshot: chiral-resolution-stage-2-feedback



chiral-resolution:stage-3 | A remote substituent change retains the local arrangement at a tetrahedral stereocenter but changes its CIP priorities. Its R/S label…

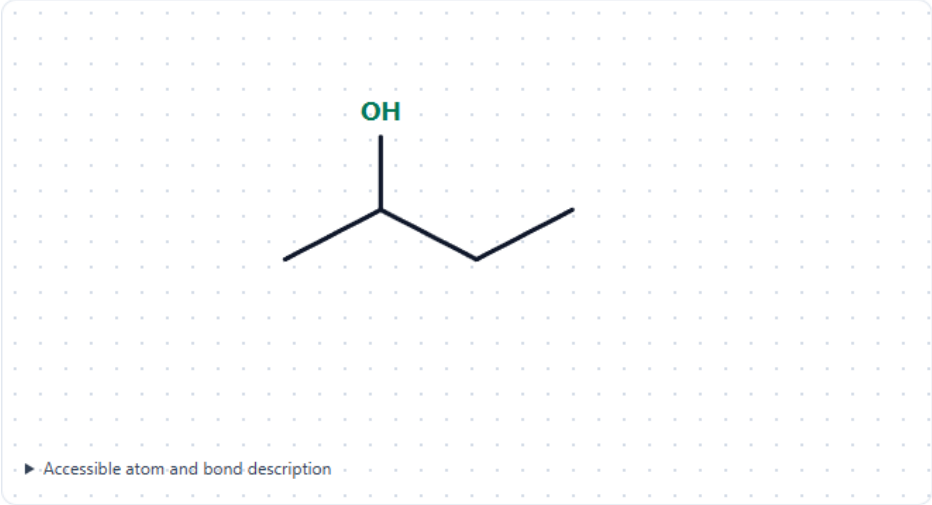
Answer type: choice

Option 0 (A): May change without physical inversion

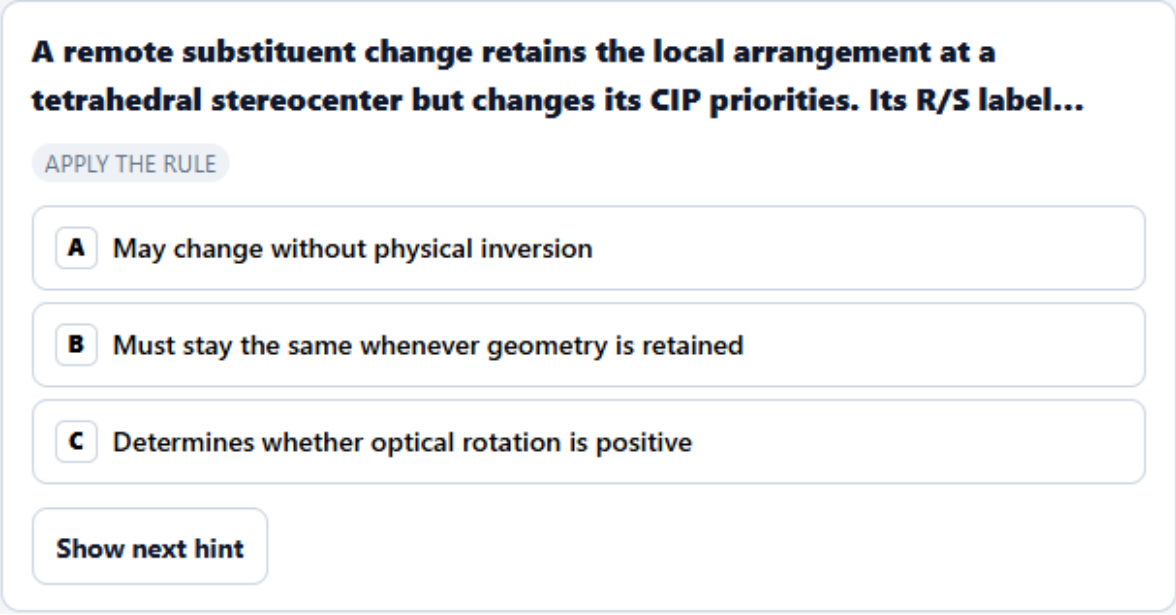
Option 1 (B): Must stay the same whenever geometry is retained

Option 2 (C): Determines whether optical rotation is positive

Screenshot: chiral-resolution-stage-3-molecule



Screenshot: chiral-resolution-stage-3-question



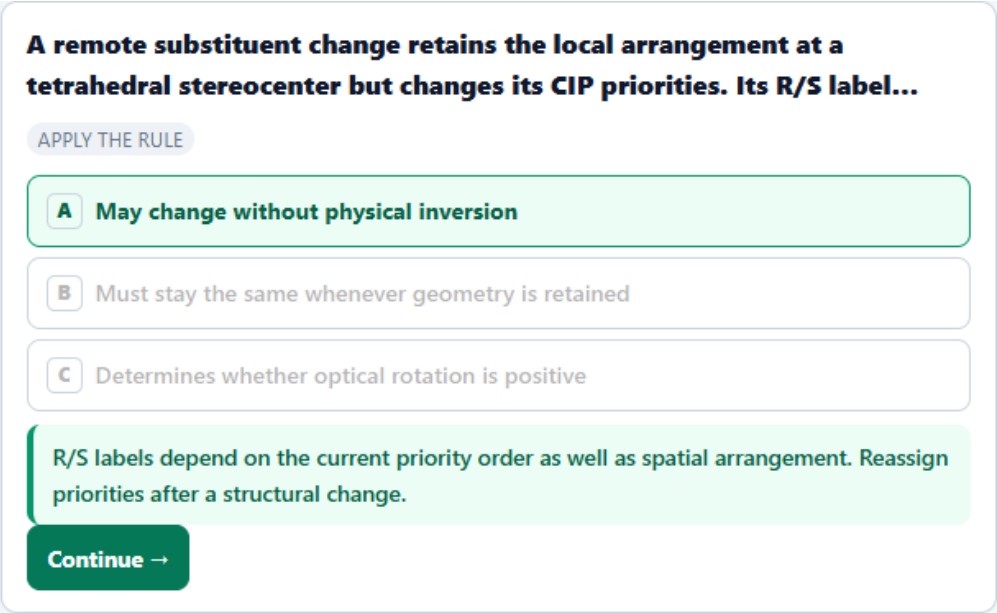
{"authorAnswer": 0, "tolerance": null, "highlightsAfterCorrectAnswer": {}}

Hint 1: Keep the geometry and priority ranking as separate inputs.

Hint 2: A descriptor change does not by itself prove inversion.

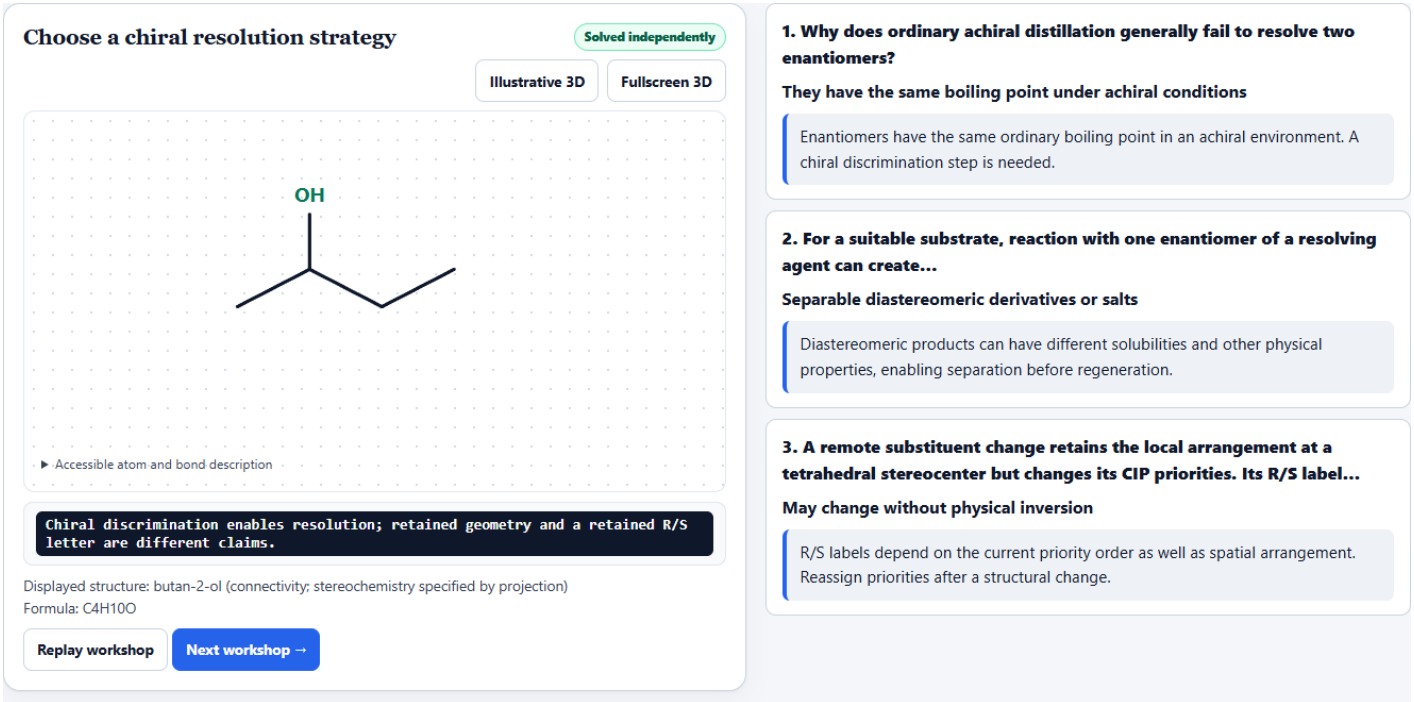
Explanation: R/S labels depend on the current priority order as well as spatial arrangement. Reassign priorities after a structural change.

Screenshot: chiral-resolution-stage-3-feedback



Completion summary: Chiral discrimination enables resolution; retained geometry and a retained R/S letter are different claims.

Screenshot: chiral-resolution-complete



Molecular repository: author claims to verify

ethanol

Author claims: ethanol | formula C2H6O | canonical composition C2H6O | net charge +0

Graph uses zero-based indices; A=0, B=1, etc. Coordinates are drawing positions, not 3D coordinates.

0 (A): C; x=50.00, y=115.00; charge=0; explicit H override=none

1 (B): C; x=108.00, y=85.00; charge=0; explicit H override=none

2 (C): O; x=166.00, y=115.00; charge=0; explicit H override=none

Bonds [atom A, atom B, order]: [[0, 1, 1], [1, 2, 1]]

ether

Author claims: methoxymethane | formula C2H6O | canonical composition C2H6O | net charge +0

Graph uses zero-based indices; A=0, B=1, etc. Coordinates are drawing positions, not 3D coordinates.

0 (A): C; x=60.00, y=110.00; charge=0; explicit H override=none

1 (B): O; x=120.00, y=75.00; charge=0; explicit H override=none

2 (C): C; x=180.00, y=110.00; charge=0; explicit H override=none

Bonds [atom A, atom B, order]: [[0, 1, 1], [1, 2, 1]]

isobutane

Author claims: 2-methylpropane | formula C4H10 | canonical composition C4H10 | net charge +0

Graph uses zero-based indices; A=0, B=1, etc. Coordinates are drawing positions, not 3D coordinates.

0 (A): C; x=50.00, y=115.00; charge=0; explicit H override=none

1 (B): C; x=108.00, y=85.00; charge=0; explicit H override=none

2 (C): C; x=166.00, y=115.00; charge=0; explicit H override=none

3 (D): C; x=108.00, y=30.00; charge=0; explicit H override=none

Bonds [atom A, atom B, order]: [[0, 1, 1], [1, 2, 1], [1, 3, 1]]

butane

Author claims: butane | formula C4H10 | canonical composition C4H10 | net charge +0

Graph uses zero-based indices; A=0, B=1, etc. Coordinates are drawing positions, not 3D coordinates.

0 (A): C; x=50.00, y=115.00; charge=0; explicit H override=none

1 (B): C; x=108.00, y=85.00; charge=0; explicit H override=none

2 (C): C; x=166.00, y=115.00; charge=0; explicit H override=none

3 (D): C; x=224.00, y=85.00; charge=0; explicit H override=none

Bonds [atom A, atom B, order]: [[0, 1, 1], [1, 2, 1], [2, 3, 1]]

cyclohexane

Author claims: cyclohexane (connectivity diagram) | formula C6H12 | canonical composition C6H12 | net charge +0

Graph uses zero-based indices; A=0, B=1, etc. Coordinates are drawing positions, not 3D coordinates.

0 (A): C; x=170.00, y=37.00; charge=0; explicit H override=none

1 (B): C; x=228.89, y=71.00; charge=0; explicit H override=none

2 (C): C; x=228.89, y=139.00; charge=0; explicit H override=none

3 (D): C; x=170.00, y=173.00; charge=0; explicit H override=none

4 (E): C; x=111.11, y=139.00; charge=0; explicit H override=none

5 (F): C; x=111.11, y=71.00; charge=0; explicit H override=none

Bonds [atom A, atom B, order]: [[0, 1, 1], [1, 2, 1], [2, 3, 1], [3, 4, 1], [4, 5, 1], [5, 0, 1]]

butanol2

Author claims: butan-2-ol (connectivity; stereochemistry specified by projection) | formula C4H10O | canonical composition C4H10O | net charge +0

Graph uses zero-based indices; A=0, B=1, etc. Coordinates are drawing positions, not 3D coordinates.

0 (A): C; x=50.00, y=115.00; charge=0; explicit H override=none

1 (B): C; x=108.00, y=85.00; charge=0; explicit H override=none

2 (C): C; x=166.00, y=115.00; charge=0; explicit H override=none

3 (D): C; x=224.00, y=85.00; charge=0; explicit H override=none

4 (E): O; x=108.00, y=25.00; charge=0; explicit H override=none

Bonds [atom A, atom B, order]: [[0, 1, 1], [1, 2, 1], [2, 3, 1], [1, 4, 1]]

butanediol

Author claims: butane-2,3-diol (stereochemistry unspecified) | formula C4H10O2 | canonical composition C4H10O2 | net charge +0

Graph uses zero-based indices; A=0, B=1, etc. Coordinates are drawing positions, not 3D coordinates.

0 (A): C; x=50.00, y=115.00; charge=0; explicit H override=none

1 (B): C; x=108.00, y=85.00; charge=0; explicit H override=none

2 (C): C; x=166.00, y=115.00; charge=0; explicit H override=none

3 (D): C; x=224.00, y=85.00; charge=0; explicit H override=none

4 (E): O; x=108.00, y=25.00; charge=0; explicit H override=none

5 (F): O; x=166.00, y=165.00; charge=0; explicit H override=none

Bonds [atom A, atom B, order]: [[0, 1, 1], [1, 2, 1], [2, 3, 1], [1, 4, 1], [2, 5, 1]]

e-butene

Author claims: (E)-but-2-ene | formula C4H8 | canonical composition C4H8 | net charge +0

Graph uses zero-based indices; A=0, B=1, etc. Coordinates are drawing positions, not 3D coordinates.

0 (A): C; x=60.00, y=50.00; charge=0; explicit H override=none

1 (B): C; x=115.00, y=100.00; charge=0; explicit H override=none

2 (C): C; x=185.00, y=100.00; charge=0; explicit H override=none

3 (D): C; x=240.00, y=150.00; charge=0; explicit H override=none

Bonds [atom A, atom B, order]: [[0, 1, 1], [1, 2, 2], [2, 3, 1]]

z-butene

Author claims: (Z)-but-2-ene | formula C4H8 | canonical composition C4H8 | net charge +0

Graph uses zero-based indices; A=0, B=1, etc. Coordinates are drawing positions, not 3D coordinates.

0 (A): C; x=60.00, y=50.00; charge=0; explicit H override=none

1 (B): C; x=115.00, y=100.00; charge=0; explicit H override=none

2 (C): C; x=185.00, y=100.00; charge=0; explicit H override=none

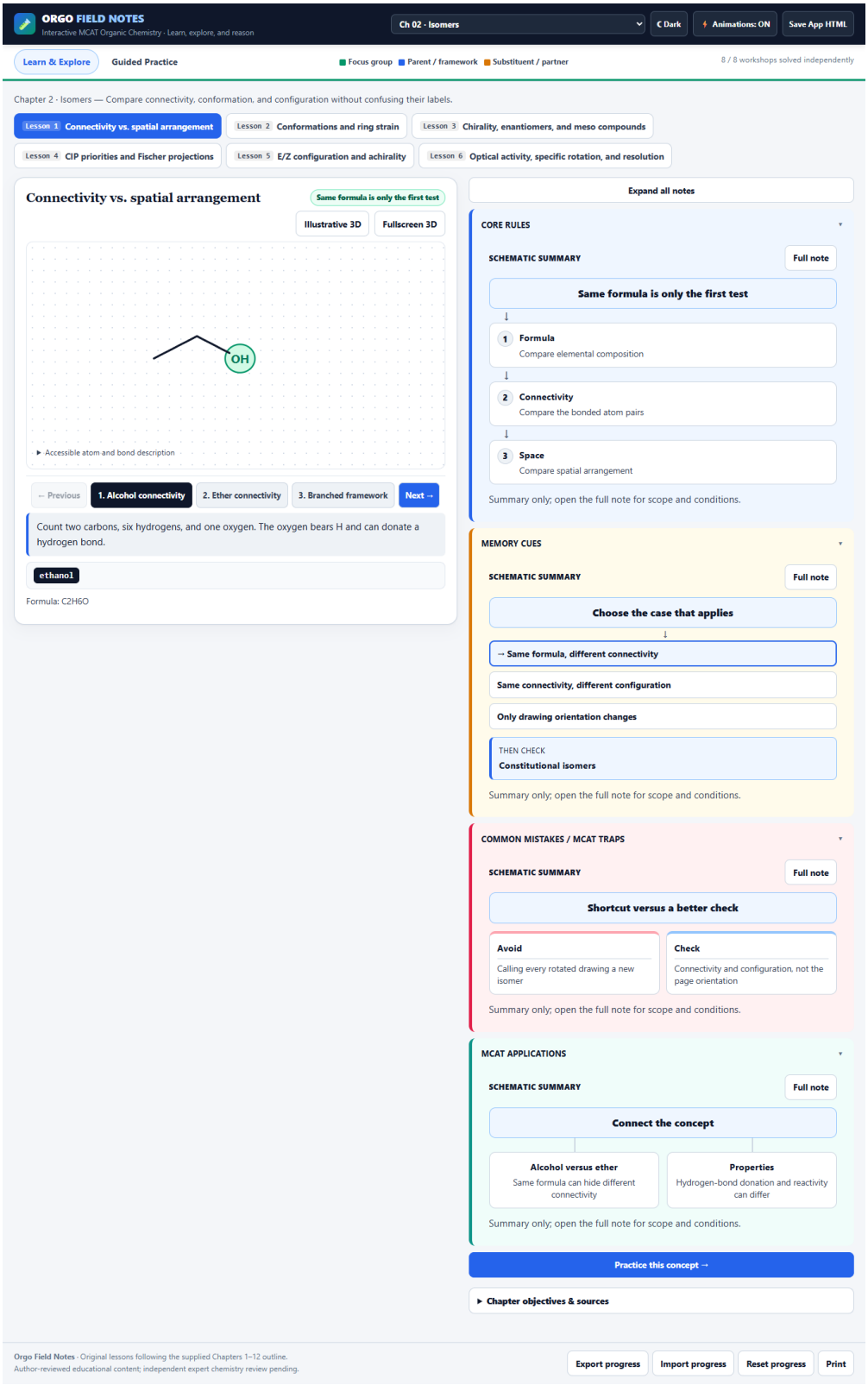
3 (D): C; x=240.00, y=50.00; charge=0; explicit H override=none

Bonds [atom A, atom B, order]: [[0, 1, 1], [1, 2, 2], [2, 3, 1]]

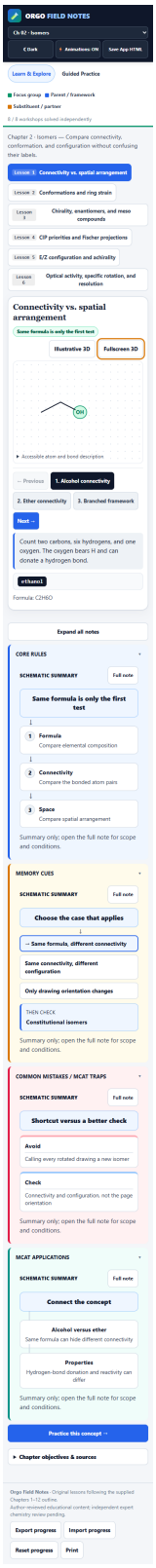
Layout samples

These overview images are scaled for page fit. Detailed structure and stage images appear earlier. Samples cover the first concept on desktop and mobile in both themes, not every responsive state.

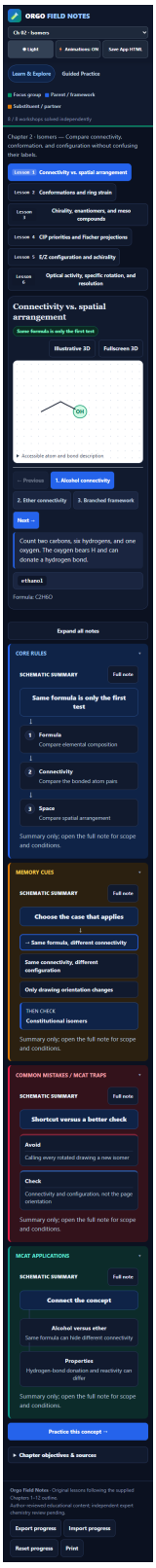
Screenshot: layout-desktop-light



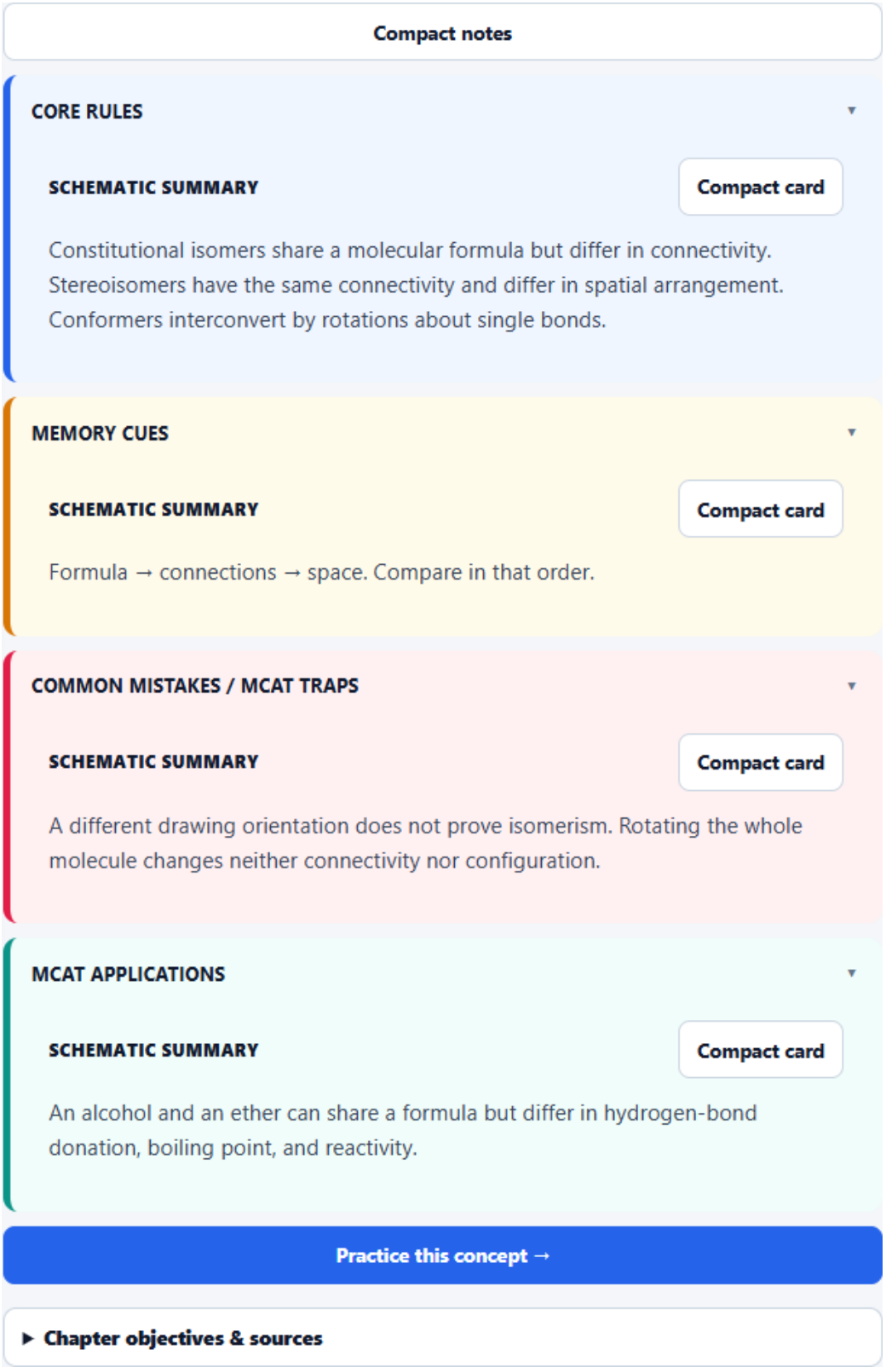
Screenshot: layout-mobile-light



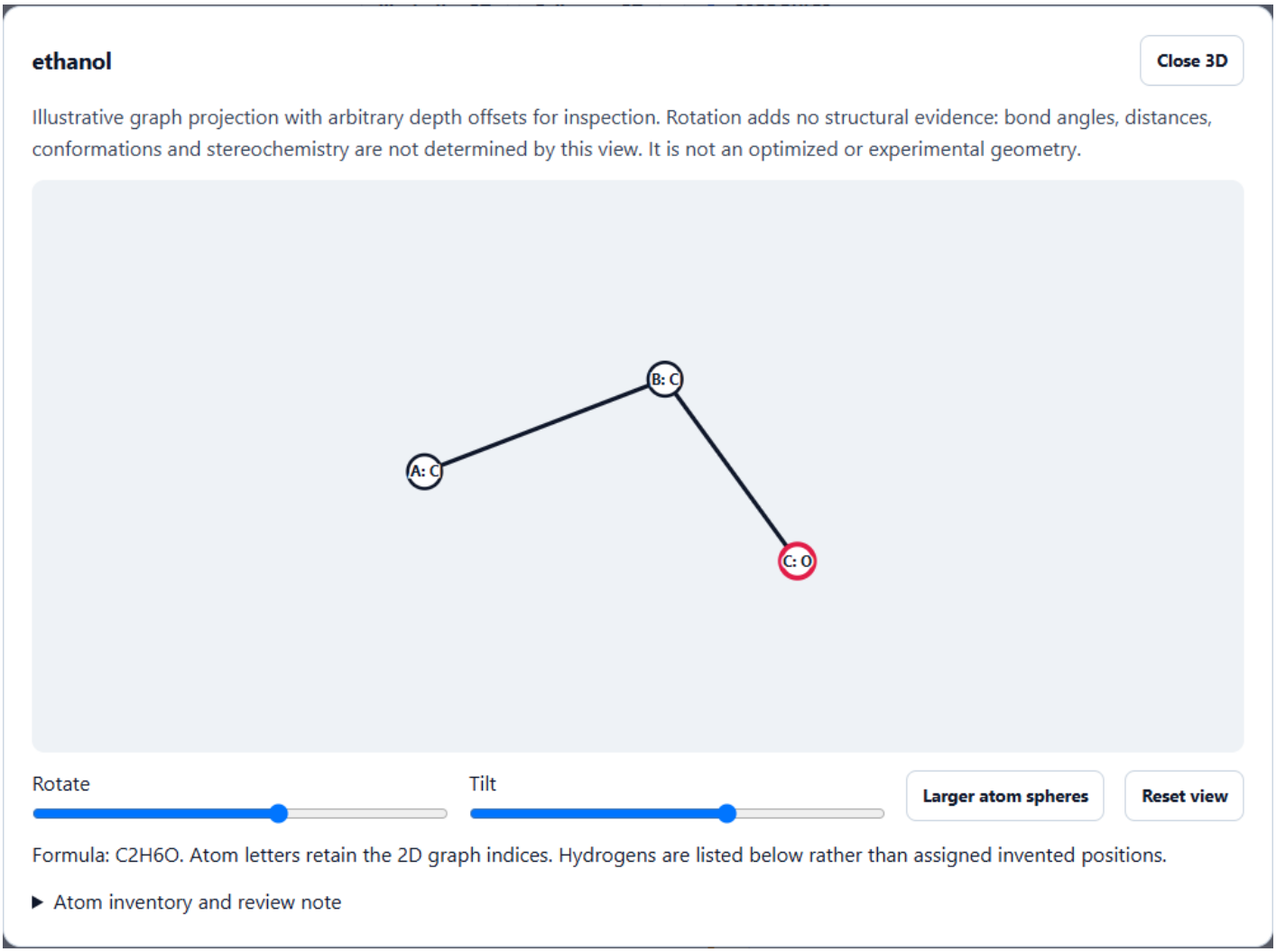
Screenshot: layout-mobile-dark



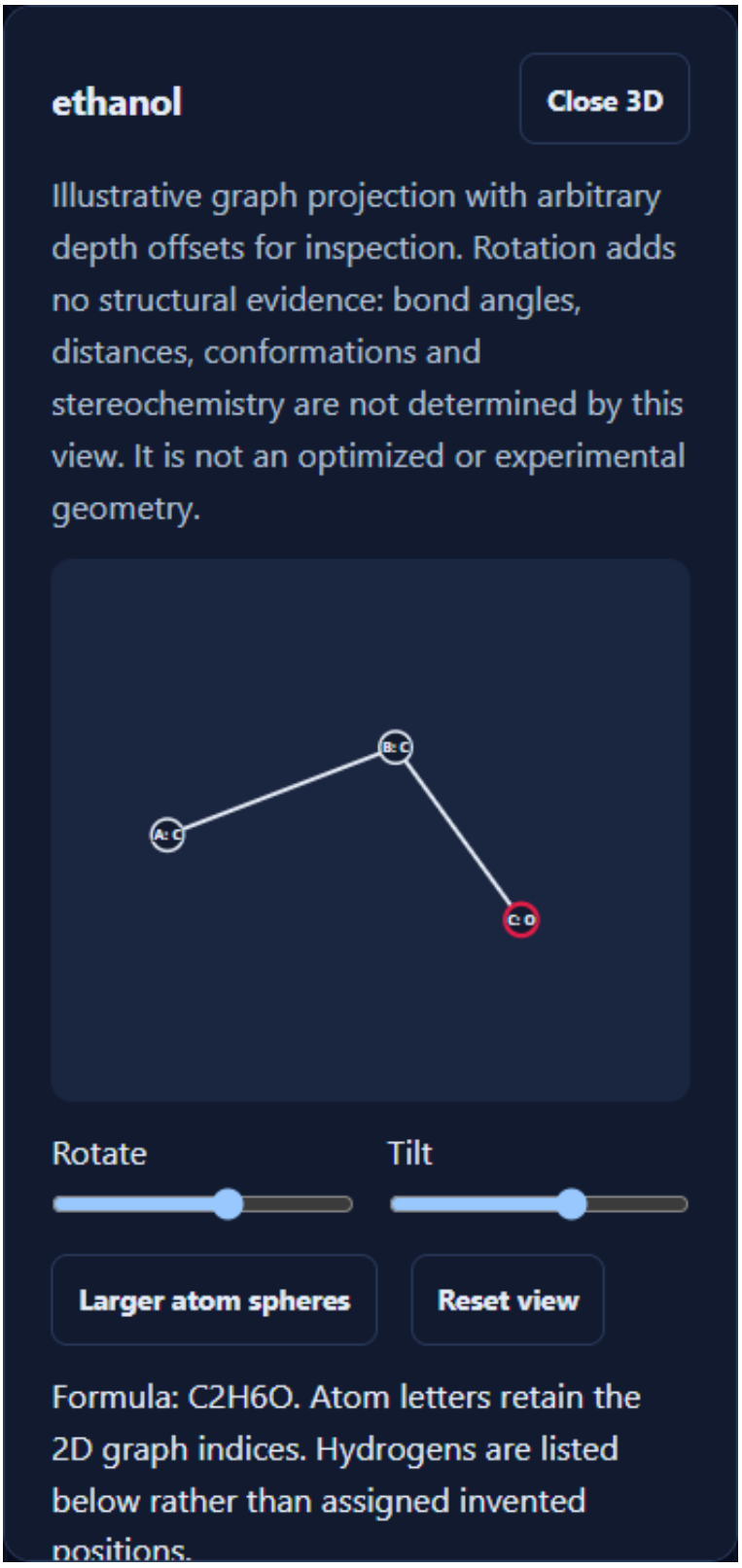
Screenshot: visual-notes-expanded



Screenshot: projection-desktop-light



Screenshot: projection-mobile-dark



Execution evidence

115 screenshots and 25 recorded interactions. Incorrect answers rejected and encoded keys accepted for captured guided stages. This checks implementation against its data; it does not validate chemistry independently. Static screenshots do not verify animation timing, arbitrary clicks, or screen-reader experience.